



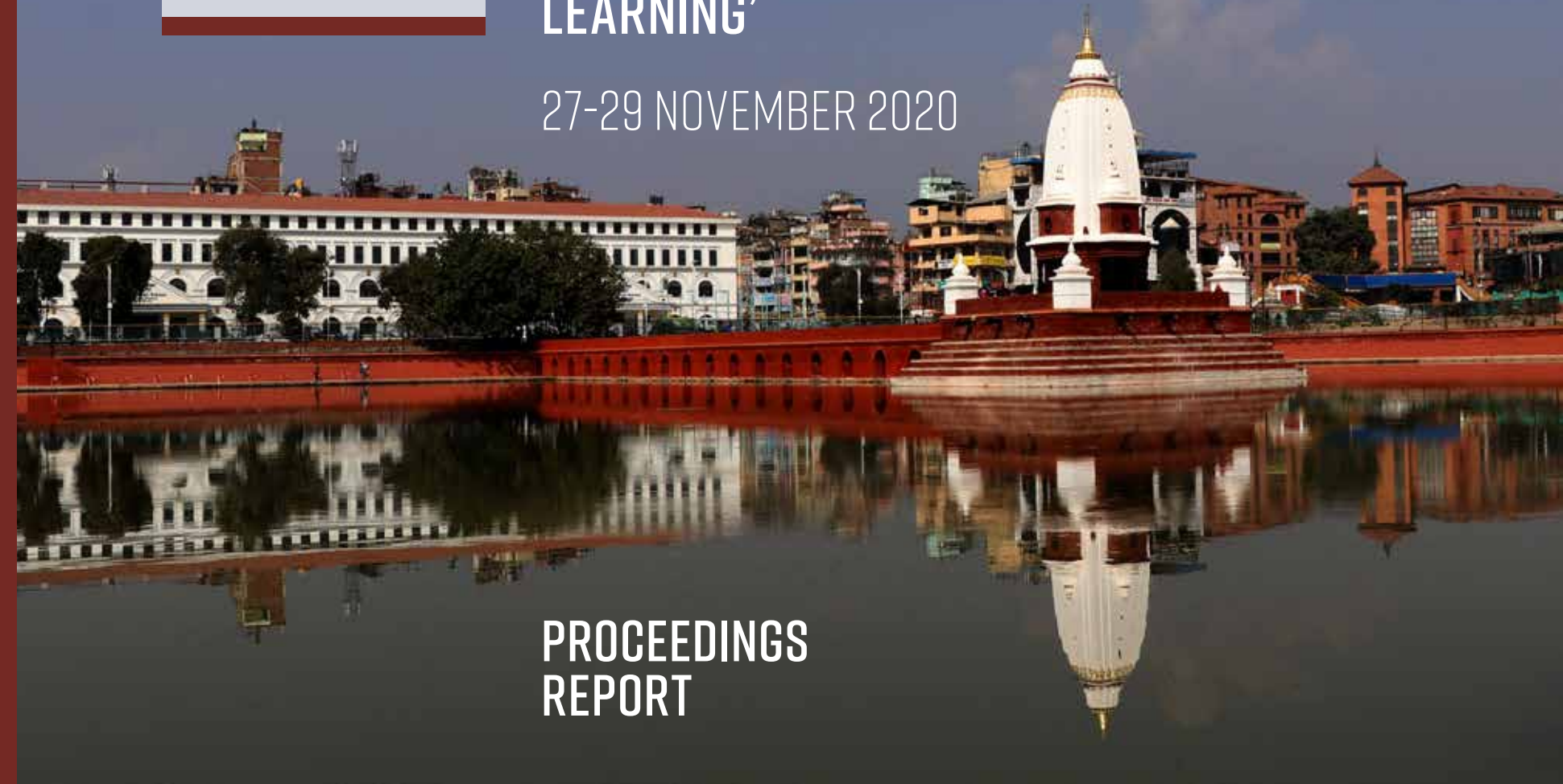
PROCEEDINGS REPORT NATIONAL SEMINAR



GOVERNMENT OF NEPAL
NATIONAL RECONSTRUCTION AUTHORITY

WEB-BASED SEMINAR ON 'NRA'S
ACHIEVEMENTS IN RECOVERY AND
RECONSTRUCTION - EXPERIENCE AND
LEARNING'

27-29 NOVEMBER 2020



PROCEEDINGS
REPORT



GOVERNMENT OF NEPAL
NATIONAL RECONSTRUCTION AUTHORITY



PROCEEDINGS REPORT NATIONAL SEMINAR





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The contents of this Proceedings Report of the Web-based Seminar on NRA's achievements in recovery and reconstruction reflect the summarised presentations and discussions of the numerous contributors, stakeholders, and participants of the event. They do not necessarily represent the official opinion or position of the NRA regarding the post-earthquake reconstruction process and outcomes



WEB-BASED SEMINAR ON 'NRA'S ACHIEVEMENTS IN RECOVERY AND RECONSTRUCTION - EXPERIENCE AND LEARNING'

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Preamble



Sushil Gyewali
Chief Executive Officer
National Reconstruction Authority



Preamble

The National Reconstruction Authority (NRA), as a ‘Special Purpose Vehicle’ (SPV), has been mandated with the coordination and implementation of Nepal’s recovery and reconstruction after the devastating Gorkha Earthquake of 2015. Together with a number of partners – Government line agencies, earthquake affected households, donors, Nepal Rastra Bank, commercial banks, and international and local NGOs – NRA has achieved huge successes, whilst overcoming numerous challenges faced along the way, during its five years tenure to date. More recently, the Government of Nepal (GoN) extended NRA’s time period for an additional one year.

NRA intends to capture and document its experiences, learning, and best practices. NRA has and continues to compile and expand a comprehensive Institutional Memory, and, in recent months, and as part of the process of ensuring a sustained legacy of its contribution to Nepal’s post-disaster history, is elaborating its own corporate document: the NRA Compendium.

In Nepal’s history, this will be the first document of its kind and nature. It was inspired by reflection and discussion with numerous stakeholders – both in and outside of government – and has been supported equally in its elaboration by these actors. It was also essentially driven due to NRA’s own experience of facing the post-earthquake reconstruction without having such

an all-encompassing set of information, guidance, and practices on which it had to base its own intervention; thus, it was felt imperative to provide this for future generations.

For taking forward the documentation of NRA’s learnings and best practices, NRA constituted the International Conference on Nepal’s Reconstruction (ICNR) Secretariat. The Secretariat conducted six thematic workshops at the outset of 2020, a National Symposium held in August, and this seminar held in November. All these initiatives maintain and sustain the dialogue around post-disaster reconstruction, engage all interested parties including both practitioners and academics, show a clear drive towards continued learning and sharing of knowledge, and promote the accumulation and compilation of these into documentary form that can play a valuable role – not just for Nepal but for other countries that like ours, suffer from naturally-triggered disasters.

We trust that readers of this Seminar Proceedings Report will not just take away ideas, but also build on them and apply the learning that they gain from this. We also trust that through further dissemination – in professional networks around the globe and in our upcoming International Conference next year we can all benefit from our experiences and learning in Nepal, and be better prepared for facing all future disasters, together. My thanks to you all!

Acknowledgements



Suresh Acharya
Secretary, NRA

Acknowledgements

After assuming the office at NRA, I am very happy to know that NRA is publishing a seminar proceeding in January 2021. In this new initiative, I would like to acknowledge the contribution of my predecessors Mr. Shivahari Sharma and Mr. Ram Krishna Sapkota for making this ICNR Seminar a success. Their dedication and continued guidance are manifested not only in ICNR achievements but also NRA's overall reconstruction activities. I reaffirm to all NRA stakeholders that I will apply my efforts to support the ICNR and all remaining reconstruction and recovery activities with a high level of integrity.

As a matter of fact, I would like to mention that NRA has not undertaken monumental tasks and responsibilities by itself; a vast array of actors and stakeholders must be accredited in line with political, financial, technical, logistical, and other support required along the way, and in its achievements and accomplishments. We would not have achieved this much without establishing cohesiveness among Governmental and Non- Governmental agencies.

First and foremost, appreciation must be given to the Rt. Hon. Prime Minister K.P. Sharma Oli, who firmly patronized the front line of the NRA, and chairs its Steering Committee. This kind publication was only realized due to an official decision provided by the NRA Steering Committee. NRA CEO Mr. Sushil Gyawali deserves a special thanks for managing such kind of events with tireless energy and commitment. Without his leadership to NRA over the past five years this kind of solidarity and smooth operation of the activities with extension of tenure for one more year would not have been possible.

At this juncture, I would also like to acknowledge the significant commitment, productivity, efficiency and organising role of the ICNR Secretariat, which is convened by Dr. Chandra Bahadur Shrestha, Executive Committee Member and supported by Mr. Manohar Ghimire, NRA Under Secretary as a Member Secretary of the Secretariat. Dr. Adi Walker, Director of the EU-funded NEARR Facility represents the Development Partners in the Secretariat, supplemented by Mr. Shyam Sundar Sharma and Mr. Sandeep Gurung, all of whom are actively engaged in this campaign through the financial, technical and logistical support by the European Union.

Additional support was provided to the ICNR Secretariat to implement this event successfully by Ms. Rita Lamsal, Section Officer NRA, Mr. Pradip Acharya, Mr. Dipak Sapkota, Mr. Vikash Bhusal, Mr. Janardan Nepal, and Mr. Ashok Sharma, as well as by the other NEARR Facility team members including Mr. Prakash Pyakurel, and Ms. Bindu Belbase.

Nevertheless, all chairs, moderators, commentators, and the participants themselves played extremely crucial roles, and all deserve our special thanks. Finally, the seminar would not have been possible without presentations made by the various speakers, supported by the ICNR Secretariat. The presentations succinctly captured and delivered the key messages as important take aways, and the ensuing discussions will certainly lead to an ultimately improved and high-quality Compendium. I am confident that this proceedings report will be a valuable resource for all the participants and practitioners of Nepal's reconstruction activities.



Foreword



Dr Chandra Bahadur Shrestha
NRA Executive Committee Member
ICNR Convener

Foreword

Following a highly successful National Symposium held in August 2020, NRA in its next venture of this kind hosted this Web-based Seminar in November 2020. Whilst the symposium focused on presenting scientific perspectives in the form of papers from a broad range of experts, practitioners and academics on the recovery and reconstruction process, this seminar was structured around the corporate learning undertaken by NRA over the past five years.

These experiences have been broken down into six main thematic sectoral areas, namely: private housing, heritage, public buildings and social infrastructure, livelihoods and economic recovery, rescue and relief, and governance and institutional arrangements. Whilst rescue and relief was not specifically within the mandate of NRA, the experiences from this period in the earthquake recovery formed the foundations upon which the work of NRA commenced. Additionally, as it is envisaged that NRA will hand over much of its responsibility and tasks to the NDRRMA, responsible in future for all post-disaster response in Nepal. Thus, the combined learnings from all thematic areas are captured and shaped as chapters within the NRA's corporate paper – its 'Compendium'. An additional presentation was made in the seminar on a planned impact assessment to be undertaken, exploring the effects of the reconstruction process on the socio-economic conditions of the affected population, that will equally feature in the NRA Compendium.

This web-based seminar provided the opportunity for lead stakeholders engaged within the different thematic areas – both from NRA and other actors and institutions – to present synthesised versions of the soon-to-be released Compendium chapters, whilst at the same time allowing for eminent and renowned figures within each sector to comment on the presentations, providing feedback and inputs that will be taken up in each compendium chapter's fine-tuning. Questions from the participants also raised key issues that will be reflected on and factored into the finally elaborated chapters of the compendium.

This 'continuous dissemination and learning practice' will ultimately culminate in presenting the compendium in an International Conference to be held from 23-25 April 2021, where NRA – together with its Development Partners – will also present their learning, experiences, developed best practices and recommendations for post-disaster response to the world. High level government officials from neighbouring countries in the region as well as prominent senior representatives from the most prominent donors and development partners will be participating. This international event will be held – Covid-19 dependent – in a hybrid format using innovative means of mediums of communication, both live and virtual.

We trust that this Seminar Proceedings Report contributes to the build-up and sharing of knowledge and continues to pave the way forward toward the international conference, and to the consolidation of NRA's documented legacy for Nepal.



Abbreviations



Abbreviations

BBB	Build Back Better	DRM	Disaster Risk Management		and Restoration of Cultural Property
CEO	Chief Executive Officer	DRR	Disaster Risk Reduction	DG	Director General
CLPIU	Central Level Project Implementation Unit	DuDBC	Department of Urban Development and Building Construction	ICNR	International Conference on Nepal's Reconstruction
CSEB	Compressed Stabilized Earth Block	EIA	Environment Impact Analysis	ID	Identity
CSSR	Collapsed Structure Search and Rescue	EQ	Earthquake	IDP	International Development Partners
DCC	District Coordination Committee	Er.	Engineer	JICA	Japan International Cooperation Agency
DDRF	District Disaster Response Framework	EU	European Union	KVPT	Kathmandu Valley Preservation Trust
DFID	Department for International Development	FGD	Focus Group Discussion	LEOC	Local Emergency Operation Center
DLPIU	District Level Project Implementation Unit	GESI	Gender Equality and Social Inclusion	LG	Local Government
DoA	Department of Archaeology	GMALI	Grant Management and Local Infrastructure	LNOB	Leaving No One Behind
DoLIDAR	Department of Local Infrastructure Development and Agricultural Roads	Arc.	Architect	M&E	Monitoring and Evaluation
DOR	Department of Roads	GoN	Government of Nepal	MIS	Management Information System
DPR	Detail Project Report	GVA	Gross Value Added	MoHA	Ministry of Home Affairs
Dr.	Doctor	HR	Human Resource	MoUD	Ministry of Urban Development
		HRM	Human Resource Management		
		ICCROM	International Centre for the Study of the Preservation		

NDCC	National Disaster Coordination Committee	NEOC	National Emergency Operation Center	NRCS	Nepal Red Cross Society
NDRF	National Disaster Response Framework	NGO	Non-Government Organization	NSET	National Society for Earthquake Technology
NDRRMA	National Disaster Risk Reduction & Management Authority	NIMS	National Incident Command System	NSNR	National Symposium on Nepal's Reconstruction
NEARR	Nepal EU Action for Recovery and Reconstruction	NPC	National Planning Commission	ODR	Owner Driven Reconstruction
		NRA	National Reconstruction Authority	CGI	Corrugated Galvanized Iron
				PM	Prime Minister

OPMCM	Office of the Prime Minister and Council of Ministers	SIA	Social Impact Analysis	USD	United States Dollar
PDNA	Post Disaster Needs Assessment	SOP	Standard Operating Procedure	GDP	Gross Domestic Product
PDRF	Post Disaster Recovery Framework	SPV	Special Purpose Vehicle	WASH	Water, Sanitation and Hygiene
Prof.	Professor	UK	United Kingdom	WB	World Bank
RC	Reinforced Concrete	UNDP	United Nations Development Programme		
RCC	Reinforced Cement Concrete	UNESCO	United Nations Educational, Scientific and Cultural Organization		





Seminar Agenda

DAY ONE:

27TH NOVEMBER 2020

Themes for Day 1

Private Housing, Heritage, Public Buildings and Social Infrastructure



Time 9:00-09:30

Program Day briefing

- Activities**
- Commencement of the Program (2 min)
 - Program briefing of the day (8 min)
 - Previous Day briefing (8 min)
 - Opening remarks (12 min)

Moderator Ms. Rita Lamsal, Dr. Chandra B. Shrestha, ICNR Convener
Er. Ram K. Sapkota, Secretary, NRA, Er. Sushil Gyewali, CEO

Time 09:30-11:00

Program Fourth Session: Livelihood and Economic Development

- Activities**
- Presentation by Dr. Bishnu Bhandari (30 min)
 - Response by Commentators
 - Response by the Thematic Team
 - Conclusion – Session Chair (10 min)

Moderator Chair: Er. Sushil Gyewali, Moderator: Dr. Chandra B. Shrestha

Commentators Ms. Jasmine Rajbhandari, WB (10 min), Mr. Vijaya Singh, UNDP (10 min)

Time 11:00-12:30

Program Fifth Session: Rescue and Relief

- Activities**
- Presentation by Er. Anil Pokharel (30 min)
 - Response by Commentators
 - Response by the Thematic team
 - Conclusion – Session Chair (10 min)

Moderator Chair: Er. Sushil Gyewali
Moderator: Dr. Chandra B. Shrestha

Commentators Mr. Leela Mani Poudel, Former Chief Secretary and Former Nepal's Ambassador to China (10 min)
Mr. Ramraj Narasimhan, UNDP (10 min)
Mr. Bipul Neupane, NRCS (10 min)

Time 12:30-13:15

Activities ■ Lunch Break and side event

Time 13:15-14:45

Program Sixth Session: Governance and institutional arrangement

Activities

- Presentation by Er. Ram K. Sapkota (30 min)
- Response by Commentators
- Response by the Thematic Team Members
- CEO's remarks (10 min)
- Conclusion – Session Chair (10 min)

Moderator Chair: Mr. Dhruva P. Sharma

Commentators Dr Marco Gemmer, EU (10 min)
Mr. Leela Mani Poudel, Former Chief Secretary and Former Nepal's Ambassador to China (10 min)

Time 14:45-16:30

Program Seventh Session: Work in Progress of the NRA Impact Study

Activities

- Presentation by Prof. Dr. Govinda Nepal (30 min)
- Response by Commentators
- Response by the Expert Team Members
- Conclusion – Session Chair (10 min)

Moderator Chair: Er. Sushil Gyewali

Commentators Mr. K. Akbar, WB (With support of Dr Nigel Fisher) (10 min)
Mr. Bishal Thapa, Economist (10 min)

Time 16:30-17:00

Program Summing-up the Day

Activities

- Summing up and way forward
- Remarks
- Concluding Remarks

Moderator ICNR Convener Dr. C. B. Shrestha (10 min)
Er. Ram K. Sapkota, Secretary (10 min)
Er. Sushil Gyewali, CEO (10 min)

Commentators



DAY TWO:

28TH NOVEMBER 2020

Themes for Day 2

Livelihoods and Economic Recovery, Rescue and Relief, Governance and Institutional Arrangements, Impact Study



Time	9:00-09:30	Commentators	Ms. Jasmine Rajbhandari, WB (10 min), Mr. Vijaya Singh, UNDP (10 min)
Program	Day briefing		
Activities	■ Commencement of the Program (2 min) ■ Program briefing of the day (8 min) ■ Previous Day briefing (8 min) ■ Opening remarks (12 min)	Time	11:00-12:30
Moderator	Ms. Rita Lamsal, Dr. Chandra B. Shrestha, ICNR Convener Er. Ram K. Sapkota, Secretary, NRA, Er. Sushil Gyewali, CEO	Program	Fifth Session: Rescue and Relief
		Activities	■ Presentation by Er. Anil Pokharel (30 min) ■ Response by Commentators ■ Response by the Thematic team ■ Conclusion – Session Chair (10 min)
Time	09:30-11:00	Moderator	Chair: Er. Sushil Gyewali Moderator: Dr. Chandra B. Shrestha
Program	Fourth Session: Livelihood and Economic Development	Commentators	Mr. Leela Mani Poudel, Former Chief Secretary and Former Nepal’s Ambassador to China (10 min) Mr. Ramraj Narasimhan, UNDP (10 min) Mr. Bipul Neupane, NRCS (10 min)
Activities	■ Presentation by Dr. Bishnu Bhandari (30 min) ■ Response by Commentators ■ Response by the Thematic Team ■ Conclusion – Session Chair (10 min)	Time	12:30-13:15
Moderator	Chair: Er. Sushil Gyewali, Moderator: Dr. Chandra B. Shrestha	Activities	■ Lunch Break and side event



Time	13:15-14:45	Activities	■ Presentation by Prof. Dr. Govinda Nepal (30 min) ■ Response by Commentators ■ Response by the Expert Team Members ■ Conclusion – Session Chair (10 min)
Program	Sixth Session: Governance and institutional arrangement	Moderator	Chair: Er. Sushil Gyewali
Activities	■ Presentation by Er. Ram K. Sapkota (30 min) ■ Response by Commentators ■ Response by the Thematic Team Members ■ CEO’s remarks (10 min) ■ Conclusion – Session Chair (10 min)	Commentators	Mr. K. Akbar, WB (With support of Dr Nigel Fisher) !10 min) Mr. Bishal Thapa, Economist (10 min)
Moderator	Chair: Mr. Dhruva P. Sharma		
Commentators	Dr Marco Gemmer, EU (10 min) Mr. Leela Mani Poudel, Former Chief Secretary and Former Nepal’s Ambassador to China (10 min)	Time	16:30-17:00
		Program	Summing-up the Day
		Activities	■ Summing up and way forward ■ Remarks ■ Concluding Remarks
Time	14:45-16:30	Moderator	ICNR Convener Dr. C. B. Shrestha (10 min) Er. Ram K. Sapkota, Secretary (10 min) Er. Sushil Gyewali, CEO (10 min)
Program	Seventh Session: Work in Progress of the NRA Impact Study		

Seminar Programme

Day One – 27th November 2020

Opening session



Ms. Rita Lamsal – Section Officer NRA, commenced the opening session of the web-based seminar by welcoming all to the programme at 9 am (Nepal time) on day one, November 27, 2020 at the Everest Hotel, Kathmandu, Nepal. She requested Er. Sushil Gyewali – CEO NRA, Er. Ram Krishna Sapkota – Secretary NRA, Dr. Chandra B. Shrestha – Executive Committee Member and ICNR Convener NRA, Mr. Dhruva Prasad Sharma – Executive Committee Member NRA and Dr. Hari Ram Parajuli – Executive Committee Member NRA to take their respective seats. She announced that the opening session would be chaired by Er. Sushil Gyewali – CEO, NRA.

Dr. Chandra B. Shrestha highlighted the objectives of the programme and briefly shared the experiences and learnings from the

NRA's recovery and reconstruction activities. He mentioned that the presentations were based on seven thematic chapters that will be making up the NRA Compendium – a document capturing its five plus years of experience in overseeing and coordinating the Earthquake recovery and reconstruction programme. He requested feedback from the audience to enrich the documents that were later presented in the thematic sessions. Further, Dr. Shrestha briefly introduced some lessons from six different thematic areas that are identified by the NRA. He also shared the highlights of the findings of the ongoing impact study of NRA elaborating on the impacts of recovery and reconstruction works on the socio-economic situation of the population affected.

Er. Ram Krishna Sapkota cordially welcomed everyone, including the organizers, compendium chapter presenters, commentators, moderators, session chairs and the participants. He mentioned that the NRA has been working massively in the recovery and reconstruction as well as documenting their work, processes and achievements, keeping in mind that at the end of NRA's tenure, the institutional memory could be transferred and kept safe for future reference. He stated that this event is also one of the contributing activities in the preparation of the upcoming international conference – ICNR which is scheduled to be organized in April 2021. In this light, this seminar could be considered as a knowledge sharing platform for interested people and communities, where



the six thematic chapters related to reconstruction in Nepal are to be presented, discussed and enriched with relevant feedback.



As his opening remarks, Er. Sushil Gyewali formally welcomed all contributors and participants for their productive engagements in the seminar. He stressed that all the knowledge and experiences achieved during the reconstruction works, as well as all the feedback and suggestions received from different sources in the last five years of NRA's existence have been well documented. Those documents have also recorded the processes adopted during reconstruction programmes and their implementation, and the issues and challenges encountered which have been resolved during the course of time. These processes and activities have remained helpful to acquire data and information as and when required to analyze and act upon. As a result, he stated that NRA has achieved huge success so far in the reconstruction and recovery process, i.e., 92% of the financing and implementation of the reconstruction of private and public houses have been completed,

with special attention paid to the marginalized and the most vulnerable survivors.

Er. Gyewali highlighted the success story of relocation of settlements on the basis of vulnerability analysis and evaluation. Following this process, the concerned households have been able to reconstruct their new houses and altogether a number of new settlements have been setup, and 5,000 households have been resettled in new places either in integrated settlements or in a place of their choice, which is safe according to the set standard criteria by the government of Nepal. This measure has ensured that every homeless or landless survivor should receive either a house or land or both according to personal/local need and set criteria. As a result, a total of 12,000 households have received the land to rebuild their houses. Similarly, another achievement may be noted that a total of 920 heritage and monuments have been identified for reconstruction, where 50% of them are in process of completion. The heritage reconstruction work takes additional time because there is a need to research traditional construction models and their compatibility with modern construction models and materials, to ensure maintenance of their traditional, cultural and religious values.

When the emergency situation occurred, all the political parties came together, promulgated a new constitution, formed NRA and began a massive recovery and reconstruction initiative.

Er. Gyewali shared the achievements in the education sector wherein of 7,553 damaged school buildings, more than 80% have been completely reconstructed and the rest are in the process of completion by the end of this fiscal year. Similarly, reconstruction of health facilities is also encouraging with 60% completion. The progress in social infrastructure is also good as 800 km road is in the final stage of completion, and 1,500 water supply projects are also in the stages of completion. He further mentioned that the progress so far achieved was not free from problems and challenges. However, NRA always tried their best to resolve every issue as quickly as possible by finding appropriate solutions. These problems included political disruption that was in its peak during the time when the earthquake hit. When the emergency situation occurred, all the political parties came together, promulgated a new constitution, formed NRA and began a massive recovery and reconstruction initiative. Similarly, Nepal's difficult geographical terrain, poor infrastructure, lacking human and financial resources, and different levels of election and political transitions from a unitary to a federal government were problematic to gain access to communities and their priorities for reconstruction work at local levels. In later times, Covid-19 brought another level of challenge that has demanded new protocols for reconstruction. Further, initially, whilst all the people and communities initiated their own reconstruction works which was commendable, there were some confusion between service receivers and providers.

Talking about learning, he inferred that the technology transfer was so smooth that people have accepted the designs of houses provided by the NRA. As a result, resilient houses are built. The banking system developed has helped beneficiaries to get money

and protect themselves from fraud and bribery. NRA reinvented the traditional technology by blending structural engineering with archeological value in the archaeological sites, and the 7 reconstructed world heritage sites are ready to welcome tourists soon. Similarly, the Prime Minister's office and Dharahara is going to be completed within a couple of months. As an autonomous entity, the government has extended NRA's tenure to 2021 to complete its assigned task through mobilization of communities and human resources from government agencies. After 2021, a newly formed agency will take over so that Nepal's whole reconstruction process, action and achievements will be kept safe as knowledge and experience for future reference. In this perspective, documentation has been started along with a number of scientific papers related to earthquake safety. In addition, the documentation of the profiles of the partners will also be useful to develop functional partnerships for the future. Lastly, on trained human resources, elected representatives and local communities who were involved in this reconstruction process are the assets of the country for making communities safe and resilient. Hence, appreciation and collaboration with these stakeholders can bring a big change in the efforts of Nepal's plan to ensure prosperity and resilience. With this, he officially closed the opening session.

Thematic sessions

There were three thematic sessions in the first day of the seminar with seven presentations under the areas of private housing, cultural heritage and public buildings and social infrastructure.

Day - 1, 27th November 2020

First Session: Private Housing

Presenter: Dr. Chandra B. Shrestha

Commentators: Mr. Kamran Akbar – World Bank (supported by Dr. Nigel Fisher), Dr. M. N. Khan – USAID (supported by Dr. R. Guragain), Mr. Ram Prasad Bhandari – JICA, and Ms. Maggie Stephenson – Independent expert

Session Co-Chair and Moderator: Er. Ram Krishna Sapkota

Session Chair: Er. Sushil Gyewali





Private housing: overview



27 November 2020

2015

- 32 of 77 districts affected by 2015 Gorkha Earthquake
- 755,549 beneficiaries identified (498,852 FD + 256,697 PD)
- 8,975 people died and 22,000 people injured
- From US \$ 7 billion losses, half was in private housing sector

2020

- EQ victims in safe permanent shelters
- 87% nearing completion (TB – 791,787)
- 90% (of 4,257) vulnerable and 12, 284 landless households resettled
- Generated at least 240 million person days: employment for 200,000 people for more than 3 years



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Challenges and issues



27 November 2020

Institutional and organisational

- Lack of local governments at the initial stage
- Preliminary level of disaster preparedness
- CBS's limited know how on running beneficiary surveys
- Political lens suspected in beneficiary selection
- Dissemination of Housing Technology particularly retrofitting
- Land and finance problem in urban housing
- High frequently changes in CEO, Secretary and other senior NRA staff
- Paucity of Engineers and Masons in the market
- Engineers absence from their designated sites
- Pre-occupied NGOs
- Almost non-existent insurance policy

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Challenges and issues



27 November 2020

Operational

- Non-functional “Reconstruction Fund”
- Defining vulnerable households a struggle
- Widely-scattered houses on vulnerable land
- Mobilization of over 3,000 Engineers and 70,000 plus masons
- Unavailability of land in mountainous districts for resettlement
- Access to finance - serious problem for urban sector
- NRA's treatment as a typical Line Ministry instead of a ‘special instrument
- Conservation of vernacular architecture



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Strategies and approaches



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- Build Back Better through available and innovative technology
- Owner driven approach
- Primarily in-situ reconstruction
- Uniformity in grant amounts
- One household - one grant
- Grievance handling – ensured justice
- Beneficiary methodology - combination of census and verification model
- Promotion of Labor barter arrangement (*Parma System*)
- Special focus on vulnerable and marginalized households
- Demand based capacity development of technicians and masons
- Engagement of local governments as far as possible
- Mobilisation of concerned government agencies in their sectors of expertise

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Lessons learned



- Reconstruction policy created reconstruction framework
- National and institutional leadership was key for successful reconstruction
- Three tranche system ensured earthquake resilient reconstruction
- Consolidated 80 dispersed settlement with livelihood opportunities
- Housing database with local governments for handling all disaster cycles
- Retrofitting has great potential to protect household shelter, save assets and heritage
- Urban housing requires intervention in land and access to finance
- Reconstruction agency with clear chain of accountability
- Extra cautious approach for supporting marginalized households
- Grievance handling – an exemplary action to maintain justice
- NGOs need proper guidelines and encouraging environment
- Revalidation of compliance – ensured housing resilience

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Lessons learned (cont.)

- Risk of politicization of reconstruction
- Communication is as important as taking decisions and implementation
- Meticulous planning to meet demand surge of all construction workers
- Division of labor between politically appointed policy makers and government bureaucrats must be outlined in the Reconstruction Act
- The SPV should be open towards new construction technology and methods
- Banking system tranche payments contributed to success in reconstruction
- Achieved political consensus through Acts and by conduct



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Conclusions



- House reconstruction in urban areas is extremely difficult
- Local governments should maintain housing database systems
- The owner driven approach was successful but requires adjustment
- Damage assessment survey lacked local government steering
- Tranche payment through banking structures should be integrated in the system
- Vulnerable households need settlement in safe locations with livelihood opportunities
- The banking system could not ensure housing loans for needy households
- Implementation through normal bureaucratic policies & systems was ambitious
- Transparent policy & fact-based decisions prevent unwanted political incursions

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Key messages

1. Local governments should establish and maintain housing databases
2. Organised resettlement policy must be initiated for vulnerable settlements
3. Building registration practice should be initiated in rural municipalities
4. Retrofitting technology needs to be disseminated nationwide
5. Organised & focused awareness campaigns are essential for housing resilience
6. A functional mechanism for access to finance needs to be assured
7. The present NDRRMA requires further strengthening
8. Empowering women and marginalized communities must be continued
9. Technological sensitization in 32 districts needs nation-wide application
10. NRA's disengagement strategy must be designed and implemented with the National Planning Commission, NDRRMA, MoUD and local governments

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NRA Compendium Web-based Seminar - Private housing

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Presentation 1: “The world’s largest owner-driven housing reconstruction programme”

Dr. Chandra B. Shrestha introduced the background, housing reconstruction programme, progress made, challenges and issues, lessons learned, his conclusion and several key messages. Talking about the background, Dr. Shrestha stated that empirical evidence proved that Nepal’s initiatives for the recovery and reconstruction is one of the world’s largest owner-driven housing reconstruction programme, covering 32 administrative districts, to recover an approximate devastation of 7 billion USD, in which 50% of losses were in the private sector. Similarly, approximately 88 – 92% people had to be re-sheltered, where NRA had a primary role. As the right to shelter is acknowledged by the government, shelter for landless and homeless people was put as a priority. This massive reconstruction work has generated an approximate 240 million person working days to complete. To date, no other projects have created such a huge number of employment opportunities at local levels. He visualized the scenario just after the 2015 Gorkha Earthquake resulting in 8,975 deaths, 22,000 injuries, 755,549 people suffering from the losses of loved ones and properties of estimated 7 billion US\$, of which half were private houses. Later, these survivors were identified as beneficiaries. This situation has changed completely by the end of 2020, as the beneficiaries are shifted to reconstructed safe permanent shelters; 87% nearing completion (TB – 791,787), 90% (of 4,257) vulnerable and 12,284 landless households resettled, generating at least an employment for 200,000 people for more than 3 years. As for challenges, in

the initial stage there were no elected representatives at local government level, and local arrangements for disaster response were not in existence. The issue of land ownership and other local disputes prevailed in the urban communities. So, NRA is a little behind in the urban areas as compared to rural areas. Also, there were challenges in managing the human resources, absence of house insurance & other operational challenges to equip NRA with laws, procedures and guidelines, defining vulnerable households, reaching out to the dispersed settlement in high altitude, etc. He concluded the presentation with some recommendations and key messages as per the preceding presentation.

Immediate response by the Co-Chair and Moderator



The Co-Chair and moderator for the session Er. Ram Krishna Sapkota summarised the important points from Dr. Shrestha’s presentation saying the presentation was very comprehensive and fruitful because it covered all important sub-themes under the main theme of private housing, highlighting the background and situation of the earthquake and the changed context of 2020. The moderator then welcomed the commentators and requested them to provide their comments.

Response by the Commentators



Mr. Kamran Akbar appreciated the presentation and its content. He emphasized that NRA had a real challenge in each and every step of the recovery and reconstruction process. NRA only began the real work after 9 months but quickly covered every aspect. He highlighted the payment through the banking system ensured transparency and is also helpful in the financial inclusion of a large number of households where 30% are women. He acknowledged that it is a good step towards achieving inclusiveness. He talked

about the smooth promulgation of regulations and amendments of laws which show the informed coordination and collaboration with and among the partners, political parties and the three levels of government. He was glad to know about that the future successor of NRA – NDRRMA, and suggested that the newly born agency needs full autonomy. He then invited Dr. Nigel Fisher for his comments.



Dr. Nigel Fisher appreciated the content of the presentation. He commended NRA on its huge achievements considering the timeline and the challenges. Developing and incorporating the provisions that were developed in reconstruction, housing, lands, property rights and ensuring that these legislations and policies are adopted, was very appreciable. Also, making provisions for the vulnerable in terms of location, marginalization and poverty, and conducting risk analysis were just a few of the challenges which had to be managed. In terms of housing design and standards, a variety of training programmes were conducted for engineers

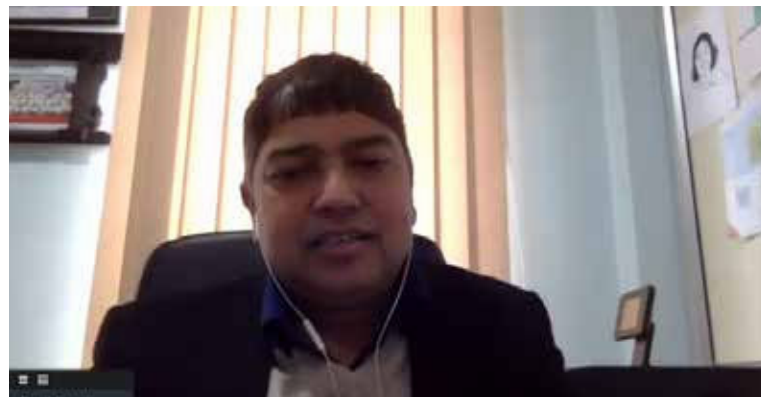
and masons. The chapter presented has the scope for knowledge generation and lessons learning for future generations. In addition, other countries may learn the lessons generated by NRA. If NRA properly documents their knowledge and experiences, the NDRRMA can take this forward which is very important for NRA's legacy.

If NDRRMA is going to take the charge, it is important to make sure that NDRRMA is actually operational to the level that NRA would like to see.



The next commentator Dr. M. N. Khan appreciated the content, lessons and messages included in the presentation. He mentioned that the whole initiative of recovery and reconstruction of Nepal is the largest housing project he had ever worked with. Now, NRA can contribute towards the sustainability of the process of

reconstruction by making sure that the lessons learned be transformed into measurable actions for the future. However, there are still issues like inter agency communication problems, cooperation, retrofitting and management of NGOs, etc. which demand a need for clear laws, working procedures and guidelines to initiate and manage emergency response activities, in case of any disaster in the future. He suggested that it is high time to have clarity on who is going to take charge when NRA's tenure will be over next year. If NDRRMA is going to take the charge, it is important to make sure that NDRRMA is actually operational to the level that NRA would like to see. Finally, he made a strong suggestion that NRA should address an important yet urgent need of how to engage the private sector in the recovery and reconstruction process, to create the markets for economic contribution and growth, and ensure a sustained and resilient reconstruction process.



Dr. Ramesh Guragain expressed his happiness for capturing the success stories and lessons learnt in a well described presentation.

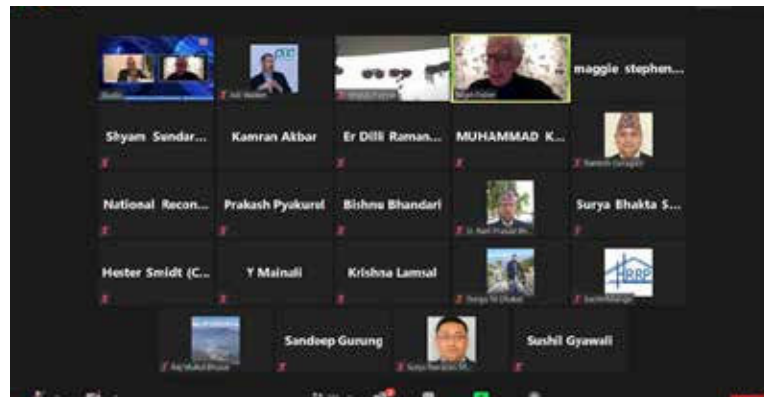
He highlighted that one of the most remarkable works done was the training and mobilization of technical engineers and local masons, who were the main pillars of the reconstruction process. He briefly listed some of the praiseworthy works of NRA: the skill/capacity enhancement in local government/ward level, and emphasis given to the revision and implementation of building codes and mandatory approval for construction/reconstruction works, etc. He shared that providing building models was good but providing them late created some confusion in selection of the most suited model. Hence, in many places the prescribed building models did not work. Therefore, reconsideration is needed to update the building models to be prescribed in future. Similarly, there is an urgent need for strict provision of building permission, its monitoring and evaluation as well as punishment for the rule violators. A provision has to be made for the mobilization of social mobilizers/facilitators together with technical engineers so that they can help people realize the importance of building codes, models and need for technical assistance while doing any construction at their level. Also, the local governments need skillful human resources for damage assessment to make sure everyone follows resilient construction procedures. Finally, he mentioned that the NRA's work was well coordinated within and outside of the agency for knowledge sharing and resource mobilization. Since NRA is an established agency it should have a plan for at least the next one year to make an easy handover of all its knowledge, experience, achievements and documents. He suggested for scaling up the programme and baseline data at local government level, emphasizing on the building permit system and building code implementation, and highlighting the usefulness of technology that worked well

and made improvements in damage assessment, ensuring easy access for service seekers.



The third commentator, Mr. Ram Prasad Bhandari started off sharing JICA's experience of reconstruction and recovery. He mentioned that JICA has provided support to 109 wards in Gorkha. Mr. Bhandari emphasized that the compendium document reflects the reality. He suggested that the word local government in third slide should be replaced by the elected government because there was an absence of elected representatives. In the presentation, one housing challenge particularly mentioned is retrofitting, but in fact technology was also a major challenge. In slide 5, Parma is mentioned as a strategy, instead it should be mentioned as a lesson learned. He stressed that monitoring is essential even after the final payment as some people are not completing their construction. Another important issue that needs to be tackled is having a housing data base. If it is done, beneficiary identification would be easier. He suggested to add community mobilization

and a data base on the demand and supply of trained engineers and masons as a lesson learned. Build back better is missing in the presentation, which needs to be mentioned clearly as NRA follows that approach. The approaches of no one left behind and core housing unit are needed to be discussed further and incorporated in the compendium chapter when finalized. Some other specific observations he made were – the number of districts was mismatched, so consistency should be maintained in future. The mobilization and motivation of technical staff were not addressed in presentation. The prescription and dissemination of housing models, retrofitting models and their processes, have been serious issues. He advised to: reinforce and reemphasize resilient reconstruction; add community enhancement for identifying problems and finding solutions in local reconstruction; match technology and masons training to promote better construction; create better databases and adopt better documentation for future references; establish market hubs and resource centers so that quality construction may be sustained.



The last commentator, Ms. Maggie Stephenson opined that Nepal's recovery and reconstruction initiative is not only the biggest own-driven reconstruction programme but also the most difficult in terms of both physical and economic reasons for the government as well as for the individual survivors. She started with the background that Nepal was not really prepared in terms of capacity in rescue and response for the survivors. Therefore, it was observed that people began to reconstruct their damaged houses before NRA came. Even though NRA had a big role, the majority of the houses were built by the people themselves, so credit should be given to the people as far as possible. She suggested simplifying the banking transaction procedures – good timing and better communication with the beneficiaries would be helpful to learn the terms and timings of action, i.e., damage assessment and planned activities. She stressed that the reconstruction may go for some more years, so reconstruction plans need to be expanded for some more years. Other issues she highlighted were work completion and quality assurance, which are not well captured in the presentation. Reconstruction accompanied with safety, energy, water and sanitation access is essential for resilient living. There are still big questions around how the transfer of skill, experience, capacity and knowledge to the next generation are going to take place successfully.

As long as complexity in habitation increases, the community support mechanisms, financial mechanisms and so on become more complex and inaccessible especially for marginalized people.

Our world is urbanizing every day and the more urbanized society becomes, the more complexity arises especially in disasters management. As long as complexity in habitation increases, the community support mechanisms, financial mechanisms and so on become more complex and inaccessible especially for marginalized people. So, cities have more challenges as well as opportunities to learn and develop strategies for safe living in future. Resilience of the Nepali population and communities seems commendable as they repaired or built shelters within the first few months. Timelines and phases of projects need to be communicated to the people, so that they get prepared for each tier or type of support. The recovery and reconstruction process may continue for several generations, so the mechanism for implementation and monitoring has to be set transparently. Similarly, housing resilience is mentioned but that is not yet combined with other housing and infrastructural support, i.e., WASH.

Response by the thematic team



Mr. Surya Narayan Shrestha from NSET expressed his appreciation for his participation and of the contents included in the compendium document as well as in the presentation. He pointed out that the smaller lessons which are not significant in numbers also need to be captured and recorded in the most appropriate way. Because a negligible amount of small work may bring greater learning and effects in the whole process. He added, a dynamic shift in policy could be beneficial to the world as well as to Nepal. Therefore, the documentation process should reach a logical end to enable this.



Mr. Anil Pokhrel – CEO NDRRMA also expressed his satisfaction with the compendium document as well the presentation. He shared his experience that some of the laws formulated immediately after the 2015 Gorkha earthquake failed to clearly identify the needs and role of recovery and reconstruction for similar disasters, likely to occur within our life span again and again. Agreeing to the comments made by others on the presentation, he added that there is a need to build capacity within the NRA itself for the

moment, and for the NDRRMA to effectively continue this kind of multi-level communication, dialogue, plan and execution of emergency responsive recovery and reconstruction in future too.

For professional development and transfer of learning, we need to work with the universities to develop curricula and disseminate through national media.

Presenter's views on comments

Dr. Chandra B. Shrestha expressed that he is overwhelmed by the number of acknowledgements and comments received. He appreciated the generous support from all the support agencies, the professionals and the academia. NRA is trying its best to engage all other agencies, like NPC, relevant ministries, NDRRMA, etc. to take forward the recovery and reconstruction process. He reminded that the third day of this seminar will witness the Asian Shelter Forum 2020, as a symbol of NRA's policy and practice for a wider engagement of partners. NRA is in line with promoting vernacular architecture for safe shelter, simplifying as per the community demand but also accommodating the aspirations of the communities. In this process, it is worth noting that NRA has so far brought a total of 800,000 households to the banking system, has recruited and trained about 3,000 engineers and has capacitated huge numbers of masons at the community level for resilient construction, but there is an issue of sustaining them and make them available when they are required. In the meantime, certification for the completion of buildings is an issue to

be addressed permanently, therefore needing a mechanism to be established for proper monitoring of the construction, evaluation of the completion of work, and recommendation for the certification for the completion of work associated to the recovery and reconstruction process.

Dr. Shrestha assured that the comments raised here will be incorporated in the proceeding report as well as in the Housing chapter of the Compendium. He summarized the suggested points to incorporate: the link up of rural people to the banking system as it helps to monetize the economy; replication of best practices to other districts is a challenge but we have a year to try; political pressure will be handled on the basis of professional grounds and ethics, even though NRA executives are appointed through the political parties; NRA will make sincere attempts to have disaster friendly policies. NRA is very sincere towards the sustainability of NDRRMA and intends to address the widespread question of whether and to what extent NDRRMA will be autonomous. The discussion is already initiated, and suggestions have started to come which surely will bring a meaningful result. In this discussion, the universities and other likeminded institutions are also being engaged.

Comments & queries from chatroom and social media

Rupesh Shrestha: Since NRA's deadline is next year, are there any plans on how to continue urban housing recovery, any strategic policy, directives or decisions happening soon?

Naresh Shrestha: Not only housing data is needed; they should prepare disaster risk reduction plans and find out vulnerable and hazard prone zones to prepare for upcoming disasters. What is the national and provincial plan or budget to prepare for DRR planning?

Deependra Sharma: Dr. Chandra, you have well highlighted the need for local governments maintaining strong housing databases. There is a strong need for enhancing the capacity of local governments in this regard. It is, however, equally important to ensure that the local governments are capable to monitor the quality of construction of owner-driven houses to promote disaster resilience. In the post EQ reconstruction, more than 3,000 engineers and other technicians were there to supervise and facilitate the process. But, as we are aware, in general, this level of monitoring is not availed by the local governments. Bridging this gap of capacity in monitoring by the LGs is one of the key issues in the housing sector that needs to be addressed.

NRA is in line with promoting vernacular architecture for safe shelter, simplifying as per the community demand but also accommodating the aspirations of the communities.

Radha Verma: Thank you Dr Chandra. It has been great for UK aid to engage on a lot aspects mentioned – housing for those most vulnerable, resettlement, retrofitting, geohazard risk mapping, grievances, re-surveying, coordination, etc. There are lots of

lessons learned and our partners continue to make sure they are housed and hooked into relevant areas/departments/NDRRMA.

Deependra Sharma: Preventive retrofitting is quite important in the Nepali context given the scale of earthquake disaster risk that the country is exposed to. The lessons learnt from post-EQ reconstruction on retrofitting should be well documented and the knowledge transferred to the MOUD, NDRRMA and the local governments.

Sachin Malego: I think for what Dr. Guragain mentioned about having damage assessment experts in place, I would advise to include this in the curricula of technical studies as well as mandatory skill training for local level government who later can cascade this knowledge further.

Sanjeeb Ghorasainee: What are some lessons learned from reconstruction of integrated settlements? I believe many of these efforts have shown characteristics of Community-driven and Donor-driven approaches – for example, two settlements in Giranchaur and Melamchi Municipalities. What implications can be drawn from these housing projects, given that the owner-driven approach was the general strategy adopted by NRA? How do the results from these DDR/CDR projects compare to ODR made individual houses?

Anonymous user: The major argument for the success of the ODR approach in the presentation came from the number of houses built in these 5 years, but what are your views on the housing quality such as the newer foreign designs adopted, size of the new houses, and housing amenities and other social infrastructures around the reconstructed houses? What learning lessons can be drawn from the very technical and bureaucratic approach of rural

housing reconstruction process which many believe left little room for creativity from within the communities, nor emergence of local leadership, and supporting tight bonds between neighbors and a sense of community that was further developed during the immediate aftermath of the earthquake?

Deependra Sharma: Government has launched a cash grant programme to replace the thatched roofs of private houses by CGI roofs. If this program could be adjusted to meet the requirements of preventive retrofitting, this could be a huge leap forward in Nepal's context in relation to promoting disaster-resilient houses.

Sachin Malego: Based on ActionAid's experience in the reconstruction project in Lamjung, I just wanted to highlight and emphasize the special attention needed for marginalized and vulnerable households to complete the reconstruction process inclusively. The social mobilization with sociotechnical support is crucial and this shall be well-captured in the reflection. Thanks for the very nice presentation and reflection.

Empowering local governments is suggested to prepare designs of ER houses and settlements as per local households, livelihood, community needs and suitability in future.

Surya Mohan Dhakal: Dismantling the old houses is a great challenge. How and who can help the people for this process. The people are not interested in such processes.

Suman Bayalkoti: Skilled manpower for retrofitting in rural area is lacking, so, may I know the further plans of NRA on this topic?

Naresh Shrestha: Maybe in the new development pattern, one clear message that went out after the earthquake is that only concrete buildings are safe. Now everywhere even in remote villages, people have started building concrete houses. Is that good or bad? One thing is, we are losing the old types of houses we used to have in the past.

Ruplal Aidi: How is NRA planning to handover the tranche process, retrofitting caseload, vulnerable caseload and urban recovery to remaining entities?

Anonymous user: NDRRMA should be autonomous under the leadership of PM. Also, can we say that NRA has successfully completed the reconstruction work in rural parts of Nepal? Is there anyone left?

Manindra Malla: Empowering local governments is suggested to prepare designs of ER houses and settlements as per local households, livelihood, community needs and suitability in future – as a pre-positioning plan.

Anonymous user: Is there any provision to transfer retrofit beneficiaries to regular beneficiaries?

Closing remarks by the session Chair



Er. Sushil Gyewali thanked all the presenters, commentators and participants. He assured that all comments and feedback will be documented and shared later in the proceeding report. He emphasized that NRA has been documenting each and every activity since inception and will continue to do so till the end. All this documentation will contribute to the eventual institutional memory which will be handed over to future generations. As NDRRMA is gradually taking over, NRA is doing everything to support and strengthen NDRRMA for a smooth transition and continuation of the legacy. NRA has itself extended for a year and there are a lot of things to do. He ensured that NRA is duly undertaking all thematic tasks and actions, and is committed to achieving the 100% completion of all mandates. Responding to Dr. Nigel Fisher's point of a development planning framework being integrated with

the overall DRR planning and regulation, he termed the point valid and informed the audiences that NRA has already designed a program for local government to prepare risk sensitive land use planning. We also need to strengthen our advocacy, lobbying and capacity building to develop mechanisms so that NDRRMA could be in the place where NRA wants to see it within the timeline of NRA. NRA is closely working on the legal framework for it. Sharing an example of Sindhupalchowk, he opined that we should not repeat the same mistakes in NDRRMA.

He informed that the preparation of planning and building by-laws is going on. NRA has already identified and decided the entity that will continue the remaining projects. Hence, the NRA road map is clear. He highlighted that there is a need of a strong building system in the local government level – needing a human resource system to be implemented and regulations to be made so that the issues can be addressed at local government level. We should be looking towards the single housing unit with overall infrastructure and environment, for which more research is needed to gather information from many successful projects of WASH, roads, drinking water and others. Many valid comments were received during this session to make NDRRMA autonomous. NRA will certainly work hard to make this a reality. He asserted that he firmly believes that NRA has implemented one of the biggest and the most difficult housing reconstruction projects in the world. The lessons learned from our housing experience should be useful to everybody everywhere because natural disasters can occur anywhere anytime. Urban regeneration is perceived as one of the major challenges in such a huge recovery and reconstruction process, therefore, NRA is working hard on this issue so that we

can hand over the entire project in a proper way to NDRRMA. He shared the news that NRA is launching three large urban housing regeneration programmes as integrated programmes in Khokana, Bungmati and sakhu.

The lessons learned from our housing experience should be useful to everybody everywhere because natural disasters can occur anywhere anytime.

He affirmed that NRA works mainly in lobbying, execution and networking and these mechanisms at three levels of government needs to be enhanced. He assured that though NDRRMA is taking time to build itself, very soon, it will also support operations at the local level. NRA has already initiated this work, i.e., in recent landslide incidents at Sindhupalchowk. He agreed that longer term development plans need to be integrated into the planning process, and NRA intends to continue this process. Now, the

regulations need to be made and the local government strengthened for sustainability and the long run so that Nepal can be well prepared for any kind of disaster in future. He informed that NRA is working on a single housing model and incorporating a number of utility schemes. He agreed that the involvement of private agencies is important, for which a policy has been developed to widen cooperation with private agencies. Supply chains were reestablished with enhanced monetary/banking mechanisms, water/sanitation mechanisms and capacitating of local governments together with private sector for planning, monitoring and evaluation. He emphasized the necessity of the reconstruction database for future urban housing – as NRA is initiating a large-scale urban resilience programme to revive towns or develop new towns. He shared that NRA is developing a network with universities, media and experts to develop a curriculum about disaster response. Lastly, the recommendations and feedback will be acknowledged while refining the policies, plans and strategies for future. He then formally closed the session.



Second Session: Cultural Heritage

Presenter: Mr. Damodar Gautam – Director General of Department of Archeology (DOA)

Commentators: Mr. Christian Manhart – UNESCO, Dr. Rohit Jigyasu – ICCROM, Ms. Guo Qianru – Chinese Embassy and Prof. Dr. Sudarshan Raj Tiwari – Independent Expert

Moderator: Dr. Chandra B. Shrestha

Session Chair: Er. Kishore Thapa



National Reconstruction Authority
Compendium theme: heritage
Mr. Damodar Gautum – DG, Department of Archaeology

Nepal's tangible and intangible cultural heritage legacy assured

Heritage: overview



27 November 2020

2015

- Destruction of heritage sites in 31 districts:
 - 920 monuments
 - 845 Buddhist monasteries
- Kathmandu Valley World Heritage property:
 - 33 collapsed monuments
 - 137 partially damaged monuments
- Numerous festivals and rituals lost/postponed (intangible heritage)

2020

- 453 monuments rehabilitated and reconstructed (49%)
- Survey and grading completed: 1297
- Monasteries: 289 repaired, 2 completed
- Master plans for all heritage sites established

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Challenges and issues



27 November 2020

Institutional and organisational

- Inappropriate Public Procurement Act and National Building Code for cultural heritage to regulate reconstruction and ensure conservation
- Governance and oversight in the heritage sector lacking: who leads?
- Conservation of monuments deserves enhanced priority
- Lack of DOA knowledge and capacity to handle 'large-scale' programme
- Lack of 'big picture' thinking (monument sites and complexes rather than just individual buildings)
- Coordination with local stakeholders
- Donors' 'conditions'
- Heritage 'lagging behind' other sectors in policy and implementation priority
- Documentation from past not maintained (no institutional memory)
- Budgetary constraints

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Challenges and issues (cont.)



Operational

- Lack of capacity in construction companies
- Lack of knowledge of traditional methods in engineering sector
- Lack of availability of traditional construction technology and construction materials (timber, bricks etc.)
- Balancing multi-stakeholder perspectives (government, UNESCO, Heritage organisations, local communities)
- Finding and applying the right blend of traditional and modern technology
- Addressing encroachment in urban settlements
- Balancing differing perspectives of architects and structural engineers
- Special training and facilities lacking
- Lack of prior maintenance of monuments a challenge 'before' the earthquake



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Strategies and approaches



- BUILD BACK BETTER
- Constructing resilient buildings
- Learning by doing (adaptation, on-the-job-training...)
- Direct funding to contractors, user committees, and through I/NGOs
- Capacity building of DOA staff and concerned stakeholders
- Community-led and participatory approaches to heritage reconstruction
- Overall coordination facilitation by NRA; day to day supervision & monitoring by DOA, and through both CLPIU and DLPIU Building
- Implementation through user committees, I/NGOs, Contractors, and mixed agents (contractor & users committee)
- Rigorous documentation, investigation, the development of technical interventions and implementation



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Strategies and approaches (cont.)

- Blend ancient and modern technologies where possible
- Develop sustainable skilled manpower and techniques in conservation
- Develop traditional material production
- Address encroachment using modern construction and amenities
- Ensure strong relationship between government and communities, including 'guthis'



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Lessons learned



- Archaeological research needs further strengthening
- Conservation guidelines essential to address restoration and conservation of heritage monument
- The practice of community involvement and a robust contracting system yields productive results
- Documentation of lessons learned to address [future] mass destruction of monuments is essential
- Producing skilled artisans is essential for stone, wood carving and masonry works, and provides legacy
- Conscious efforts are required for transformation of traditional technology to the future generations
- Raising awareness to conserve the tangible and intangible heritage is vital

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Lessons learned (cont.)

- Understanding and taking on board local interests and pride
- Maintaining wider stakeholder engagement ensures buy-in
- Preservation of local technology, skills and traditional construction materials is vital for Nepal's tangible and intangible heritage



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Conclusions



- Timely amendment of the 'Ancient Monument Preservation Act, 2013' (1956 AD) – is essential and it must be enforced
- Modern technology should be applied without compromising the heritage attribute
- Results of scientific damage assessments must be incorporated in master plans as well as individual site rehabilitation and conservation plans
- Communities must be involved from the outset in consultation, planning, and where possible implementation
- Preservation of local interest, sentiments, and wider stakeholder engagement are key to cultural heritage reconstruction
- Reconstruction of cultural heritage is a long-term process

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Key messages

- 1. DoA's capacity must be bolstered for preserving Nepal's heritage**
- 2. Conservation of heritage without strong engagement of and sensitivity to community and their values is unthinkable**
- 3. Public Procurement Act should harmonise with heritage requirement**
- 4. Nationwide inventory of monuments and heritage is crucial**
- 5. Federalisation needs to be reflected in DoA's organisational structure**
- 6. Traditional technology has to be reconciled with earthquake resilient technology**
- 7. Division of heritage monuments between DoA and Guthi has created confusion which has to be streamlined**
- 8. Conservation standards and norms needs to be finalised through a consultative process which needs to be strictly adhered to**

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Presentation 2: "Nepal's tangible and intangible cultural heritage legacy assured"

Mr. Gautam initiated the presentation highlighting the destruction of heritage sites in 32 districts, in which a total of 920 monuments – among them a total of 33 monuments of Kathmandu Valley World Heritage property – collapsed, and 137 monuments were partially damaged. As a result, numerous festivals and rituals (intangible heritage) are lost. In this situation, the international community is interested in supporting heritage reconstruction with integrated traditional and new scientific technologies. Among the total 920 monuments and 845 Buddhist monasteries collapsed by the 2015 earthquake, a total of 453 (49%) monuments are already reconstructed and rehabilitated.

NRA moved ahead with the concept of Build Back Better (BBB) and now those cultural heritage sites are being reconstructed with the blend of better versions of traditional and modern technologies.

The progress is across 1,297 sites through surveys and grading, with 289 monasteries repaired, and master plans developed for all heritage sites. Then, Mr. Gautam highlighted why the transitional government was a challenge at that time: on the one side there was no clarity on who will take the lead on reconstruction processes in terms of maintaining traditional, cultural, archeological as well as religious values, conservation standards and donors' conditions for funding for reconstruction. On the other side, the capacity of contractor companies to reconstruct the heritage sites was lacking. With these difficulties, NRA moved ahead with the concept of Build Back Better (BBB) and now those cultural heritage sites are being reconstructed with the blend of better versions of traditional and modern technologies. From this experience, NRA realized that cultural heritage reconstruction is a long-term process, and it needs archeological research for further strengthening of the proper reconstruction processes for preserving cultural values.

Second Session: Cultural Heritage (cont.)

Presenter: Dr. Rohit K. Ranjitkar – Country Director, Kathmandu Valley Preservation Trust

Commentators: Mr. Christian Manhart – UNESCO, Dr. Rohit Jigyasu – ICCROM, Ms. Guo Qianru – Chinese Embassy and Prof. Dr. Sudarshan Raj Tiwari – Independent Expert

Moderator: Dr. Chandra B. Shrestha

Session Chair: Er. Kishore Thapa



Heritage: overview



2015

- 7 monuments in Patan Darbar Square totally collapsed
- 20 monuments partially collapsed and structurally damaged
- Impact on intangible Heritage - many temples closed (structurally weakened)

2020

- 4 Collapsed monuments rebuilding completed, 2 under reconstruction
- 6 Partially collapsed monuments restored, 2 under reconstruction
- 2 Structurally damaged monuments strengthened, 2 under reconstruction
- Intangible heritage revived

Challenges and issues



Institutional and organisational

- Disaster preparedness for future generations
- Insurance for the heritage monuments
- Insurance of artisans
- Honouring of artisans

Operational

- Not an 'overnight job'
- Limited funding for documentation
- Not enough trained technical support for site supervision and monitoring
- Inappropriate interventions
- Proper use of materials and resources



Strategies and approaches



- Regular maintenance by proper artisans
- Trained site Supervisors & Artisans – having a long lead-time to train new artisans as well as technical personnel
- Regular monitoring during restoration to ensure quality
- Time and fund for research before the actual work starts



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Lessons learned



- Structural weaknesses are often hidden
- Department of Archeology's active support and guidance
- Choices required between old and new materials for reinforcement (not to replace existing material)
- Insufficient skilled craftsmen as well as technical experts
- Nepali timber supply difficult to procure, and is costly
- Rebuilding needs to be structurally sound
- Exacting standards required
- Community involvement is crucial



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Conclusions



- HONORING Craftsmen
- Need to use traditional artisans rather than cheap labor
- Need to train more artisans and technical personnel
- Need to document all the process (drawings, photographs, reports...)
- Strong collaboration between government, local community and institutions doing the reconstruction is essential
- Research is vital, and implementation should be based on this
- Awareness programmes need to be carried out with community, and with local stakeholders

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Key messages

1. For human safety, rebuilding must be structurally sound
2. Communities understanding of their heritage will avoid inappropriate interventions
3. Strong collaboration between central and local government, and community and institutions involved is paramount
4. Procurement Act need to be 'heritage friendly'
5. Funds available for conservation need augmentation
6. Regular maintenance ensures sustainable legacy
7. Monuments are not only for tourist development: living culture / active monuments ... improve our urban quality of life

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Presentation 3: “Kathmandu – home to seven of the world’s cultural heritage sites”

The presenter Dr. Rohit K. Ranjitkar, said that the earthquake of 2015 totally destroyed the 7 monuments in Patan Darbar Square, and 20 monuments are partially collapsed and structurally damaged. This has resulted in a huge impact on both tangible as well as intangible heritage with the closure of many temples which were identified as structurally weak. During the time of rescue immediately after the shock, because of a lack of coordination, many bulldozers and heavy trucks ran over the damaged and collapsed monuments to rescue people and to clear the space. At that time the rescue authorities and the general public were giving priority to rescue human lives without giving attention to the recovery of damaged of historical and archeological artifacts. As such archeological sites have high cultural, traditional, religious as well as touristic value which showcase the history of the country, hiring cheap and unskilled construction workers for their temporary preservation and reconstruction made them worse as there was no regulatory authority overseeing or monitoring quality on those sites. The presenter briefly informed about the situation of Durbar Squares in 2015 and 2020, including the challenges and issues in the institutional, organizational and operational levels, followed by the strategies and approaches that they adopted and then lessons learned. He further mentioned that with the hard efforts of NRA and the local people, now in 2020, changes can be observed. The reconstruction of 4 collapsed monuments has been completed, two sites are under reconstruction, six partially

collapsed monuments are already restored, and other two monuments are under restoration. Similarly, another two structurally damaged monuments are now strengthened, and a further two are under the strengthening process. With many intangible heritages revived, temples are under way to open for daily pray, and some religious functions have restarted. In conclusion, he expressed that NRA as a government authority should understand the importance of traditional artisans, and value of the preservation of their skills. Research and complete documentation of the entire archeological site with their importance and plans to preserve them in a sustainable way is the urgent need of today. Originality could be maintained in such places, provided there is regular maintenance. The concerned authority must have standards for the reconstruction and renovation of such important sites. Importantly, collaboration between locals, local government and federal government is required, together with giving primary responsibility to the local people because locals are the primary users and preservers. However, all these all stakeholders should work cohesively together to keep proper documentation of all the sites and their renovation process for future reference.

Response by the commentators



Mr. Christian Manhart began with high appreciation to the presenters for comprehensive and specific presentations. He said that heritage in Nepal is a cultural and religious identity as well as income generator from the internal and external visitors. He continued, “we held a damage assessment and implemented a pilot project retrofitting in Pashupati and Hanumandokha and we are also planning to establish a heritage information and management system to allow Nepal to have its own inventory and have access to online information easily. Developing some applications and other activities are in the pipeline.”

Research and complete documentation of the entire archeological site with their importance and plans to preserve them in a sustainable way is the urgent need of today.

UNESCO organized a meeting five days after the earthquake to prioritize assessments, and then continued with site surveys, documentation, technical assistance, coordination and heritage rehabilitation pilot projects. Re-assembling broken elements retrieved from 26 dumping sites were brought back to Hanumandhoka. He mentioned that he fully agrees with the presenters that if not managed right, the challenges of reconstruction and renovation of cultural heritage will only grow bigger and there is a chance of having more destruction of these assets/monuments and archeological sites in near future. He opined that high-level restoration is only possible if we have highly qualified professional workers in this area. Unfortunately, it is not happening enough to date, so sufficient resources and enhancement of human capacity is urgently needed. He highlighted that due to unplanned infrastructure development, the preservation of several archeological areas/sites are at risk. He expressed his opinion that whilst significant work has been done in such a difficult and crucial moment, we have a long way to go.

He mentioned some of the key issues and problems that have been hurdles for a long time. On one side, we are talking about the lack of specific skills required to work in conservation and renovation of such historical sites, and on the other side, the government wage rates being too low to hire highly qualified craftsman. High

quality cannot be achieved without highly qualified contractors and builders. More efforts are needed to support highly qualified craftsmen who are available in Kathmandu. Similarly, it is observed that there is not enough effort to ensure quality control from government agencies for craftsmanship, materials and restoration methods, as they have not been given sufficient funding, manpower, autonomy and decision making to fulfill their roles. There are unique technical complexities in the monuments in the Kathmandu valley as no monuments are duplicable. Therefore, there is a strong need to enhance better communication and coordination between the Department of Archeology (DOA), NRA and local communities along with various national and international partners to have common understanding and processes to preserve such historical places. Another important area that needs to be addressed is how local governments should consider the preservation of such historical sites while giving building permission to local people to construct their houses in close proximity of the sites, with the risk of destroying the aesthetic view of the sites. Therefore, rebuilding of traditional houses in villages and urban areas should respect these sites and as far as possible promote traditional architecture and materials. The administrative issue that is related to (and a burden on) quality control is frequent change of leadership in the Ministry of Culture, Tourism and Civil Aviation that creates delays in decision making. For example, there is a serious concern and risk to lose traditional villages in Khokana and Bungmati due to unplanned infrastructure, but no authority has taken strong action to preserve the sites there. He concluded saying it is in fact long-term work. With the help of software and proper documentation, the information may be easily accessible

online for the wider public and can be used as a modern science management tool, accessible for all the stakeholders. He assured that UNESCO is always ready to cooperate and provide support to NRA and any other authorities in this regard. Finally, he ended with the hope that the government authority in cooperation with local municipalities and people of the area will work together to look into these issues seriously and find permanent solutions to the restoration and maintenance of heritage monuments.

As there is a growing concern about the balance between heritage and safety, the value and safety assessments should be done side by side as values are also changing.



Commentator Dr. Rohit Jigyasu fully agreed with the presenters on the importance of cultural heritage and efforts made so far to revive these sites. He stressed, “we have to reinforce that recovery is not only about the outcome but also the process which is important as the end result is part of decision making and the implementation process in the long term.” So, documentation must be seen in multiple levels with detail and definition. Craftsmen and professionals should work together as they have been doing this work for a longtime. Another dimension he highlighted was a balance between renovation and reconstruction of the heritage sites and their safety. He expressed that we should look at multiple options and try to maintain heritage value, but sometimes evolution is also necessary as regards the availability of building materials. We should have faith in our traditional engineering technology and that also needs to evolve along with time. We can mix traditional and modern construction methodologies and come up with suitable models. We can take an example of retrofitting, where we can renovate the sites protecting the old artifacts with new strong technology and materials. Therefore, it is high time to learn better technology and be better prepared for the next emergency. He also suggested that the research and documentation at multiple levels is immediately required to have records and information for future reference. As there is a growing concern about the balance between heritage and safety, the value and safety assessments should be done side by side as values are also changing with time. Finally, he concluded by saying that it is good to have this type of platform to bring professionals closer and to encourage them to work together.



Commentator Ms. Guo Qianru started off with appreciation for getting an opportunity to be a part of the seminar and put forward her views on behalf of the Chinese government on Nepal's recovery and reconstruction efforts. She commended the presentations and all the achievements made so far by the Nepali Government. She highlighted the support provided by the Chinese Government in the efforts of Nepal's recovery and reconstruction process. She asserted that immediately after the Gorkha 2015 earthquake, the Chinese Government sent rescue teams and has continuously been supporting the post-earthquake reconstruction processes in Nepal. After a year of the earthquake, a survey was conducted to assess the damage in the historical sites. Following the results, the support in renovation of different sites has been started, for example in the very historical and famous Basantapur Durbar Square and the nine-story old palace. In this project, internationally recognized and accepted concepts and technology for the renovation

of cultural heritage was used and followed in every quality and standard while implementing the renovation and reconstruction of the palace. The concept, technology and process were decided in consultation with the government of Nepal. As far as possible, traditional materials were used and local labor was hired for the restoration keeping in mind that doing so, the traditionalistic features may be preserved, and the technology exchange and transfer may take place successfully. Due to the COVID19 pandemic, it is temporarily difficult and has become a little late, but it is being managed without any serious problem. Ms. Qianru expressed her satisfaction with the progress cited in the presentations and efforts made so far to revive these sites. She concluded emphasizing on the processes and results.



The last commentator Prof. Dr. Sudarshan Raj Tiwari congratulated and thanked the presenters for accommodating the status of reconstruction of heritage sites. Appreciating the content in

the presentations, Prof. Tiwari put forward some key questions: what is the purpose of presenting this document? Why ToRs are not included in the presentations which is important to know for the audience? Did we analyze the structures associated to heritage conservation and renovation before starting the work? Are we actually strengthening or weakening those existing structures in the name of compatibility? What is the level and nature of challenges? He went on to say it would be better to have them specified, so that the appropriate measures can be taken for future. He then talked about some of the challenges faced during the initial stage in 2015 and asked whether they are still persisting. Everybody is talking about technological issues and the idea of BBB, but they are not quite compatible in the case of cultural heritage renovation. If these questions were answered in the presentations, it would have given a complete picture. Summarizing his observation, he said that the issues presented by both presenters highlighting the limits of hiring of experts would create difficulty in responding to the real needs, because this type of work demands extensive research and if we limit ourselves in finding alternative solutions then how can we improve our interventions. In fact, the experience of the earthquake reconstruction process addresses the current needs and lessons learned for improvement in the future. What is happening in this area is lacking in the presentation. He further stated that the question raised about not having proper guidelines and the DoA not following the existing guidelines are alarming. He posted a serious concern to the DoA to follow standard procedures while undertaking rescue in the archeological sites, and to make sure of informed community participation. The number of buildings restored are counted but lacking is the analysis of investment

and achievement, how the problems were solved, at what level the need is addressed and what level is yet to be addressed, how much resources are received and expended, and how they were mobilized, and projection for the future prospects.

Similarly, recording of how the work was done and what are the real problems and challenges encountered in the heritage reconstruction and restoration would be more useful for future reference. In any instance, the issue of procurement was also raised but what exactly was the barrier is not mentioned. If it was that level of problem, then why is the problem still not addressed – it is a big question! He further argued that the lessons learned previously were not used in restoration of the sites, therefore the method and technology that was used in the renovation of heritages sites this time needs a revisit and to be documented properly. We need to ask the question, ‘are our guidelines appropriate to address present day’s needs?’ Why were guidelines not followed by DoA? Research is important, which is raised by almost everyone, but how and who initiates it in what area is not answered. Another buzz word used here is community participation – since communities are primarily the people who live in affected areas, and they are the primary users and preservers, therefore they should be consulted and involved in the whole processes of renovation or reconstruction. He questioned that over the 5 years period, how often and how many experts were consulted with? Had any renovated area been used for training purposes so far as a best practice? We do not have answers to these questions? Therefore, heritage friendly urban planning is to be initiated not only during the recovery and reconstruction phase but also as a continuing process in every rural and urban municipality. He concluded saying,

“not having expert craftsmen was the issue at the initial stage but in 5 years, a huge number of experts should have been developed. However, still required is to reassess the situation, develop a database of people with stagnant skills and people who developed skills during the recovery and reconstruction process itself and put them into the network with the communities so that they can contribute to the heritage reconstruction and conservation process for a longer time.”

Heritage friendly urban planning is to be initiated not only during the recovery and reconstruction phase but also as a continuing process.

Presenters’ views on comments

Presenter Dr. Rohit Ranjitkar appreciated all the comments received from the commentators. He assured to incorporate and address the issues raised in the heritage chapter of the compendium according to their importance and relevance. He shared a few things about strengthening of monuments. All the broken, collapsed or fallen down monuments are rebuilt because there is no other option. Even in a new construction, the originality of the site and monuments are kept intact as far as possible without making any changes in model and shape. In the case of stone pillars broken from the middle portion, we had no traditional solution other than to replace with modern technology, but replacement was made giving a traditional look as far as possible. He concluded with the assurance to take all the comments and feedback

positively and use them in his work in the preservation of cultural heritage monuments and sites.

Comments & queries from chatroom and social media

Deependra Sharma: It is obvious that reconstruction of heritage takes time not only because we have to adopt the conservation strategy of ‘minimal intervention’, but also because we lack the skilled artisans and materials. To add to these, the decision-making process for the use of new materials and technology involves a lot of discussions and approvals. Seeking socio-cultural consensus is also time-consuming. There are many other factors that make the reconstruction process slow. Hence, the physical progress in reconstruction of heritage reconstruction has been the least of all categories. I see a lot of work remaining after the end of this fiscal year, after which the NRA will initiate its disengagement strategy. So, I see a strong role for the Department of Archeology for a few more years to come in post EQ reconstruction. Adequate provisioning of financial and human resources in these years is a must for achieving the targets. NDRRMA could be given the responsibility for coordinating the remaining works. I strongly recommend that the NRA’s exit strategy takes these factors well into account. Since cultural heritage reconstruction and conservation is a unique endeavor, flexibility is required for the contract management including cost estimation and procurement management. This should come as a strong recommendation as the lessons learned with regards to cultural heritage reconstruction as well as conservation.

Sachin Malego: Thank you for the very interesting session. Will the presentations be shared following each day? The informative slides would be very useful for participants learning and reporting.

Remarks by the CEO NRA

Er. Sushil Gyewali thanked all presenters, commentators and participants in the session. He particularly focused on some of the key observations, comments and questioned based on the diversified knowledge and experience gained from tangible and intangible heritage reconstruction and preservation. He exclaimed that NRA was expecting even more heritage areas to cover. He appreciated the points that came up during discussion, which are the key take-aways of this event, one of them being the level of incompetence that was experienced within the contractors for melding the traditional and modern construction models as required for the renovation of heritage sites. The knowledge and experience of the community on traditional construction technology and the knowledge and experience of modern contractors on modern technology has to be melded to produce an appropriate model for Nepalese archeological site renovation and reconstruction. For this a tripartite mechanism or network of the local communities, contractors and the government agencies may be helpful. He highlighted the occasional confusion and misunderstanding that took place between structural engineers and archeologists in technology and materials to be used in historical places for their renovation. There should have been specific guidelines developed to engage the community in such types of heritage maintenance and conservation, with specified roles and responsibilities of different level of governments as well as communities and responsible authorities

following the set monitoring and evaluation procedures. Regarding lacking capacity of contractors, we need to enhance the capacity of engineers, archeologists and conservation experts together with local craftsmen and masons. In addition, special training packages may be developed and delivered to the contractors as well.

Tangible and intangible heritage conservation should go hand in hand. At every step, the importance of engagement of the communities is mandatory because they are the primary protectors and users.

To resolve the tension between structural integrity and archeological conservation, value and safety assessment should be done simultaneously, and the results should be adopted to make heritage sites resilient structures. The local craftsmen and masons should be honored and provided with training to enhance their skills to tie up with the current needs. Tangible and intangible heritage conservation should go hand in hand. At every step, the importance of engagement of the communities is mandatory because they are the primary protectors and users. As of now, the realization is there for the documentation of monuments and heritage sites which needs to be initiated very soon. He concluded stating that the recovery is not a one-shot act to produce one output, but it is a process to achieve multiple results and outputs on the long run.

With this, the moderator Dr. Chandra B. Shrestha requested the session Chair Er. Kishore Thapa to share his remarks over the presentations and comments, and formally conclude the session.

Closing remarks by the session Chair



The Session Chair Er. Kishore Thapa began his remarks with great appreciation to the presenters, commentators and the participants for their invaluable contributions. The two presentations successfully highlighted the ground issues of heritage conservation, restoration and reconstruction of damages caused by the Gorkha 2015 earthquake. The procurement issues have been coming up often but from his long experience of contracting and procuring the services in the government, he clarified that the problem lies in the interpretation of rules and regulations. The procurement law and procedure are not the obstacles because there are other options for tendering process, i.e., user committees. If the procurement law was a barrier, why is there still a problem even after it has been amended 10 times in the past five years? The answer is clear that the problem is not the regulatory process itself; instead,

it is intentionally created by people for their benefit. So, it is high time to think and act in the public interest keeping in mind that rules and regulation are not meant to be obstacles, rather they are for the facilitation of the process. So, if we have a positive attitude, there are always multiple options to facilitate the process. Another debate is about the traditional vs modern technology of construction. In this subject, we should respect both age long construction techniques and the modern construction models. We should look for a scientific and wise balance. We should always go for progressive construction models balancing between both. For this, we have to capacitate both DoA and the construction companies. This is also high time to undertake research on the role and capacity of the user committees, their knowledge, experience, strengths, capacities, abilities and limits because many of them are working as middlemen/brokers. Once findings are available,

we can revisit guidelines and manuals to use the user committees in renovation and reconstruction works. He concluded with the note that several works have been undertaken in the last five years. Among this, some noteworthy work is the completion of the heritage conservation area where the sites present a space for live cultural and religious activities, which are becoming an attraction point for tourists. Therefore, we should take the recovery stage as an opportunity to recover and revive our historical and archeological sites to make our history come and stay alive.

It is high time to think and act in the public interest keeping in mind that rules and regulation are not meant to be obstacles, rather they are for the facilitation of the process.



Third Session:

Public Building and Social Infrastructure

Speaker:

Er. Bharat P. Aryal – Joint Secretary, NRA

Commentators:

Mr. Lu Yin – Shanghai Construction Group Co. Ltd Overseas Business Department, Mr. Krishna Lamsal – JICA, Dr. M. N. Khan – USAID (Supported by Dr. Arjun Koirala) and Mr. Purna P. Kadariya – Independent expert

Moderator:

Dr. Chandra B. Shrestha

Session Chair:

Dr. Hariram Parajuli



National Reconstruction Authority
**Compendium theme: public buildings
and social infrastructure**
Mr. Bharat P. Aryal – Joint Secretary, NRA

**Enabling spaces for state-
citizen engagement**



Public buildings and social Infrastructure: overview

*(excluding information from
CLPIU Education and Building)*

27 November 2020

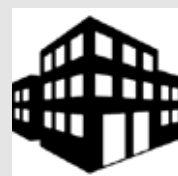
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2015

- Health buildings reconstruction required: 1197 (544 CD, 653 PD)
- Security buildings damaged: 440; reconstruction from NRA: 216
- Water and sanitation schemes damaged: 3212
- Irrigation: I/NGOs approved schemes for repair: 26



2020

- Health buildings completed: 698 (58%)
- Security buildings (NRA) completed: 207 (96%)
- Water and sanitation systems completed: 791 (25%)
- Irrigation systems completed: 23 (89%)



Challenges and issues



27 November 2020

Institutional and organisational

- Capacity building of implementing agencies missing
- Dilemma in selection of executing agency
- Expectation and quality gap divergence
- Delay in donors' concurrence

Operational

- Capacity of contractors lacking
- Availability of land for Health projects
- Provisioning of periodic maintenance
- Lack of availability of labourers & skilled human resources
- Problems with accessibility during flooding and landslides

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3

Strategies and approaches

- BUILD BACK BETTER
- Technology transfer at different level
- Demand-driven and standardized structures
- Participatory and consultative engagement with key stakeholders
- Policy instruments for Reconstruction of resilient buildings and systems (Revision of NBC 1994)
- Procedural guidelines formulated (procurement, NGO, UC, etc.)
- Implementation through SMCs, I/NGOs, and Contractors
- Overall coordination facilitation by NRA; day to day supervision and monitoring by central (CLPIU Building) and district offices (DLPIU Building)



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Lessons learned

- Quality assurance in both process and result is vital
- Wider consultation (stakeholder engagement)
- Satisfaction and trust of public must be maintained
- Adequacy of human resources from the outset
- Multiple approaches and modalities adopted
- Adoption and adaptation of land-use plans and building codes
- Risk-based land-use planning essential



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Conclusions

- Military, health, water and transport sectors are of prime importance and require special attention for disaster risk reduction
- Detailed prior assessment and strengthening of the structures is needed
- Integration of DRM for disaster resilience in development works
- Effective enforcement of technical instrument (e.g. building codes)
- Special attention to contract management



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Key messages

1. **Implementing agency and users of structures- identified immediately after damage assessment**
2. **Conservation of historical structures - line of demarcation**
3. **Standards, codes of practices- promulgated at the earliest**
4. **Overall strategic planning- results into economic gain**
5. **Reconstruction priority – front line/others**
6. **A multi-stakeholder approach requires additional attention paid to inter-agency communication**

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Presentation 4: “Enabling spaces for state-citizen engagement”

Er. Bharat P. Aryal began his presentation assessing the damage incurred during the Gorkha earthquake 2015. He mentioned that a total of 1,197 (544 CD, 653 PD) health facility buildings, 440 security buildings, 3,212 water and sanitation schemes and 26 Irrigation schemes were damaged – of which up to 2020, the reconstruction of a total of 698 (58%) health facility buildings, 207 (96%) security buildings, 791 (25%) water and sanitation systems and 23 (89%) irrigation systems have been completed. During this time, even though there was support from several donors, some internal coordination weaknesses were observed. Also, the lack of land availability on time for the health facilities, lack of capacity of contractors, and lack of skilled manpower for the reconstruction process hurdled the work.

The BBB approach was used in which technology transfer was done following a demand driven approach. He highlighted military, health, water and transport sectors as being of prime importance and requiring special attention in disaster risk reduction. If these facilities remain intact – especially for the future – the emergency response becomes easier. While undertaking such huge reconstruction initiatives, prior detailed assessment and strengthening of the structures is vital. Similarly, integration of DRM for disaster resilience in other regular development projects can play a very important role in resuming the services more quickly. Effective enforcement of technical instruments (e.g., building codes) and special attention to contract management may facilitate more efficient project implementation.



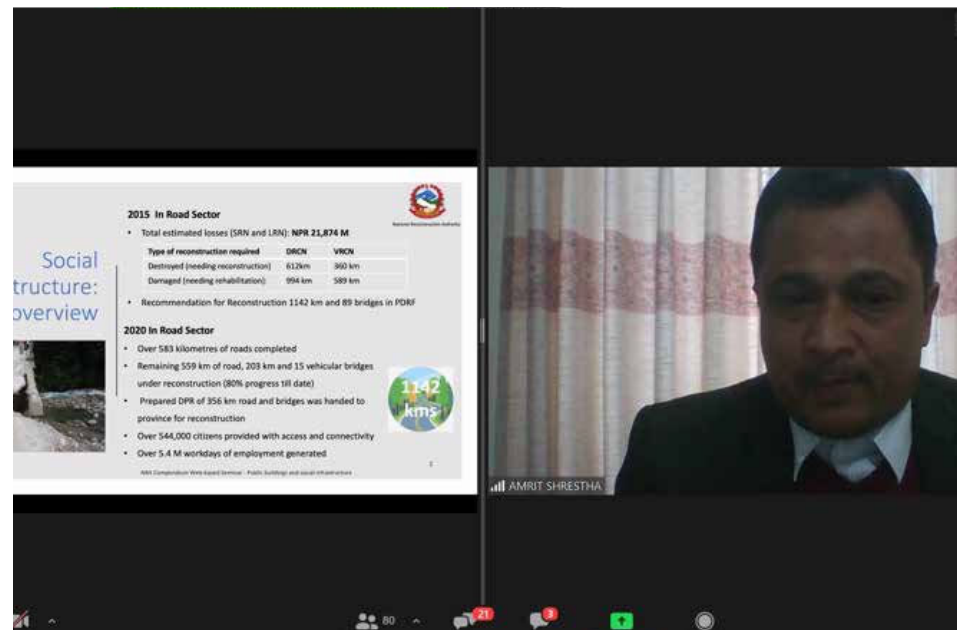
Third Session: Public Building and Social Infrastructure

Speaker: Er. Amrit Shrestha – CLPIU GMaLI

Commentators: Mr. Lu Yin – Shanghai Construction Group Co. Ltd Overseas Business Department, Mr. Krishna Lamsal – JICA, Dr. M. N. Khan – USAID (Supported by Dr. Arjun Koirala) and Mr. Purna P. Kadariya – Independent expert

Moderator: Dr. Chandra B. Shrestha

Session Chair: Dr. Hariram Parajuli



Social Infrastructure: overview



27 November 2020

2015 In Road Sector

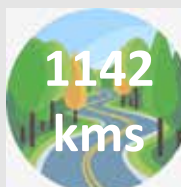
- Total estimated losses (SRN and LRN): **NPR 21,874 M**

Type of reconstruction required	DRCN	VRCN
Destroyed (needing reconstruction)	612km	360 km
Damaged (needing rehabilitation):	994 km	589 km

- Recommendation for Reconstruction 1142 km and 89 bridges in PDRF

2020 In Road Sector

- Over 583 kilometres of roads completed
- Remaining 559 km of road, 203 km and 15 vehicular bridges under reconstruction (80% progress till date)
- Prepared DPR of 356 km road and bridges was handed to province for reconstruction
- Over 544,000 citizens provided with access and connectivity
- Over 5.4 M workdays of employment generated



2

NRA Compendium Web-based Seminar - Public buildings and social infrastructure

Social Infrastructure: overview (cont.)

27 November 2020

2015

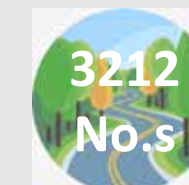
WASH Sector

- Total losses and damage : **NPR 11,400 M**
 - Water supply Schemes: Major damage: 2,317; Partially damage: 5,424
 - Private toilets 388,000
 - Recommendation for Reconstruction 3,212 schemes

2020

WASH Sector

- Water and Sanitation (W/S)Schemes:
 - completed 791,
 - ongoing 670
- Remaining w/s schemes handed over to DWSSM for reconstruction



3

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Challenges and issues



27 November 2020

Institutional and organisational

- Gaps in appropriate organisational set up
- Gaps in human resources for implementation
- Late start due to delay in finance processing and approval
- Gap in inter-agency coordination and support

Operational

- Problems with tree clearance and implementation resettlement plan to provide access to contractor for reconstruction works
- Yearly landslides hindering rehabilitation and reconstruction process and causing new damage
- Contract Management capacity
- Handover of completed project

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4

Strategies and approaches

- Build Back Better designs applied
- Environmental and climatic consideration included
- Improved standards applied to rehabilitation for resilience
- Multi-stakeholder approach applied: Central, provincial and local governments engaged



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Lessons learned



- Reconstruction projects require their own independent funds organisational structure and human resources
- Approval of multi-year plans for procurement and preparation (i.e. tree cutting, RP) essential and should be delegated to competent authority
- Applying improved standards contributes to resilience and sustainability
- Include experts within project structures for smooth implementation of social safeguarding activities
- Establishment of effective monitoring (implementation schedule and human resource deployment chart , equipment, PMIS) ensures progress

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Conclusions



- Selection of sub projects should be quantitatively undertaken by experts
- Contracts should only be awarded after establishment of resettlement plans, land acquisition and required tree cutting
- Contract periods should be established on type of road and work volume
- Mechanisms for transferring lessons learned on pre- and post-disaster management between communities at risk should be designed and adopted
- Coordination among line agencies should be clearly stated with roles, responsibilities and timeframe in project preparatory document

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7



Key messages

1. **Road network reconstruction to ‘build back better’ standards reduces travel time, allows for increased freight volume, in turn enhancing reconstruction and recovery processes**
2. **Rehabilitation and reconstruction works generate local level employment opportunity to support vulnerable households**
3. **Improved connectivity provided easy access to market and public service center increases economic activity, and reduces migration trends**

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Presentation 5: “Providing connectivity and WASH”

Er. Amrit Shrestha started off his presentation with background information. He informed that over 1,142 km roads and 89 bridges were damaged by the earthquake of 2015, that was the equivalent of NPR 21,874 Million, is a huge cost in terms of investment for reconstruction. Once NRA was formed, the process of reconstruction began smoothly though there were some obstacles initially. As of 2020, 583 kilometers of road has been revived by reconstruction, and of the remaining 559 kms of road, 203 kms is under construction with 80% progress to date. Similarly, a total of 15 vehicular bridges are also under reconstruction. Additionally, DPR of 356 kms road and bridges was handed over to the concerned provincial governments to commence the local reconstruction process. With the reconstructed roads and bridges, over 544,000 citizens have been enjoying access and connectivity. At the same time, the construction work has created over 5.4 M workdays of employment at a local level.

With the reconstructed roads and bridges, over 544,000 citizens have been enjoying access and connectivity.

He concluded his presentation with some learnings and recommendations: 1. Selection of sub projects should be quantitatively undertaken by experts 2. Contracts should only be awarded after establishment of resettlement plans, land acquisition and land clearance 3. Contract periods should be established based on type of road and work volume 4. Mechanisms for transferring lessons learned on pre- and post-disaster management between communities at risk should be designed and adopted 5. Coordination among line agencies should be clearly stated with roles, responsibilities and timeframe in the project preparatory document.

Third Session:

Public Building and Social Infra-structure

Speaker:

Dr. Youb Raj Paudyal – Deputy Project Director, CLPIU Education

Commentators:

Mr. Lu Yin – Shanghai Construction Group Co. Ltd Overseas Business Department, Mr. Krishna Lamsal – JICA, Dr. M. N. Khan – USAID (Supported by Dr. Arjun Koirala) and Mr. Purna P. Kadariya – Independent expert

Moderator:

Dr. Chandra B. Shrestha

Session Chair:

Dr. Hariram Parajuli



National Reconstruction Authority

Compendium theme: social infrastructure

Dr. Youb Raj Paudyal – DPD, CLPIU Education

Building Nepali children's education and future from the ground up



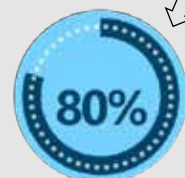
Social infrastructure: overview

2015

- Damaged and destroyed: 7,923 schools (49,681 classrooms) and 70 university and college campuses
- Estimated costs according to the PDRF: 1806 US \$ Million

2020

- Schools merged: 370
- School reconstruction completed: 6,085
- On-going school reconstruction: 1,468
- University and college campuses reconstructed: 32 (38 DPR ongoing)



27 November 2020

NRA Compendium Web-based Seminar - Public buildings and social infrastructure

2

Challenges and issues



27 November 2020

Institutional and organisational

- Starting from scratch (no institutional memory)
- Institutional set up (central level to local level) – MOUD to NRA
- No local authority (main-actor/stakeholder) at the beginning
- Coordination / stakeholders (political/ local/ social commitment / engagement / coherence, I/NGOs, DPs)
- Legal framework development (policy / guidelines / codes)

Operational

- Damage assessment and need identification
- Resources: financial, manpower, materials
- Resource distribution (equitable and GESI, one door policy)
- Geography / access
- Scope and scale
- Technical (new design and retrofitting appropriateness etc.) capacity and quality assurance

NRA Compendium Web-based Seminar - Public buildings and social infrastructure

3

Strategies and approaches

- PDNA/PDRF
- Vulnerability Assessment (SIDA)/Recovery Plan
- Establishment of special instrument NRA, CLPIU and DLPIUs
- Multi-stakeholder engagement (GoN, DPs, I/NGOs and community) mechanism
- Establishment of policies and guidelines (Architectural and Structure Design, R&D for remote school design and SMC modality), modular/type designs
- Principle: 'Build Back Better' (Sendai Framework of Action 2015)
- Principle: Holistic planning and implementation (child friendly, GESI inclusive)
- Principle: 'Sustainability' (SMC/Local Government-MOU) of the created facilities
- Use of national and local capacities (contractors, engineers, masons, materials)
- Human resource capacity development (technical/social/financial)
- Decentralized allocation of responsibility (Engagement and role of School Management Committee and Local Government)
- Coordination systems and monitoring and evaluation mechanisms
- Establishment of model designs for schools



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Lessons learned

Institutional and organizational

- Clear Responsibility and Authority (promptness on decision-making)
- Development of plan and program (scale of planning)
- Resource management (financial and human)
- Transparency and accountability (central to local)
- Communication and coordination mechanism among the stakeholders (Government/DPs/community)
- Planning and monitoring processes for quality assurance (web-base platform)
- Should have established additional entity for Universities and Colleges



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Lessons learned



Operational

- Code of ethics was instrumental for effective management
- Need based infrastructure through involvement of local experts & manpower
- Responsible School Management Committee and Local Government for sustainable school infrastructure
- Application of web-based system (PMIS) enhanced effectiveness of construction supervision
- Capacity development of engineers, masons and local stakeholders enhanced quality and sustainability
- Disaster sensitive designs and program enhanced local knowledge on DRR
- Quality of education enhanced with child-, environment, gender friendly school



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Conclusions

- Political consensus a priori event
- Strong leadership, and harmonized policy of the government and coordination among the stakeholders vital
- Recovery / priority investment plan through vulnerability assessment
- Holistic approach of planning and implantation (reconstruction, retrofitting and DRM) to enhance resiliency and sustainability
- Conducive teaching learning environment helps increase students' enrolment and enhances quality education
- Program for quality education is inevitable
- Resilient, affordable and replicable design helps for fast recovery
- Capacity building programme for technical persons, masons and contractors helps increase quality and fast recovery process
- Creating resilient society through school recovery process (construction, retrofitting and DRM program)



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Key messages

1. **Vulnerability assessment-based investment plan was effective approach**
2. **Introducing NBC Standard/design and planning guidelines were instrumental for resiliency- Build Back Better**
3. **Local technology and materials must be promoted by innovative research and development for cost effective and sustainable school building construction**
4. **Quality enhancement through capacity building and reporting mechanism (such as PMIS)**
5. **Availability of land in a number of locations was a great hurdle**
6. **Proto -type school design sometimes not necessarily fulfil the local condition**
7. **Community/local participation helps in ownership building and hence sustainability of the created facilities**



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Presentation 6: “Building Nepali children’s education and future from the ground up”

Dr. Youb Raj Paudyal informed that the damaged and destroyed school buildings during the Gorkha earthquake were 7,923 in which a total of 49,681 classrooms were completely destroyed. Similarly, 70 university colleges and campuses were badly affected. The approximate loss of this destruction was estimated at 1,806 Million USD, which is not only huge but beyond the capacity of Nepal. As an effect of this destruction, many education institutions remained closed for many months; many schools operated in temporary shelters and 370 Schools were merged. In such a huge destructive situation, reconstruction projects were started. He further informed that so far, a total of 6,085 school buildings have been reconstructed and the reconstruction of 1,468 school buildings is going on. 32 University colleges and campuses have been reconstructed and DPR is under way for another 38 campus buildings. He then highlighted some of the lessons learned during this whole process. Political consensus is hard but once the consensus is reached even hard work becomes smooth. This is a big learning in the process of reconstruction in Nepal. Similarly, strong leadership and harmonized policy of the government and

coordination among the stakeholders are vital. Recovery/priority investment planning through vulnerability assessment is found very supportive.

Strong leadership and harmonized policy of the government and coordination among the stakeholders are vital.

Taking a holistic approach of planning and implementation (reconstruction, retrofitting and DRM) is very effective to enhance resilience and sustainability. Providing a conducive teaching and learning environment helps to increase students’ enrolment and enhances quality education after having the infrastructure in place. Putting capacity building programmes into place for technical persons, masons and contractors helps increase quality and fast recovery process and creates resilient society through the school recovery process (construction, retrofitting and DRM programme).

- Third Session:** Public Building and Social Infrastructure
- Speaker:** Arc. Raju Neupane – Deputy Project Director, CLPIU Building
- Commentators:** Mr. Lu Yin – Shanghai Construction Group Co. Ltd Overseas Business Department, Mr. Krishna Lamsal – JICA, Dr. M. N. Khan – USAID (Supported by Dr. Arjun Koirala) and Mr. Purna P. Kadariya – Independent expert
- Moderator:** Dr. Chandra B. Shrestha
- Session Chair:** Dr. Hariram Parajuli



National Reconstruction Authority

Compendium theme: public buildings

Mr. Raju Neupane – DPD, CLPIU Building

Enabling better state service provision and citizen orientation

Govt. Buildings: overview

2015

415 Govt. buildings affected

- Completely Damaged: 162
- Partially Damaged: 253

NBC 105:1994 was not updated for 21 years

2020

NBC 105:1994 Revised (and endorsed by Cabinet)

361 (87%) buildings Reconstructed/Maintained/Retrofitted

- Maintenance/retrofit completed: 245
- Reconstruction Completed: 116

Transitional Offices: 288

Community Shelters: 50



Challenges and issues



Institutional and organisational

- New federal set up and institutional arrangements confusing: utilisation of buildings
- High turnover of project managers
- Financial constraints – timing and disbursement
- Exit strategy lacking

Operational

- Unknown territory of retrofitting works
- Research lacking for robust and valid approaches: reconstruction or retrofit?
- Contract management lacking
- Supervision and monitoring of 'turnkey projects' missing
- Lack of availability of labourers & skilled human resources



Strategies and approaches

- Investment for resilience
- Continuity of services
- Prioritising of works
- Principles: Accessibility, Function and Aesthetics
- Integration where possible
- Consultation with experts/stakeholders
- Presence in urban landscape



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Lessons learned



Institutional and organisational

- Adoption of the sector modality was appropriate
- Consultation with experts/stakeholders reaped dividends
- Architectural expression of central buildings should also be given prime importance
- Use of ICT in Project Management was crucial



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Lessons learned (cont.)

Operational

- Packaging of contracts is crucial from management point of view
- Contract enforcement is a challenge that needs to be tackled
- High capacity of implementing agency on MEP works essential
- Supervision consultant proved crucial to implementation
- Monitoring mechanisms need to be established and implemented for DLP



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Conclusions



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- Investing in preparedness for resilience is key (NBC 105: 2020)
- Service continuity during reconstruction leads to reinforced trust
- Dedicated institutions to oversee and implement contracts are vital
- Integration where possible is a modus operandi to follow
- Improved quality of service delivery should be targeted (BBB)
- Creation of better workplaces can translate into greater staff satisfaction and productivity
- Architectural presence in the urban landscape must be ensured

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National Reconstruction Authority

Key messages

1. **Build Back Better (Safety, functions, aesthetics and accessibility)**
2. **Promote Safer Building Construction (implementation of Building Code and regular revision)**
3. **Service continuity should be ensured during reconstruction phase)**
4. **Dedicated Implementing agencies should be entrusted with reconstruction works**
5. **The restoration & retrofitting of historical buildings and structures in their original character should be a priority**

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Presentation 7: “Enabling better state service provision and citizen orientation”

The last presenter for the day, Arc. Raju Neupane began with background information about the damage of public buildings during the 2015 earthquake. A total of 415 Government buildings were affected, where 162 were completely damaged and 253 were partially damaged.

As of 2020, a total of 361 (87%) buildings have been revived either by reconstruction or maintenance or retrofitting, which were undertaken on the basis of priority damage assessment. Similarly, 245 buildings were maintained with retrofitting, 116 were reconstructed and 288 structures were built as transitional offices along with 50 community shelters. Some of the key learnings that he cited were: investing in preparedness for resilience is the key for resilient living; service continuity during the reconstruction phase leads to reinforced trust; dedicated institutions to oversee and implement contracts are vital; integration where possible is a modus operandi to follow; improved quality of service delivery should be targeted particularly using the Build Back Better (BBB) approach; creation of better workplaces can translate into greater staff satisfaction and productivity, and architectural presence in the urban landscape must be ensured for resilient construction.

Response by the commentators



The first commentator, Mr. Lu Yin from Shanghai Construction Group, appreciated that all four presentations were informative and rich enough with information and lessons from the massive recovery work. He shared his thoughts and experience, mentioning that many projects have been completed and more are in process of completion. He shared that his project has hired about 5,000 local people for the reconstruction work. In his experience, land acquisition is difficult in Nepal so the ring road design should not exceed the right of way. Many issues were faced during the

preliminary planning of the project, so he suggested that the Nepalese parties should provide the machinery and high-level government officers engagement. During his tenure, he encountered several issues like robbery, deforestation, demolition, resettlement and other constraints which were not mentioned in the presentations. As he was involved in the ring road and durbar high school projects, he highlighted that one of the main achievements was the local worker's involvement, which was good for technology exchange or transfer. As far as development of projects was concerned, land is a primarily required component. Land acquisition is a big and complicated issue in Nepal. Therefore, he suggested that policies and procedures should be clear, with strict law enforcement required for smooth operation of the projects. In his view, the Nepali parties and local governments should be provided with more resources for the projects and assure timely completion of the projects. Further, he suggested to have better documentation of the works done and in maintaining standards, as well as on site monitoring and evaluation, so that the work will not be delayed by any disruption caused by a strike or road closure etc.

Transferring skills to the local community for sustainable maintenance of services and schools is another good message to share.



The second commentator Mr. Krishna Lamsal (JICA) expressed his appreciation for being a part of the session. He said that although the presentations have well documented the progress, there is a need for consolidating the document and events like these, as they are good platforms to generate knowledge and share information for common understanding. There was some dilemma on how and where to start just after the earthquake, but later with the formation of NRA, these works took momentum and the reconstruction and recovery work started to take proper shape. This has opened the way for cooperation and the commencement of interventions, with quick survey and damage assessments for immediate rehabilitation and foundation for a long-term planning. He opined that the presenters have incorporated overall achievements. According to him, working with local and national contractors has taught a lot of good lessons. A number of bridges, office buildings, schools and water schemes were reconstructed to resume service provision as

soon as possible. A holistic approach was adopted for the reconstruction of school buildings including social aspects. Transferring skills to the local community for sustainable maintenance of services and schools is another good message to share. He stated that the combined approach of designing and providing hardware and software in projects is more effective.

Technical support for the design, provision of supervision/construction, and contract management -combined with both national and international technical expertise (if they have opportunity to work together) – would be effective. Similarly, some interventions like quick surveys of the damaged schools through technical cooperation paved the way for their immediate rehabilitation. Effective procurement of local contractors leads to efficient contract management. Different integrated types of interventions like quick impact projects focusing on heavily affected districts – Gorkha and Sindhupalchowk – were also effective to resume the services on time. Gender and disabilities- friendly facilities were considered as resilient and realized through the reconstruction of physical, social, economic and institutional aspects, based on BBB. Facilitation support to ensure proper utilization and maintenance of facilities could be very precious. In addition, harmonization and dissemination of messages, best practices, and innovations discussed through various platforms need to be widely disseminated sector wise.



Dr. M. N. Khan stated that vulnerability of infrastructure still persists in a large sense throughout all Nepal. While working for school and health facility construction, the access to service to each of these facilities for critical service provision was very critical. These infrastructure development projects are an environmental issue as well. Another issue today is COVID-19. How are we dealing with these issues? We have to learn, document and implement. We frequently talk about limited capacity of the contractor, but accountability is also an issue. If it is not addressed, the envisaged quality of construction may not be achieved. Agreeing to the comments made by the JICA representative, he posited that it was good to learn that a significant number of health facilities have been reconstructed. Whatever has been done so far, and how we will proceed further, determines our long run goals. One of the important components is to make the infrastructure environment friendly. If local companies are encouraged, they will

perform better in the long run. We have identified some companies and worked with three of them. They are ISO certified and their works have expanded to other districts as well. At the end, he mentioned that thinking differently helps to achieve a lot of things. Consideration of health and safety protocols in the workplace during construction is very vital. In order to extend these learnings, we need to think in a broader way of how we can reduce vulnerability beyond the earthquake districts.



As a supporting commentator to Dr. M.N. Khan, Dr. Arjun Koirala began his comments mentioning that he agreed with his comments. He stated that although disasters are destructive once they happen, they also bring opportunities to make communities more resilient and safer. He posited that the issue of contract management has been rightly pointed out in the presentations. It needs to be properly looked into. During this whole process so many people were trained but now we have this issue of how

to retain these skilled workers so that we can utilize them in future? There is a need to strengthen the safety practices in both hardware and software in the health, education and other social sector infrastructures. He summarized some of the important action undertaken in the past as: standards and codes having been reviewed and improved; resilient sites have to be selected not only for reconstruction but for the normal construction as well; contract management significantly impacted the period of performance; huge construction generates a huge number of technical resources but how we retain and re-engage them is a challenge; capacity and gaps of different institutions at various levels should be discussed because it this not only limits the institution but limits its practices. Capacity of beneficiaries should be strengthened so that they can be involved in operation, maintenance, designing and construction. Delivery of services from the public infrastructure includes the provision of crucial hardware, but we need to focus on strengthening software as well. Having a database of the disadvantaged community would be very helpful in identifying the proper beneficiaries.



Mr. Purna P. Kadariya congratulated NRA for such a huge achievement in a short period of time. He highlighted that NRA has achieved great results in spite of continuous political transitions. The experts are in the driver's seat and they know their destination, that's why it is successful. He pointed out that the presenters did not highlight the transparency of NRA. The whole resource allocation and mobilization should be made apparent for the wider audience. We have not gone to the specialization and performance-based amendment so we should make provision for capacity building, gender equality and social equity. Reconstruction is possible with an educational mandate. When an agency is created to deliver a service of making a structure resilient, not only the building matters but the service matters too – software aspects should not be disturbed while going through the reconstruction process. We discussed on symptoms like problems with contract management and procurement but did not discuss the root cause of the

problem. Therefore, we need to focus on performance-based maintenance. He concluded with the note that the policies have to be streamlined. Experts who know the issues and can give options should be kept in the loop. NRA has a very good team of experts; hence it needs to follow the standard guidelines and transparency that results in good delivery and achievement. We could start by finding root causes of issues and by making provision in contractual arrangement while engaging the contractors.

Many people were trained but now we have this issue of how to retain these skilled workers so that we can utilize them in future.

Presenters' views on comments



The first presenter of the third session, Er. Bharat P. Aryal responded that there are guidelines for the case of environmental safety and NRA has been following EIA, IEA or both as defined by these guidelines. He confessed that we have to develop a better maintenance culture because there have been issues of only periodic maintenance in the past. We have been following transparency as well – there are public auditing and public reviews done regularly in NRA.



Dr. Youb Raj Paudyal extended his appreciation towards the government of China for supporting the reconstruction of Durbar High School. Regarding quality aspects, he stated that we need to have good coordination mechanisms and institutions. For instance, during the initial stage of planning of the reconstruction of the education sector, we created a platform involving government agencies, development partners and donor agencies to discuss, plan and work together using the same guidelines

and implementation mechanisms. That had really helped. During planning, we need to follow integrated and harmonized planning – inter development partner coordination and inter agency coordination is very important for effective planning and for quality implementation as well. We agree that our testing facility is less, but we have learned a lot and recruited many engineers and applied site specific supervision procedures and set up different laboratories too in different areas for testing. That helped a lot in assuring quality. As for the transparency and integrity from NRA, we have special guidelines, coordination mechanisms and information sharing mechanisms with our key stakeholders. We always share available information before signing agreements and before doing activities at the local level. And later, we inform them about the progress as well and do proper handovers at the end of the projects. More importantly, we have much vulnerable infrastructure, mainly buildings outside of the 32 districts. For them, the government needs to make an integrated programme to assess those vulnerabilities and make a phase-wise or priority-wise programme to support them.



Arc. Raju Neupane stated that he agrees with most of the comments made by the commentators, especially with Mr. Kadariya who said we should be focusing on creating resilient society, not only resilient buildings. And, he stated, that is exactly one of the CLPIU Building's achievements. During the establishment of CLPIU Building, one thing we wanted to change from the very beginning was the NBC 105:1994. It took a lot of time, energy and resources to discuss and come to consensus of revising the code. We have done the software part, now, the challenging part of implementation lies ahead of us.



Regarding quality, presenter Er. Amrit Shrestha stated that we all have to check the design period itself to control quality. We have been monitoring designs for controlling the quality. We also have laboratory tests and field test designs during the document preparation phase.

Comments & queries from chatroom and social media

Rupesh Shrestha: (To Amrit Shrestha) In private housing reconstruction, beneficiaries did not prioritize the private toilets nor gave WASH infrastructure any priority. Any suggestions on how to complete these elements in coming years?

Deependra Sharma: Reconstruction of public buildings and social infrastructures were managed by the staff of DOLIDAR (now DOLI) and the DUDBC, as evident from the number and

extent of permanent staff of these institutions being deployed at the CLPIU/DLPIU previously under the concerned line ministries and afterwards, under NRA. The institutional arrangements and achievements made in reconstruction would not have been possible without these experienced managerial staff availed within a very short span of time. The scope and size of these institutions are now being redefined in the federal context. Their services will not be as easily available in the case of another disaster. I see this to be a seriously challenging issue in case of future disasters. This also shows how important it is to plan and invest in the capacity building of local governments to manage future disasters. Also, I understand that meeting Occupational Health and Safety (OHS) standards was an important aspect of housing, heritage, public and social infrastructure reconstruction. It is advisable to mention some of the lessons learnt with regards to OHS during reconstruction. Also, to make contract management less complicated, it is advisable to handover the ongoing reconstruction of health buildings to DUDBC.

Krishna Bahdur Raut – Joint Secretary, OPMCM: This is an excellent webinar at this juncture of NRA. It is helpful for knowledge creation and management. It will be instrumental for Build Back Better reconstruction and rehabilitation which will ultimately contribute to creating resilient communities.

Remarks by the CEO NRA

Er. Sushil Gyewali began his remarks responding to the presenters and commentators. First, he commented on the presentation mentioning connectivity as very important, not only in reconstruction

of roads, but also in telephone and internet connection in all local governments' areas. For reconstructing local government buildings there is a fund available so this will be done by local governments on their own. He responded with agreement to Mr. Lu Yin's observation of clearance of the right of way (environmental clearance, land acquisition, utility network clearance, etc.) for any reconstruction, and that this needs to be addressed. Responding to Mr. Lamsal's comments, he said that the engagement of good consultants is important for quality construction, so it will be continued. Similarly, responding to Dr. Khan on vulnerability assessments of social infrastructure, this should be taken into consideration for resilient public infrastructure in future. Environmental compliance guidelines are there and are being used as this is very important. The pandemic situation is highlighted in the housing sector and it will be highlighted in other sectors as well. Talking about occupational health and safety, he mentioned that there is a need to have information on these to record the level of success. He further said the private sector engagement is vital, and we will have to look into this and develop appropriate guidelines and procedures. He also responded Dr. Koirala's comments saying the cost vs quality perspective is to be looked upon very seriously so that the issue can be addressed properly. He concluded with the note that he is in favour of providing a license to trained masons to honour their work and contributions. He then thanked all for enriching the presentations and making valid comments. He added one point about connectivity that the roads were not only constructed as roads for transport, but for the mobility and connectivity purposes of the people. He assured that information will be collected from all Palikas and wards about the status of local buildings because NRA

has not been able to reconstruct the local government buildings due to political transitions and federal structures. The Shanghai Group from China has completed a number of reconstruction projects like the Ring road and Durbar High School; there is so much we can learn from them. JICA has highlighted the engagement of consultant contractors which is highly appreciable. Vulnerability assessment at the sites is very important, as prioritized by the World Bank. This needs to be incorporated in our future projects. Fast track land acquisition for infrastructure projects is very essential with resilience components. And finally, quality assurance is very vital in each and every project under reconstruction.

Connectivity is very important, not only in reconstruction of roads, but also in telephone and internet connection in all local governments' areas.

Closing remarks by the session Chair



The session Chair Dr. Hariram Parajuli thanked all presenters, commentators, participants and NRA experts for sharing their knowledge. He pointed out the comments and queries posted in the chatroom and social media and assured that the answers to those queries have predominantly been answered. He also assured that all the relevant comments and feedback from everyone including the CEO of NRA will be further discussed upon and incorporated in the Compendium and the proceeding report. He summarized that NRA has completed the reconstruction and rehabilitation of more than 4,000 water supply systems, and have completed reconstruction of 583 Kms roads, more than 6,000 school buildings, restored 32 irrigation systems, and constructed 698 health facilities and 415 government buildings. In the meantime, we revised the building code, and it is already functioning. Looking back at the last five years, it took more than six months for

setting up, formulating the guidelines – hence, NRA’s effective time has been four and a half years only. There were administrative and political disturbances and multiple changes of leadership which really hindered the path of NRA. However, considering everything, NRA has indeed achieved a lot. NRA has always maintained quality of work everywhere and it will keep up the good work for another one year before handing over to another agency. He concluded his speech and the session assuring that the reconstruction works completed by NRA can be compared to the works of Japan and Western countries for quality assurance.

Closing Session



The programme host Ms. Rita Lamsal – Section Officer, NRA moderated the closing session. She thanked everybody involved in the programme and assured all participants that proper documentation

in the proceedings report will be made for dissemination. She called upon all present NRA executive members and officials to take their respective seats for the closing ceremony. She then called upon the Chair and the Co-Chair of the seminar Er. Sushil Gyewali – CEO NRA and Er. Ram K. Sapkota – Secretary, NRA to take their seats. The host then requested Dr. Chandra B. Shrestha, Er. Ram K. Sapkota and Er. Sushil Gyewali respectively to deliver their closing remarks.

Closing remarks



Dr. Chandra B. Shrestha briefly shared his observations of the day. He stated that all the sessions went well. The housing sector received critical feedback and suggestions on how best to apply NRA experiences in future, how the best practices can be brought into use, and how the learnings of NRA can be incorporated into the policy framework, etc. He also appreciated the acknowledgements

received from all development partners. He added there are some limitations in the heritage sector, but we have to take initiatives for sustainable heritage conservation. For public buildings, the projects are going on and they have to be carried forward. We have well received and recorded the suggestions to explore the root causes for every problem and we will address them appropriately. Licensing to mason is a good suggestion, and NRA has already initiated the establishment of a database of trained masons and engineers, which will be taken forward. We are looking to prepare a roster of such human resources and professionals. Retrofitting of the heritage sites is also a good suggestion and it is well received but will the retrofitting really preserve the traditional heritage characteristics? All historians and archeologists must ponder upon this. These are the main take-aways from day one.

What is the strategy to strengthen local governments so that they can take over reconstruction responsibilities more fully? In this regard, NRA is in the process of developing a roadmap for its disengagement strategy.

Er. Ram K. Sapkota congratulated all for their active participation and full contribution. He commended all colleagues and stakeholders for the smooth running of day 1 and for completing every presentation and session right on time producing the desirable outcome. Looking back at the sessions there were many common issues like the urgency for reconstruction, urgency for service providing, political transition in the initial phase of reconstruction, etc.; all are applicable to all thematic areas. There were also

many specific issues discussed during the presentations which were later diligently dealt with. Similarly, when we talk about the strategies adopted by NRA in the reconstruction process, many are common because the bases are common, like the legal framework, transparency, accountability, responsiveness to people, justice, grievance handling, etc. All these strategies and approaches point towards the execution of better communication, coordination and collaboration at all levels. This is what NRA stands for. Lastly, looking back at the lessons learnt, there are again many common lessons, like BBB which is universal and applies anywhere in the world, and there are specific lessons as well. Seminars like this are the platforms where we can discuss those specific challenges and learning in detail. He said that he is very satisfied with the comments and feedback raised during discussions which is being recorded at the moment and will be properly documented later for future reference.



The programme Chair Er. Sushil Gyewali began his remarks by thanking everyone who had contributed to the seminar sessions and the post-presentation discussions. He informed that day 1 of seminar was a success and that the thematic teams have received many observations, feedback and suggestions which will all be carried on into the remaining tenure of NRA and beyond. He also pointed out that he does not have many remarks to share because he has been giving concluding remarks during all presentations and sessions. One recurring query that he highlighted was, what is the exit strategy of NRA and how the lessons learned of NRA will be sustained in the future? In addition to that, what is the roadmap of NRA for strengthening NDRRMA to take over all the responsibilities? Similarly, since all the works of NRA in housing and public buildings involved local government, it is frequently asked – what is the strategy to strengthen local governments so that they can take over reconstruction responsibilities more fully? In this regard,

NRA is in the process of developing a roadmap for its disengagement strategy. Among the three levels of the government, the NRA has to engage with local governments as they have accurate information on housing stocks and socio-economic information. Such information will be regularly updated. If disasters happen in the future, the local governments can easily obtain the information and start taking immediate and necessary actions. He shared his perception of the main take-away messages from day 1: for the housing sector, how to manage post NRA issues? How can the best practices be integrated within other relevant institutions? How to strengthen NDRRMA? For heritage, how to carry out assessment research before restoring a concerned heritage monument or site? For public buildings, how to transfer knowledge to the local governments and how to strengthen them? Finally, he thanked everyone again and closed the session and the first day.



Day Two - 28th November 2020

Opening Session

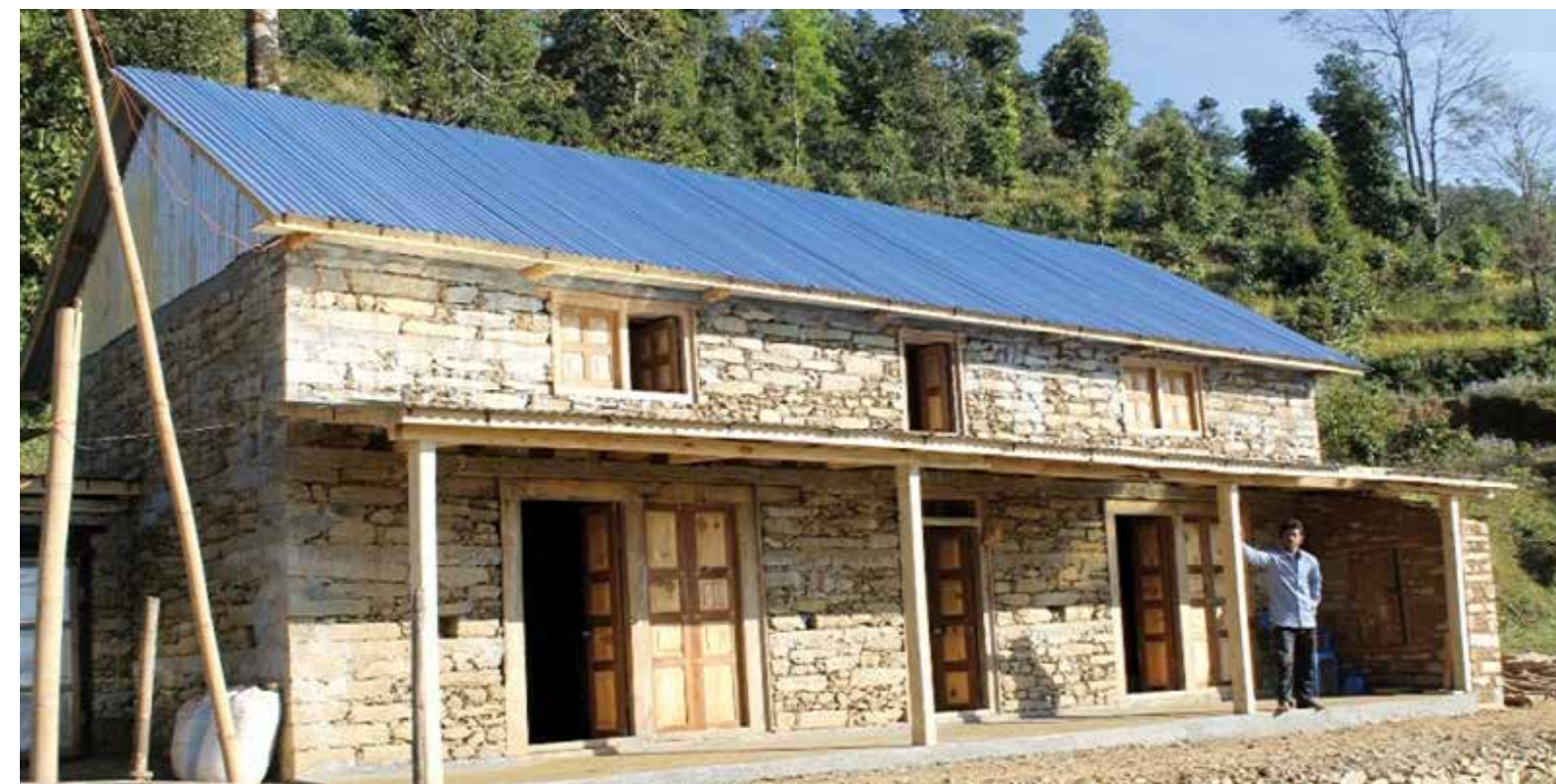


The second day of the web-based seminar commenced with Ms. Rita Lamsal – Section Officer NRA, the host, welcoming all to the programme. She requested all executive committee members, the Secretary and the CEO of NRA to take their respective seats. She announced that the Secretary Er. Ram Krishna Sapkota and the CEO Er. Sushil Gyewali will respectively reprise their roles as the Co-Chair and the Chair of the programme. She called upon Dr. Chandra B. Shrestha – Executive Committee Member and ICNR Convener NRA to briefly shed light on the sessions scheduled for day two of the seminar. Similarly, she called upon the Co-Chair Er. Ram Krishna Sapkota to make a brief recap of day one. She then called upon the Chair Er. Sushil Gyewali to deliver his opening remarks.

The Chair Er. Sushil Gyewali briefly summarized day one. He shared that the presentations of day one highlighted the NRA's strengths, challenges, programmes and progress in private housing, cultural heritage and public building reconstruction and recovery. NRA's priority and focus has always been on benefitting people affected by the earthquake, and also to involve local communities in the reconstruction process for a resilient and sustainable future. He mentioned that even in the given hard geographical terrain, NRA reconstructed 800,000 houses and completed reconstructing 8,000 school buildings with an average of 80% achievement, and with the full engagement of all political parties in different committees of NRA developing consciences for ownership at all levels. He informed that NRA is in the process of establishing operation centers to operate emergency and reconstruction services in case of disasters in all sectors in the future, at all levels of government including the ward level. Therefore, NRA is working closely with all the concerned agencies to maximize efforts to achieve desired results. He also talked about economic recovery informing that there was 4.8 Million USD invested from the government of Nepal and a total of 3.18 Billion USD has already been invested in reconstruction and recovery works. There is also a huge support from the private sector which has created huge numbers of employment and added value in the GDP of Nepal. NRA has always given the importance to community and house owners

in the reconstruction programme. In this regard, 70% of school reconstruction is managed by communities themselves. This has strengthened social cohesion in our society. All things considered, Nepal's reconstruction process has not all been smooth; we should not forget about challenges, political transitions, time & resource

constraints and so on that greatly impacted on the whole process. Finally, he welcomed the moderators, presenters, commentators and participants and expressed his happiness on the productive discussions held on day one. He expected that the same quality will be maintained in day two and throughout.



Day - 2, 28th November 2020

Fourth Session: Livelihoods and Economic Recovery

Presenter: Dr. Chandra B. Shrestha

Commentators: Mr. Kamran Akbar – World Bank (supported by Dr. Nigel Fisher), Dr. M. N. Khan – USAID (supported by Dr. R. Guragain), Mr. Ram Prasad Bhandari – JICA, and Ms. Maggie Stephenson – Independent expert

Session Co-Chair and Moderator: Er. Ram Krishna Sapkota

Session Chair: Er. Sushil Gyewali



National Reconstruction Authority

Compendium theme: livelihoods and economic recovery

Dr. Bishnu Bhandari – Executive Committee Member, NRA

Taking care of the vulnerable and promoting citizen resilience

Livelihoods and economic recovery: overview

2015

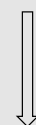
- 94 million workdays lost affecting 5.6 million workers
- 32 districts with 2.8 million households affected
- About 74,500 workplaces completely damaged
- Loss of US\$ 171 million

2020

- 166 million workdays generated
- 77 thousand masons trained (16% women)
- Distribution of materials & direct interventions
- GESI works on the ground
- Line Ministries and Partner Organisations reached out to 460,000 households investing NPR 20 billion



94 M
Workdays
lost



166 M
Workdays
created

Challenges and issues



Institutional and organisational

- Replicability of good examples and cases missing
- Linkage to market forces not reinforced
- Outcome of interventions not assessed
- Longer term employability of trained graduates lacking
- Soft loans not easily accessible to the affected population

Operational

- Continuity of livelihood activities not supported
- Unavailability of alternate livelihood opportunities
- Lack of skilled workers at the outset
- Provision of livelihood opportunities in integrated settlements lacking



Strategies and approaches

- Labor-based construction technology
- On-the-job training to masons & training of engineers
- Special support for vulnerable and marginalised groups
- Addressing livelihoods using a three-pronged strategy: through government, NGOs, and de-facto through the reconstruction activities
- Establishing NGO Guidelines and partnerships



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Lessons learned



Institutional and organisational

- Livelihood should be linked with all areas including housing at all rungs of reconstruction & recovery process right from the outset
- Cooperatives are the best platform for people's access to markets & finance
- Transparency and accountability are not only about sharing information but the involvement and meaningful participation of the people
- Disability is a development issue; it is a right issue; it is a health issue: livelihoods provide solutions to resolve or minimize these issues
- Government – especially local government - can play a vital role in scaling up and rolling out good livelihood models and practices
- Private sector should be more involved in developing access to and supporting market outlets



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Lessons learned

Operational

- Cash-for-work scheme was appreciated by many communities
- The traditional method of *Hat Bazaar* should be strengthened
- Involvement of local governments is key to an increased sense of ownership and sustainability
- Special livelihood modalities should be adopted for vulnerable groups, especially those people with disabilities
- GESI is key to successful, fair and equitable reconstruction and recovery



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Conclusions



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- Maximum outputs with minimum resource
- Many good cases examples but scattered and on small scale not rolled out broadly
- Tripartite collaboration is a unique way of mobilising POs
- Further support is required to develop marketing infrastructure and linkages
- Livelihoods themselves benefitted from being aligned to the reconstruction process with a huge number of workdays being created
- Valuable accomplishments have been achieved following GESI approaches
- Economic recovery needs time and takes time

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Key messages

1. **Livelihood is sustainable when it goes together with marketing**
2. **Sense of ownership exists when cost-sharing is provisioned**
3. **Rural people always diversify their livelihood options**
4. **Tripartite collaboration has proven effective in addressing livelihoods during the reconstruction and recovery process**
5. **An end-line study is the best way for self-assessment and evaluation of successes**
6. **Skills & incentives help to ensure application of the ideas learnt**

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Presentation 8: “Taking care of the vulnerable and promoting citizen resilience”

The presenter informed that his presentation mainly focuses on 3 sectors of agriculture, industries & commerce, and tourism because those sectors were mostly affected by the 2015 Gorkha earthquake. He also informed that several projects are being run to address the losses in those areas, and now they are gradually progressing. He shared that in 2015 because of earthquake, a total of 32 districts with 2.8 million households were affected, 94 million workdays were lost affecting 5.6 million workers, with about 74,500 workplaces completely damaged which made a total loss of 171 million USD. Once NRA began to work, as of 2020, 166 million workdays have been generated, 77,000 masons have been trained (16% women), line ministries and partner organizations reached out to 460,000 households investing NPR 20 billion.

This has created new hope and a message that the people and country of Nepal has revived and is moving forward. He also mentioned several institutional and organizational challenges and issues that were encountered by NRA during the recovery and reconstruction process, including the sustainability of the employment opportunities of the trained human resources. Continuity of livelihood activities and opportunities were not supported as there were many operational challenges faced by NRA. As a strategy, training was given to local staff to reach out to target groups with the fullest utilization of resources of government and partner organizations, so that they develop the skills and utilize them to

develop incomes. During these projects, the teams realized that the basic necessity of people should be delivered to the affected people and linkages to markets gives people the means of living. He further mentioned that NRA and partners incorporated GESI in the projects and advocated for inclusion at all levels. As a result of following inclusive practices and direct involvement of the local government in the programme, he concluded that the sense of ownership has increased and contributes to sustainability in the livelihoods initiatives.

Response by the Commentators



The first commentator Ms. Jasmine Rajbhandari expressed her happiness to be a part of the session as a commentator. She stated that the presentation covered a wide range of information, from achievements to lessons learned throughout the implementation of the projects.

She recalled the work she had done earlier at NRA and suggested that there should be a distinction made in the chapter around cash – cash support, conditional cash, cash forward, etc. and that this needs to be further clarified because there seems to be some mix up in the concepts. She opined there are three stages of fragmentation, where the most vulnerable and economic resilience of vulnerable people need further clarification. In addition, a comprehensive strategy needs to be applied in the sectorial plan for the short term, medium term and long-term. It would be good if the framework for this were shared. As mentioned, diversifying livelihood and employment opportunities are the longer-term strategy which should not be forgotten during planning and evaluation. The comprehensive survey data collected after the earthquake are also evidence, which can be utilized for the livelihoods chapter of the compendium. The social integrated approach could be effective for this project and establishment of employment service centers might be helpful for future initiatives. She highlighted the social registry in which socio-economic data are linked with citizens IDs, and it is useful to identify vulnerable people in case of disasters and emergencies. Numbers are essential but quality also matters i.e., it is not just the number of days to earn generated and how much they earn in terms of supporting their livelihood. Segregation of data would be interesting to look into, to explore more in this livelihood aspect of disaster survivors. She expressed her hopefulness

that the WB and NRA will do much more together in future for the benefits of the targeted people with economic vulnerabilities.



The moderator Dr. Chandra B. Shrestha thanked Ms. Rajbhandari for her comprehensive comments. He acknowledged her for highlighting many important issues – the desegregation of data and making sure that the target beneficiaries are really getting benefits out of the projects. The appropriate criteria for identification of the vulnerable groups is a discourse which is still being discussed and debated. He informed that an employment service center is going to be launched jointly by the World Bank and Nepal government soon, and it would be appreciated if NRA could join the programme. He then called upon the next commentator to deliver his inputs.



Mr. Vijaya Singh started off with appreciating the presentation which he found very comprehensive in content and coverage. He agreed with the issues highlighted by the presenter. For him, the process adopted to engage the people to improve people's livelihoods in the reconstruction process was commendable. He stated that the livelihood component should focus on helping the escape of survivors from financial vulnerability, and to make them capable enough to fight against any disaster in future. Examples are there where we can work better, not only in the 32 target districts but in other areas as well where the component of livelihood may help people to come out from debt traps through increase in income. In the meantime, if local people are engaged in the reconstruction of social or public infrastructure, it works in two ways – it brings back the mobility and services intact, and people benefit through earning from the work. Mr. Singh further informed that there are some models to make livelihoods inclusive. In the

beginning, livelihood was taken as reviving daily life or return back to normal, by reviving job business or earnings; however, later on, it was taken as a part of life. He shared his experiences mentioning that in his context, the livelihood actions started a little late; almost two years later than other recovery actions. Now, the UNDP is consistently supporting in reviving agriculture. Similarly, disabled people are engaged in making bamboo furniture, women are going for agro and cattle farming for cash focused livelihoods, with support from dairies and cooperatives. The idea is to target comprehensive activities and help make income to protect from debt traps with specific focus on people living with specific needs through access to finance, and access to work and earning. These should be incorporated into the recovery and reconstruction process where it is appropriate. He posed a question – what is the best modality of delivering recovery outcomes? Is it directly through government agencies/systems coordinated by NRA which was the system adopted? Or, is it delivered by NRA with line agencies reporting to NRA which was actually thought of as per the spirit of the NRA Act? He concluded his comments mentioning three points – 1. A targeted comprehensive approach is needed for economic recovery so that reconstruction and livelihood recovery can go hand in hand

Livelihoods are just not about counting the volume of employment generated or the income that has been produced out of it, but about protecting the people from debt traps and also from other forms of disasters.

3. Targeted interventions for various groups of people like people with disabilities should be increased, and also, we should always start the livelihoods implementation or interventions from the very beginning; for that it requires access to finance and access to cash which are very important aspects of livelihoods.

Presenter's views on comments



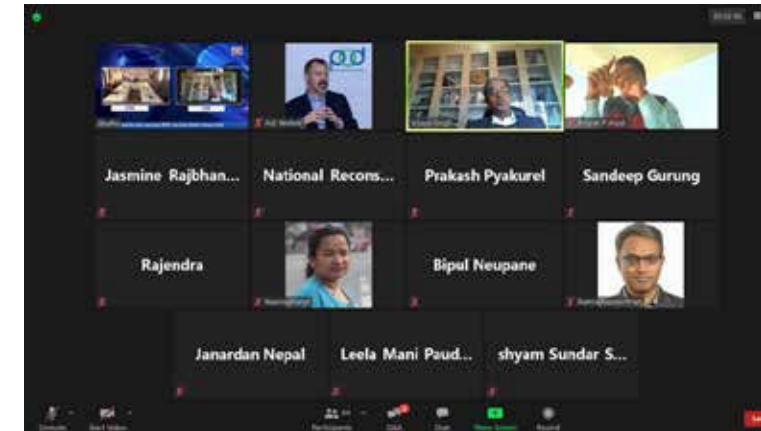
The presenter, Dr. Bhandari, expressed his satisfaction on the comments he received from the commentators. He stated he would try to respond to as many comments as possible. Taking a holistic response to reconstruction as Jasmine mentioned is exactly what NRA is doing at the moment. There are so many cross-cutting issues and NRA is careful about them at every step. PDNA has clearly said three things on livelihoods: 1. Short-term for immediate support on agriculture, tourism, industries, etc. 2. Mid-term for involvement of the private sector 3. Long-term for

involvement of the whole society. NRA was not reinventing the wheel: NRA was building on the success of what the line agencies and partner organizations had already done. Regarding the vulnerable, people above 70 years old, people living with disability and child-based households were clearly defined and identified. NRA has been trying to reach out to those people and been successful in this regard. Regarding GESI, during the initial phase, NRA had brought some necessary changes in the approach as suggested by the study done by UN women in three target districts. He agreed that social registration is a good idea. He appreciated Mr. Singh for his pragmatic experiences and knowledge. Livelihood is fully linked with other aspects of life. Reconstruction and recovery should not be looked at from the engineering perspective only but also from social, economic and every other perspective. Livelihood is a very versatile and cross-cutting subject. As far as the livelihood approach is concerned, the loan facility is there but people are still going to local money lenders who charge them huge interest. This is something we need to seriously work on.

Comments & queries from chatroom and social media

Dinanath Bhandari: As per NRA experience, when will the right time be to initiate livelihood and economy recovery programmes? Does the season/month of disasters like earthquakes affect livelihood recovery? For instance, what if it strikes during mid-July? or in winter? Also, do you think there is something we can do in advance (prevention and preparedness) such that livelihood and economic recovery is smooth and in better order in the aftermath

of mega disaster? If NRA has any applicable lessons, these would be worth taking forward by NDRRMA and all actors in future.



Responded by Jasmine Rajbhandary: As mentioned in my presentation, the concept of an integrated social registry could expedite identification of the post-disaster beneficiaries.

Responded by the presenter: The season is very important in agriculture. We always went to farmers in Mangsir (December). We have always tried to reach out to them in the right season so that they could learn, and we could learn too.

Dinanath Bhandari: In addition, institutional and organisational structures and operations are of utmost importance. Based on the learning from NRA's work (which is worth praising), do you believe the current committee-based institutional and organisational structure for NDRRMA at all levels (Federal to Local), and lack of accountability to any agency and an episodic response approach, will be able to address future challenges?

Responded by the presenter: Committees are everywhere, and they are not always either responsible or held accountable. The administrative unit is mainly responsible. Committees are there for policy judgement and for giving advice. Committees are important but they don't go to the ground for implementation.

Bijaya Bajracharya: Providing results data on GESI will be important; are they accessible?

Responded by the presenter: We have the results. We can send them to you.

Shanta Rana, UNOPS: There is still much confusion regarding the top up or the additional support for vulnerable groups as many have not received it yet. How is this being addressed by the impact study?

Responded by the presenter: There is no confusion really. If you are qualified against the three criteria, then you will 50,000 NPR. However, due to procedural factors if somebody has not received the amount then we would like to expedite this.

Suman Bayalkoti: Have really vulnerable people got benefits? How can we confirm this?

Responded by the presenter: Yes, they have. Even with the limited resources, we have reached out to most of the vulnerable groups.

Youb Raj Paudyal to Everyone: Excellent presentation! There has been tremendous progress in the reconstruction, however the progress in this sector seemed low. I may be wrong; however, it is not easily visible. There are huge opportunities to make our economy vibrant in a sustainable way through this sector. We missed to set up and support in the construction industries

(materials production and construction), particularly in rural areas in a sustainable way. We are still unable to track vulnerable groups effectively and bring them into revenue generation situations.

Responded by the presenter: Construction based technology is possible. One of our partners, Practical Action, has focused all their activities on applying construction-based technology. They have made compressed bricks, aggregation center, etc. Hence, it is possible.

Bijaya Bajracharya to all panelists: Have we also analyzed the situation in terms of construction technologies (e.g., CSEB vs fired bricks) for the long term (environmental, climate change, local micro-enterprise development potentials, etc.)? Some organisations (supported by DFID, LNOB) initiated inclusion of the most vulnerable households in an MIS. Can we explore whether that can be utilised for analysing data from the vulnerability criteria? How have we created an enabling environment for women masons for continuing their employment? There was much evidence that trained women masons could not continue their work. What factors will make women masons continue and achieve resilience?

Responded by the presenter: We have not been able to do SIA or EIA, but we have seen the direct result of the activities. We can say that they can be further expanded but to go on a large scale, there are many tests they would have to go through. We will certainly consider this.

The moderator Dr. Chandra B. Shrestha thanked the presenter Dr. Bishnu Bhandari for his responses to the comments. He informed that NRA is going to initiate a livelihood project though the agency at the twilight of its existence. He requested the Chair Er. Sushil

Gyewali to conclude the session highlighting the proposed livelihood projects.

Closing remarks by the session Chair



The session Chair Er. Sushil Gyewali thanked everyone for contributing to make the session productive. He began with highlighting the three approaches that were adopted by NRA, through which an approximate 166 million workdays have been generated. Besides the main activity of massive recovery and reconstruction, NRA also has implemented some focused programmes, along with its partner organisations. The Livelihood project is one of them, through which NRA has also integrated livelihood components into other programmes being implemented. He stated that he agrees with Ms. Jasmine's point that specifically explains that the analysis of problem and achievement should be done from the perspective of GESI. He was also in favor of the establishment of the integrated

social system and analysis of annual workdays. Responding to the queries from Bijaya and Shanta (chatroom), he said that the grant is delivered to the people through the local governments. NRA tried its best to give employment to those who are badly affected by the earthquake and is advocating to have a similar mechanism at the local government level as well. Er. Gyewali shared that NRA works together with NGOs within tripartite agreements in housing and local infrastructure project implementation, and in enhancing entrepreneurship. The livelihoods thematic team will look into the comments made by Ms. Jasmin to find the way to address these issues, and how the suggestions can be incorporated to make the MIS GESI-friendly.

Responding to the questions from the audience, he said that NRA acts on the basis of its mandate and with the local government of Nepal. As discussed earlier, the government and the World Bank are going to launch the employment service center very soon,

which will be really helpful in the livelihood sector. Another issue is debt trap. Why are people going to local landlords for loans even though we have a mechanism to provide 50,000 NPR in a first installment and up to 300,000 NPR in total? Actually, NRA tried its best to provide help to survivors, but there are many people who do not want to come into the banking system. There is a gap between provided opportunity and the demand/access, Why? Maybe because of the lack of information, lack of linkages and lack of facilitation in the process? The answer has to be explored through research. Now, the municipal ward office can facilitate the needy people to get soft loans provided by the government. Another aspect is access to markets for their products. In each of the local governments, such mechanisms should be developed. Through this approach we can address the priorities and concerns of needy people. In the future, our partners and government will develop such mechanisms so that every needy person receives timely support. With this note he closed the session.

Fifth Session: Rescue and Relief

Presenter: Er. Anil Pokhrel – CEO NDRR-MA

Commentators: Mr. Leela Mani Poudel – Former Chief Secretary and Former Nepal’s Ambassador to China, Mr. Ramraj Narasimhan – UNDP and Mr. Bipul Neupane – NRCS

Moderator: Dr. Chandra B. Shrestha

Session Chair: Er. Sushil Gyewali



Rescue and relief: overview



- Response to the 2015 earthquakes among the strongest in history
- High number of deaths (8,975); high number of injured (22,000); 800,000 houses collapsed
- National Disaster Response Framework (amended 2013)
- Prime Minister Natural Disaster Relief Fund Regulation
- International assistance within a few hours from neighboring countries: 34 countries, 141 teams, 4521 members
- Strong Security Forces and Civilian coordination
- Civil and Public response: Youth, professionals, I/NGOs, Development Partners and Private Sector

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Challenges and issues

Institutional and organisational

- Difficulty in conducting needs assessment
- Inadequate search and rescue capacity
- Limited capacity of airports, systems and airlift capacity
- Status and limited stock of emergency services – the first HSA had recently been operational
- Lack of adequate policy, technology, tools and equipment, and skills to dismantle the damaged structures
- Lack of office space for government operations as most government buildings had collapsed



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Challenges and issues (cont.)



Operational

- High altitude of the affected areas, the mountainous terrain, and the almost complete destruction of infrastructure in an area covering 32 of 75 districts (around 30,000 square kilometers)
- Landslides destroyed roads and other infrastructure
- Deteriorating weather conditions
- Temporary settlement of displaced families
- Managing space for rubble
- Providing care to children who lost their parents and families
- Gaining accessibility: Immediate repair of roads and transportation
- ICT and electricity networks not providing full coverage for communication, hindering emergency response efforts

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Strategies and approaches



Immediate response

- Within two hours of the EQ the Central Natural Disaster Relief Committee (CNDRC) meeting called
- Subsequent CNDRC held its meetings on 25, 27, 30 April and 3, 10 and 12 May 2015 and response put on the ground
- Within four hours, the Council of Ministers held emergency meeting
- The meetings were followed by meeting of Secretaries under the Chair of Chief Secretary of the GoN
- Regional Disaster Relief Committees (REOC) and District Disaster Relief Committees (DDRCs) planned and implemented activities

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Strategies and approaches (cont.)

Organisational structures and actors

- A Central Command Post formed under MOHA Secretary
- NEOC, 5 PEOCs, 49 DEOCs & HEOC
- Multinational Military Coordination Centre (MNMCC) and Onsite Operations Coordination Centers (OSOCC) operational
- Operation '*Sankat Mochan*' and Operation '*Drid Sankalpa*';
- Command Centre at Nepal Police
- Network of NGOs and INGOs rallied to support rescue and relief efforts
- Volunteer groups, especially youth and professionals engaged
- Emergency Clusters put into action



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Strategies and approaches contd.



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• Critical infrastructures, utilities and supplies

- Schools
- Hospitals and health facilities
- Electricity
- Water-supply
- Telecommunication
- Food supply systems

• Roles and responsibilities of ministries, Departments, DDMCs

• Emergency shelters, Temporary Learning Centers

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Lessons learned



Institutional and organisational

- Need for strengthening capacity of National Search and Rescue Team specially
 - Security forces (Nepal Army, Police and Armed Police Force)
 - Formation of trained local volunteer units at the local level
 - Establishing an Incident Command System
- Establish mechanisms for international support for Search and Rescue relief and recovery
- Strengthen capacity of National Search and Rescue Team specially focusing for Security forces
- Establish mechanisms for international support for Search and Rescue relief and recovery
- Use of social media for reaching out to the unserved areas

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Lessons learned (cont.)



Operational

- Develop SOPs for incoming multi-national HADR assistance teams
- Formation of trained local volunteer units at the local level
- Mechanisms to address the needs of injured, disabled, elderly and women
- Mechanisms to distribute food and nonfood items in remote areas
- Need of geo-referenced disaster information management system
- Provision of helipads in municipalities
- Provision of emergency warehouse and adequate stockpiling of appropriate supplies and equipment
- Establishment of operations centers in each municipality
- Open spaces! In Kathmandu Valley – allocated as sites for temporary shelters and debris management- insufficient for the management of debris

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Conclusions



- Need for a robust and practical response plan & SIMEXs
- DRRM Governance System: Federal, Provincial, DDMCs and Local governments need strengthening
- Nepal's critical built infrastructure needs enhancing
- Newly endorsed building codes need to be implemented
- Need for proportional increase of emergency facilities compared to the increasing urbanization rate / change in risk
 - Prepositioning of appropriate equipment (heavy plant, bridging, transportation and other logistics stores) at ideal locations
 - Management of relief stores required for immediate response

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Conclusions (cont.)



- Need for priority inclusion of vulnerable groups in high-risk location planning and programming
- Facilitation of remittances, subsidizing transfer fees, bank accounts
- After huge investment in rescue and relief, ground prepared for BBB
- Clarity on roles and responsibilities of different agencies at onset of disaster is essential, along with improved emergency preparedness
- Need for one central oversight agency with full powers; NDRRMA established now based on this

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Key messages

1. Relief items need to be culturally appropriate
2. Cash assistance is effective for affected families
3. 2015 EQ lessons led to the formation of NDRRMA: Capacity strengthening needs and continuous engagement with DDMCs, LGs, and communities for response, needs assessment, emergency drills
4. Need increase of emergency facilities and systems compared to the increasing urbanization rate / change in risk
5. Need additional open spaces to manage emergency shelter, debris and other waste during emergency phase
6. Incoming humanitarian organisations need to fit with local approaches
7. Requires a proper communication plan and strategy during emergencies
8. Response and relief are effective through joint actions of government, non-government and private sector

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Presentation 9: “Saving lives and paving the way to Build Back Better”

The second presenter of day two Er. Anil Pokhrel claimed that the Build Back Better initiative is the greatest response in the history of Nepal after the 2015 earthquake. He informed that PDNA came with the assessment report that shows a huge number of deaths (8,975), injuries (22,000), 8,00,000 houses collapsed, etc., that brought the whole country to a standstill for some time. Within a few hours, international humanitarian assistance arrived in Kathmandu from many countries including neighboring country India. Altogether 34 countries, 141 teams and 4521 members supported Nepal in the aftermath recovery. The government of Nepal also established the Prime Minister’s Natural Disaster Relief Fund Regulation, established a strong security forces and civilian coordination with a high-level positive attitude of civil and public response with huge numbers of youth, professionals, NGOs, development partners and private sector, mobilized. He described needs assessment as one of the milestones in the recovery and reconstruction initiative in Nepal. Although there was a real challenge to conduct such a need assessment in the emergency situation, it was also not possible to generate the resources needed without doing this. Once PDNA was done, it gave a picture of the loss of lives and property. By then, NRA was formed, and the work had gradually started. Lots of operational issues and challenges were confronted, including reaching high altitude affected areas to low altitude areas, with transportation and communication almost

disrupted. In many places, roads were blocked by landslides and the weather conditions were deteriorating. Providing care to the children who had lost their parents and families, was another challenge as temporary settlements were not functioning well and relief materials were not efficiently reaching the affected areas. In such situations, an immediate response was given within hours by the Central Natural Disaster Relief Committee, which coordinated the utility agencies and proceeded for repair and maintenance of the systems, establishing temporary shelters with minimum facilities, food and water. The schools, hospitals and health facilities were used as temporary shelters for the survivors.

Altogether 34 countries, 141 teams and 4521 members supported Nepal in the aftermath recovery.

Then he spoke about the progress as of 2020. NRA had struggled a lot to reach to this level, and today, about 90% reconstruction work has been completed. All services have resumed. However, there are so many areas that need to be improved. Among many issues, Nepal’s infrastructure needing enhancement is a must; newly endorsed building codes need to be implemented; there’s a need for proportional increase in hospital emergency facilities considering the increasing urbanization; management of stores are required for immediate response; prepositioning of appropriate

equipment (heavy plant, bridging, transportation and other logistics stores) at ideal locations is required; there's a need for priority inclusion of vulnerable groups in high-risk locations while planning and programming; facilitation of remittances should take place; subsidizing transfer fees should be taken up; efficient banking system needs to be engaged; clarity on roles and responsibilities of different agencies at the onset of the disaster along with improved emergency preparedness is essential, and there is a need for one central oversight agency with full authority.

Response by the commentators



The first commentator Mr. Leela Mani Poudel appreciated the presentation and expressed gratitude for the opportunity to engage in the NRA seminar. He appreciated the documentation prepared by NRA. He stated that such documentation will help a lot as reference. He confessed that the documentation part is very weak

in our government sector. Reflecting the days when the Gorkha earthquake hit, he said, everything was doomed, and the scale and magnitude of response had to be very big. People expected chaos in law and order, but nothing happened. The post disaster situation was properly handled. Even though the government had limited resources, the quality and quantity control convinced the donor organisations, and they became ready for the immediate response. Beside this, the PDNA had also created room for humanitarian assistance and partnerships. How to learn from the past? We needed to make a very detailed account of every activity every day and record everything, but who did what? How were things handled and taken care of? He added that the pandemic is creating an additional burden on NRA at the moment. So, NRA has to reflect on the scenario of how they had responded in the early days of the Gorkha disaster. Because we learn from our efforts and gradually develop capability. We did tremendous work on the given scenario of disasters. Taking detailed accounts and records will serve as a very important reference while developing response plans for any disaster and emergencies in future. For instance, on the disaster day, the NDCC meeting was supposed to be called by the Home Secretary but I was the first government officer to reach the NDOC. The situation was out of control at one point and the responsible government officials were not being able to deal well with the alarming situation. It took two hours to finally calm ourselves down and call the first meeting with all ministers and stakeholders.

Taking detailed accounts and records will serve as a very important reference while developing response plans for any disaster and emergencies in future.

According to our disaster response strategy which was already approved with international aid agencies under the leadership of UNDP. UNDP had to coordinate all those things, but they were short-handed. Moreover, the Resident Coordinator of UNDP was present at the NDOC building the day after the 2015 earthquake asking what is happening here. I asked him to arrange relief materials like tents and tarpaulins for temporary roofing for people but received water bottles. Many uninvited rescue operation teams arrived at the international airport of Nepal who didn't even have the knowledge of CSSR. We needed an SOP so that we can make people in power accountable for at situations like that. He further added, when he reached Sindhupalchowk, he saw NGOs occupying all the buildings and consuming available food stuff without having started any real action against the disaster. Even they didn't have any specific strategy of what to do and what not to do. What we needed to do is that – the government should have coordinated efforts for providing relief materials so that we know where the help has reached and where not. Instead, what happened was – a flood of NGOs and others reached to all affected areas before the government and they provided 1,000 NPR worth of relief materials and made a propaganda worth of 100,000 NPR. Moreover, they criticized that the government was not doing anything. There was a mention of food scarcity in the presentation but there actually was no food scarcity. There were only two scarcities – roofing materials (tents and tarpaulins) and

mid-term roofing materials like CGI sheets. The presenter rightly mentioned that the government officials were trying to procure these things, but they didn't have the capability and sources. At that time, he said that he had to fight with the cabinet ministers and the PM himself and convinced them to distribute cash, which actually worked. This was done because the government could fall into the procurement scandal if it had taken the responsibility to procure these Relief items from the market. This is why, if government issues an order for coordinated efforts in future for providing relief materials through the coordinated channels, then this must be followed strictly. As already mentioned, many NGOs and officials of aid agencies were protesting against the government saying the government is corrupt and the PM relief (PRF) is being misused. He also requested NRA to make robust mechanisms regarding handling of media, which is equally important in such situations. He suggested NRA to contact Dr. Swarnim Wagle (Former Vice-Chair NPC) who had presented a paper in an international forum about damage assessment in Nepal during the 2015 earthquake, and to also contact the former members of NDOC for more information. He concluded by saying that many government officials should be held accountable because many school buildings had collapsed at that time which were claimed to be earthquake resilient. Therefore, there should be a mechanism established to look after such negligence. Moreover, proper investigations should be carried out and the guilty parties should be punished.

If communities are strengthened with required capacities, they themselves can look after their local level needs.



The second commentator Mr. Ramraj Narasimhan from UNDP thanked and congratulated the presenter for such a comprehensive presentation. He very specifically stated that from PDNA to this current stage a lot of progress has been achieved. However, what were the recovery needs? How much was distributed? How and where was it mobilized? These things are not covered in the presentation. He suggested that it would be good to know about these things for the audience in a transparent way, and it would be good for future reference as well, especially before collaborating with those partners who had supported earlier and others who are willing to support later. He agreed with the presenter on the need of capacity enhancement of institutions and staff, but he also strongly suggested enhancing the capacity of communities as well. He opined if communities are strengthened with required capacities, they themselves can look after their local level needs. So, search and rescue capability should be taught, and resources should be increased at local levels as local people

are the first responders in case of any emergencies. If Local Emergency Operation Centers (LEOC) are established and operated, the local people can be available throughout the year, even in the non-emergency situations for other development works, and the local government can put them on standby, so that they can be mobilized any time in case of emergencies. In fact, resource generation and utilisation are shared responsibilities primarily within three tiers of governments as well as with the development partners. If it is done maintaining transparency, the partnership may last longer. Lastly, he pointed out that there was a total of 82 million USD recovery amount designated, but what happened later on is questionable. Regarding how much resources came in and how much was expended in what particular area, there is no proper data, so nobody knows. In such circumstances, long-term partnerships become questionable. Whatever the case, there is a strong need for enhancing search and rescue points.

The third commentator, Mr. Bipul Neupane appreciated the presentation and the comments made by the first commentator Mr. Leela Mani Poudel, which he believes are very good for future reference. In his view, rescue and relief largely depends upon the scale of disaster and the level of our preparedness. If we are prepared for mega disasters, small emergencies may not cause big disruption and the recovery process also becomes faster with less harm and loss. Hence, if preparedness is effective, we can save lives and property in case of a disaster. The next frequently asked question we hear is, for how long does rescue and relief takes place? It also depends upon the impacts of disasters. So, rescue and relief are more of a management job; if it is well coordinated and managed, it will go smoothly, and the survivors will get necessary services.

If not, the situation becomes more complicated. We can take an example of our relief distribution experiences as mentioned earlier by Mr. Poudel. He has cited the real ground level examples of how aid agencies overlapped with each other. Therefore, a strong mechanism is needed to monitor and act immediately in case of any fraud, mismatches or overlaps, and then relief support can go to the survivors' hands. Regarding the challenges, almost everything was chaotic and difficult to control – as the airports were jammed, relief materials were parked on the ramps and not properly cared for by the concerned authorities, humanitarian organizations themselves created chaos in many instances and the relief packages overlapped in distribution points, etc. He also pointed out that the provision of cash assistance was hailed as effective, however, 'how' was it effective is not well mentioned. He concluded emphasizing that the terminologies rescue and relief come together in most instances, but actually, they are different things and should be addressed differently.



Presenters' views on comments



The presenter thanked all commentators for providing their invaluable comments which are being recorded and will be utilized in future. NDRRMA is 11 months old now and in the initial months of 2020, we faced big landslides and floods that made the situation very critical. In the transition within the last five years, there has been some major changes not only from the government perspective but also from the institution's perspective. Considering everything it can be bluntly said that the resources have been inadequate. NDRRMA has not been able to reach out to far rural areas where disasters really have a terrible effect. A lot of those challenges are also due to the monsoon season and difficult geographic terrain in Nepal. A few priority areas that NDRRMA has started working on are: 1. drafting of volunteer bureau formation and mobilization guidelines. During the 2015 earthquake, close to 18,000 people were rescued by communities and local

volunteers. Hence, NDRRMA finds this bureau formation very important. 2. To strengthen the emergency store warehouses and relief centers. We have the national humanitarian seizing area right next to the KTM airport, but we have been able to launch 2 similar sites in Nepalgunj and Dhangadi. We are planning for Birgunj and Biratnagar sites as well. All these places will have provincial store sites and warehouses. 3. To have mobile sites in all 77 districts and at the municipal level itself. The PM himself had highlighted the importance of having at least one warehouse in all ward offices too. So, NDRRMA, since its inception has been working tirelessly along with development partners like UN, WFP, DFID and so on to enhance the capacity of these warehouses. We have also recently come up with a training manual to train the three security agencies so that they follow the standard protocols while carrying out search and rescue. We are also in the final approval stage of purchasing standard search and rescue equipment. The national incident command system (NIMS) has been clearly ensigned as a national act and this is how the NDRRMA would like to go. Establishing an online system will be a main focus to understand where things like search and rescue equipment, food & non-food items, and their suppliers, etc. are. It is a real challenge to map these things within every agency and municipality. But NDRRMA is trying and hopefully we will have all the information established and categorised soon. NDRRMA is also working to graduate from the weather based early warning system to impact based forecasting system which will greatly help the responders.

Comments & queries from the chatroom and social media

Shankar Dhakal: Regular landslides, and deadly rockfalls along Kodari highway, in particular are common features. I think we should categorise rockfall separately, rather than generalizing it with landslides so that it receives necessary and specialised attention. The recommended solutions can be implemented elsewhere too. Also, for local culturally appropriate solutions, I guess we need to establish local units who would input appropriate infrastructure and development needs of the community by considering the indigenous technologies and vernacular material. This has been largely missed during our reconstruction campaign.

NDRRMA is also working to graduate from the weather based early warning system to impact based forecasting system which will greatly help the responders.

Dinanath Bhandari to Anil Pokhrel: Looking at learning from the 2015 earthquake response and recent landslides, what would be an appropriate institutional structure at federal and provincial levels to carry out day to day preparation and response (in need) in a coordinated manner in the spirit of the jurisdiction allocated by the constitution of Nepal? Also, Cash is better than materials in many senses, as you presented. But why, in your opinion, it is not institutionalized properly and what could be done to adopt cash-based approaches, minimizing underlying risks and management challenges?

Youb Raj Paudyal: Excellent presentation Anil sir! Trained rescue teams play an important role to minimize the losses. However, we need to make it systematic in terms of coverage and type of disasters. Based on population and nearby re-occurring disasters, we need to set up institutions with trained manpower in strategic locations and plan the whole country accordingly. What is NDRRMA's view on this?

Er. Ram Prasad Bhandari to Anil Pokhrel: Excellent summarization. Reflecting back on my involvement in preparing the PDNA and having some level of quick studies, the messages to be taken forward are: 1. The National Disaster Response Framework was effective, it worked at the central level 2. It reached somehow at district level, but DDRF was not fully functional; district level agencies worked based on their experiences rather than following the framework prepared 3. Volunteerism was commendable, but the need felt was to connect volunteers and community level groups together with the mainstreaming response. Indeed, the volunteers were ready, but they were merely dancing to their own music as there was no mainstream music. This needs to be substantially strengthened by preparing SOPs and conducting periodic drills. For this to happen, having a new Act and NDRRMA in place is a good start, but this institution needs much more autonomy and capacitating.

Surya Mohan Dhakal: Every palika has to identify open land for response and relief activities. NDRRMA would surely need this, and it would also support coordination for the disaster response and relief.

Responded by Leela Mani Poudel: Rural municipality open space is not an issue, but for urban areas it is important, and certainly more problematic.

Dinanath Bhandari: Thank you and gratitude to NRA for bringing up rescue and relief issues as lessons. I assume the chapter on 'rescue and relief' incorporates low profile government officers, community responders, security personnel (Nepal Army, Nepal Police and APF) and victims' perspectives who were seen working in the sites, together with high level bureaucrats and dignified personalities to whom we often listen to?

Suman Bayalkoti: Rural municipalities need to train manpower for urgent rescue in the case of disasters. Community organizations also need to have a plan for disaster risk reduction for on time rescue and relief.



Nigel Fisher: In the first hours - and even days - of every post-disaster rescue and relief response, neighbours and community members

are always the first to respond, before others (national or international) arrive to help from outside the community. Is thought being given to strengthening this first community response capacity? If so, what are examples of some supportive interventions? To take one example, in the case of local entrepreneurs and shopkeepers, it should be feasible to create an emergency insurance facility, so that they can immediately release their goods to the population, without fear of loss or bankruptcy themselves.

Rajendra Chiluwal: What is the organisational setup of NDRRMA?

Responded by Leela Mani Poudel: NDRRMA should be under the OPMCM; this should be an inter-ministerial mechanism, not under MoHA alone.

The moderator informed that, due to time constraints, all online comments and queries will be delivered to Mr. Anil Pokhrel so that he can respond to them in detail. He assured that these discussions will be well captured and recorded for future reference. He then requested Mr. Leela Mani Poudel to deliver his way-forward remarks.

Mr. Leela Mani Poudel once again emphasized on open spaces that must be protected for safety against disasters and people must have knowledge of them. The government must be always ready with a certain minimum level of relief packages preferably in every municipality, so that no one can be left behind while distributing relief packages at the initial stages of emergencies. In this way, vulnerability issues can be addressed in disaster management. Having said that, the next important aspects that we should be careful about are communication, power and WASH fallibilities. Overcrowding of people weakens the networks and

signals of telecommunication in such situational emergencies, therefore certain hotlines should be designated for official and trustworthy information flow. With this note, Mr. Poudel ended his way-forward comments.

Open spaces must be protected for safety against disasters and people must have knowledge of them.

Closing remarks by the session Chair



The session Chair Er. Sushil Gyewali began his remarks expressing his satisfaction on the rigorous discussion over the paper presented, and the comprehensive comments received from the renowned expert personalities. He appreciated the point highlighted by Mr. Poudel that the detail record of the activities needs

to be captured and documented from the first minute to the last minute of each and every action. Secondly, handling media is very important; if not handled properly, things deteriorate with misinformation dissemination. He agreed with the point raised about the arrangement and competencies of the DRR facilities. He posited that it is an issue that needs to be evaluated. This issue should also be highlighted in this compendium chapter. Similarly,

he said that the community response is very important in rescue and relief which is a major part of disaster management. Further, he also agreed with ensuring simulation exercises at different level in different time periods re essential to remind people that disasters may happen any time and we have to keep ourselves prepared to cope with and respond properly for safe and resilient



Sixth Session: Governance and Institutional Arrangements

Presenter: Er. Ram K. Sapkota – Secretary NRA

Commentators: Dr. Marco Gemmer – Head of Co-operation EU and Mr. Leela Mani Poudel – Former Chief Secretary and Former Nepal’s Ambassador to China

Moderator: Dr. Chandra B. Shrestha

Session Chair: Mr. Dhruva P. Sharma – Executive Committee Member NRA



National Reconstruction Authority

Compendium theme: governance and institutional arrangements

Mr. Ram K. Sapkota – Secretary, NRA

Enabling deliver of reconstruction through organisation, process and funding

Governance and institutional arrangements: overview

2015

- PDNA / ICNR commitments
- New Federal Constitution; Madesh unrest; embargo; elections
- NRA created as ‘special purpose vehicle’ (SPV) by sunset law; “blanket mandate”, with gaps

2016

- Transitioning in; PDRF; housing survey; perceived delays
- Diverse work programmes; formulated legal documents / procedures

2020

- Considerable policy evolution
- Significant progress in all fields, despite constraints
- Systems fully established in twilight organization...
- Planning for disengagement



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Challenges and issues



Fund availability, allocation, flow, channel, mechanism, approval, timing constraints

Institutional and organisational

- Dysfunctional “Reconstruction Fund”
- Funds availability: on/off Budget and Treasury
- Inadequate budget (impact of COVID-19, DP’s and GON’s priority to address current Issues): needed US\$ 9.38 billion – received US\$ 4.88 billion
- Federalism, new central government, devolution to local governments
- SPV’s ‘special institutional arrangements’ for fast tracking did not materialize
- Dual accountability of CLPIUs at the initial stage
- Delay in establishing NRA, staffing, formulating policies and procedures, while initiating reconstruction activities
- Timely approval from DP’s side
- Political transition, frequent changes of CEO, Secretary & other staff
- HRM - Managing 3,000-plus Engineers
- Hybrid political/bureaucratic model – internal and external stresses - Policy & operational modifications become apparent



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Challenges and issues (cont.)

Operational

- **Additional (COVID -19 Impact)**
- Additional responsibility for DLPIU Building
- Enforcement of COVID-19 health protocol at construction sites
- Unprecedented scale of reconstruction challenge (ranging from private housing to heritage reconstruction)
- CBS survey – 3 tiers undertaken, and very time consuming
- Delay in establishing NRA, team formation, formulating policies and procedures, while initiating reconstruction activities
- Mindset and nature of government officials used to ‘regular work’, where this is ‘special work’ requiring different characteristics
- 2015/16: two monsoons; public pressure on housing grants
- Lack of elected local bodies at the starting phase
- Decentralization: DDCs, NRA teams, uneven coverage of NGOs
- Coordination/oversight or implementation?
- Fielding NGOs equitably to all across 32 districts



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Strategies and approaches



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- The ‘specialist organisation’ (SVP) can deliver more effectively
- Grievance handling mechanism (4 channels of communication)
- Strong leadership and management structures (AC, SC, EC, SMT, DACFC, CLPIUs, DCCs, DLPIUs etc.); participatory work culture
- Monitoring on- and off-budget projects
- Endorsing three-tranche disbursement (despite political pressure for one)
- Establishment of NGO Guidelines and partnerships
- Enabling CBOs to implement up to NPR 100 million
- Communication: to all stakeholders, including media (perception change)
- Establishment of systems and processes: financial, MIS, procurement, HR, M&E, project management

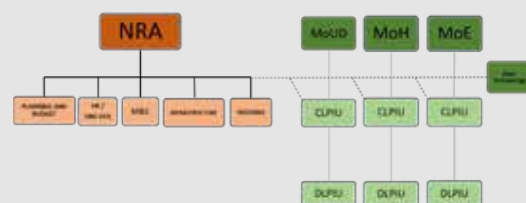


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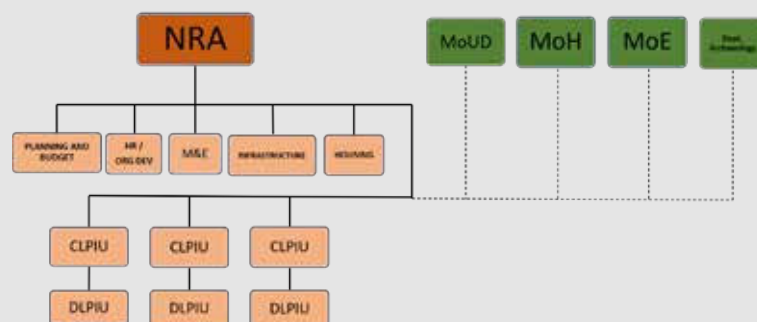
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Strategies and approaches (cont.)



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- Adaptation: remaining 'fit for purpose'; observation, monitoring and evolution of policies, procedures, and guidelines
- Collaboration: with other government agencies, DPs, LGs, partners, NGOs/INGOs
- Decentralization and devolution: from DDCs to municipalities



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Strategies and approaches (cont.)

Focus on Sustainability

- Coordination and collaboration with GON Ministries and Departments (inc. NDRRMA), and local governments on national DRM
- Revision of NBC 105 and coordination for LG implementation
- Earthquake Museum – Dharahara; Centre for Art and Culture - Keshar Mahal; Earthquake Research Centre - IoE, Pulchowk
- Preparing 'NRA Compendium': institutional memory, lessons learned
- Preparing 'International Conference on Nepal's Reconstruction' (23-25 April 2021)
- Preparing 'Disengagement Strategy': implementation from mid-2021 with NDRRMA, LGs, other actors; support from DPs

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Lessons learned



Communication, coordination and consensus are prerequisites for success

- Data management, technology, digitization
- Partners: supportive, but with their own pressures
- Such SPV needs to be mobilized immediately after the disaster
- ‘The Authority’ should be fully empowered in terms of authority and finance
- Political consensus is a must: need early whole of government consensus on mandate, policies, functional modalities
- Ensuring donors and NGOs align with government objectives
- A monolithic organisation with proper full command system essential
- Strengthening local governments for leading disaster response is sin quo non
- Programmatic adaptability essential in volatile context
- Ownership building is a difficult task – requires tireless efforts
- Institutional stability – especially of leadership & key personnel

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Reconstruction collaboration and ownership building is vital

Conclusions

- The approach of a ‘SPV’ was appropriate – it delivered! It can be a DRM institution that can be rolled over for reconstruction – now through LGs
- Disaster response and reconstruction needs an organisation with a clear accountability mechanism

This requires:

- Establishment of pre-disaster precision in agency mandate, policies and SOPs
- Incorporate essential mandate foci (e.g. devolution, poverty & vulnerability, gender, retrofitting, land issues, displacement & resettlement etc.)
- Prior clarity in special (non-bureaucratic) admin/mgt systems, e.g. authority, financial, procurement, MIS, M&E, HR planning, beneficiary participation etc.
- Full powers and institutional arrangements for successor organisation as those’ intended’ for NRA

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Conclusions (cont.)



Continuous review, analysis and update
of systems, legislation, and maintenance
of networks of stakeholders essential

- Incorporate analysis of social, cultural, environmental contexts
- Disaster/hazard risk-informed analysis in all development planning
- Develop cadre/roster of personnel functional/experienced in crisis management, DRM, DM, response & recovery

In light of above:

- Review and amend relevant legislation, policies and procedures
- Prepare and resource municipalities to respond efficiently/effectively
- Development partners: support development of 'trusted' national systems, protocols, procedures and priorities
- Ensure technical assistance is always accompanied by socio-, cultural and economic support, and backed politically

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Key messages

1. **Special purpose vehicles like NRA are essential for special types of work – autonomy in funding is required**
2. **NRA's best practices to be integrated in the relevant institutions**
3. **Research/analysis-based preparedness and adaptability**
4. **Agency stability matters: permanence, experienced personnel, career path**
5. **Money makes things possible, but humans make things happen: HRM is crucial**
6. **Such SPVs require fast track in procurement, recruitment while maintaining transparency**
7. **Partners to support 'strengthening country system'**
8. **Learn the lessons and apply the lessons learned**

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Presentation 10: “Governance and Institutional Arrangements”



The session Chair Mr. Dhruba P. Sharma started off with a brief speech regarding the governance and the institutional arrangements of NRA. After welcoming everyone to the sixth session of the NRA seminar, he pointed out the importance of the governance chapter of the compendium due to its cross-cutting nature. During the inception of NRA, many NGOs and international development partners (IDP) were already engaged in the field helping the people. NRA had the huge task of managing all that but was hopeful about the way things were going. However, very soon after establishment, NRA faced many consecutive political transitions (state restructuring process, election, etc.) which really disrupted the organizational structure of NRA. Meanwhile, NRA also witnessed many staff changes including the CEO and the Secretary which also disrupted the smooth functioning of NRA. NRA had only

three documents at that time – PDNA, the Act approved by the parliament, and the agreement between the government and donors. With these documents, NRA started their activities on December of 2015. Only after two years of establishment, some visible results of the works done by NRA were observed. So, NRA had so many issues and challenges from the very beginning which have been included in this compendium as lesson learning along with key messages and various strategies. The governance compendium chapter is being prepared based on inputs from numerous sources, including Dr. Adi Walker and Mr. Shyam Sundar Sharma from the EU-funded NEARR Facility, former secretary Mr. Pratap Kumar Pathak, and Dr. Nigel Fisher – currently a consultant to the World Bank, and which will be submitted in the ICNR-2021. Since the document is still in its developmental phase, it is requested that everybody contributes to enriching it and making it complete. He then called upon the NRA Secretary Er. Ram K. Sapkota to make his presentation on the theme of governance and institutional arrangements.

Er. Ram K Sapkota began his presentation with an overview on governance and institutional arrangements around NRA and the overall recovery and reconstruction process. He stated that immediately after the formation of NRA in 2015, there were two documents: a need assessment called PDNA, and ICNR commitments. Debate on new the constitution, Madhesh unrest, embargo and election talks impacted on everything including NRA; there was so much uncertainty at that time. In such a turmoil, NRA was

created as a ‘special purpose vehicle’ (SPV) by the sunset law and given a “blanket mandate”. By 2016, while still in transition, the PDRF, housing surveys, and lots of other processes for the preparatory work were completed in NRA. The movement was then very slow. This is the time when NRA prepared the entire legal documentation to move ahead. As of 2020, systems are now fully established, with a considerable amount of policy evolution observed, and, despite constraints, a significant progress in all fields has been observed. So, from the Institutional and organizational point of view, NRA was formed – did preparatory work – implemented the massive reconstruction work – monitored, evaluated and achieved almost 90% progress, which is quite extraordinary considering.

As of 2020, systems are now fully established, with a considerable amount of policy evolution observed, and, despite constraints, a significant progress in all fields has been observed.

He also presented the key challenges starting from the dysfunctional “Reconstruction Fund” and funds availability in terms of on and off budgetary provisions and inadequate budget, i.e., the required budget was 9.38 billion USD, whereas NRA only received 4.88 billion USD. Elections and the formation of governments took time, with political priorities being diverted. The SPV’s ‘special institutional arrangements’ for fast tracking did not materialize. Delays in establishing NRA, staffing shortages, the initial slow formulation of policies and procedures, all had to be handled while

initiating reconstruction activities. Political transition, frequent changes of CEO, Secretary & other staff, managing 3,000-plus engineers, etc. was not an easy job. He also proudly explained that NRA and its leadership succeeded to create political consensus and resolved all the issues and moved forward. He mentioned that the lessons they learned are the approach of a ‘SPV’, which was widely appreciated. It should now be a DRM institution that should be rolled over for reconstruction, including through LGs. Disaster response and reconstruction needs an organization with clear accountability, which requires a pre-disaster precision in agency mandate, policies and SOPs, and incorporating essential mandates i.e. devolution, poverty & vulnerability, gender, retrofitting, land issues, displacement & resettlement, etc. It must have clarity on non-bureaucratic (humanitarian) as well as administrative and management systems. It requires autonomous financial and procurement authority, MIS, ME, HRM, beneficiary participation, etc. He concluded with a strong recommendation for full autonomy and institutional freedom for the successor organisation of NRA.

Response by the commentators

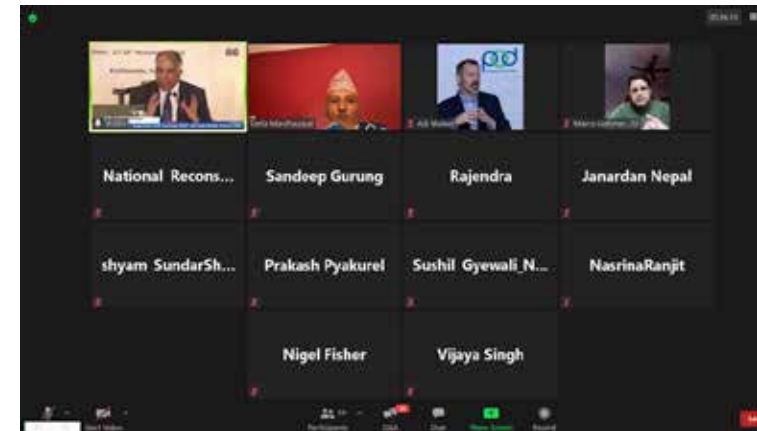
The first commentator Dr. Marco Gemmer began with an appreciation for NRA in organizing the seminar and inviting him as one of the commentators. He spoke specifically on the government situational arrangement, functioning and sustainability, leadership, teamwork and methods, approaches, partnership and mobilization of trained human resources. He began with the thought that he does not really want to comment, but rather give some inputs.



He said that everybody is talking about the reconstruction and recovery at the moment, but people have short memories and if there is no disaster in a year or two then people will start forgetting about all this because it is a human nature. As a result, the very institution created to respond to disasters will become passive, resources will become minimal and preparedness will also become inactive. In such a situation, if any disaster strikes, suddenly we will be in panic mode and start doing things from zero. To break this cycle, he suggested to continue preparedness activities regularly; the institution to be kept intact with HR, resources and simulation exercises carried on with the people, etc. This message should be given to the concerned ministries through regular presentations. He pointed out that the presentation shows the frequent changes of CEO and Secretary throughout the life of NRA. To get benefit from NRA experiences, one area could be the grievances management. Trust building and forgiveness can be considered

as a robust model of knowledge management and transfer to the new institution NDRRMA. He stated that the EU always encourages such work, and the EU will be happy to be a part of such an initiative in future too. At the end, he mentioned that as the presentation highlighted 5 messages, we may forget the incidences as any disaster does not repeat every year. If we forget, we do not keep systems at work and cut the budget. So, in every incident we will lack funding, capacity and so on. Therefore, continuation of funding and functional systems is very much required. For this, NRA could ensure the adoption and use of all their best practices in appropriate institutions from federal to local government level, right down to community level.

NRA did good work in transforming itself within a political transformation in Nepal to the new setup of federal arrangements.



The second commentator Mr. Leela Mani Poudel began with the background of evolution of NRA recalling the debate about the existence of NRA. Many were in the view that MoUD should have been given the authority and resources or form an organization as the government of India did in the past. Either way, NRA was formed in a hybrid shape, and there are some lapses in this hybridity of NRA. The government had an idea that such a new agency should have capability and knowledge about earthquake management and inter-ministerial coordination with full resources, ensuring faster but more flexible services. He also mentioned that he sees some lacking in the presentation, so he suggested the presenter to get back to him to see the document once again and remove confusions by updating the document if necessary. He opined that the right institution must hold appropriate authority, with transparency. If there is a problem of staff retention, NRA can outsource or depute from other ministries. Either way, NRA should

not be like a huge administrative entity, but rather a smart and professional agency. You are not merely occupying space to produce documents (many committees, manuals, policies, etc.), which in many instances, overlap one another. NRA did a good work in transforming itself within a political transformation in Nepal to the new setup of federal arrangements. Acknowledging all the past efforts and current achievements, he concluded with some notable suggestions which can be summarized as, a hybrid organization would be the best model to work during an emergency situation. His suggestions also included, that agencies should have capability and knowledge so that they can facilitate other agencies; centrally deposited funds should be channelized through authorities; role responsibility and capability enforcement should be specified; staff allocation norms should be clear; high turnover of leadership i.e. CEO, Secretary and senior staff should be minimized; recruitment should not be handled by the government, but rather they should be brought in from the market through an outsourcing approach for specific jobs. He contoured that NRA was not conceived as an executing agency in the beginning, instead more as a policy formation and coordinating body, and as mentioned earlier too much occupying of time and resources in producing documents and guidelines does not makes sense; documentation should be made a part of the implementation process.

Presenters' views on comments



The presenter Er. Ram K. Sapkota thanked the commentators for their constructive comments and fruitful suggestions. He highlighted one suggestion from Dr. Marco Gemmer that is very important – forgetting and funding. If we forget, we will not receive funding anymore. So, in order to retain that, the NRA must continue working hard till the last day and must document everything and must internalize. Then he focused on the comments made by the second commentator Mr. Poudel. He acknowledged Mr. Poudel as somebody who has a wealth of knowledge in public administration. He summarized Mr. Poudel's comments as a brief description of the history of NRA; its experiences and challenges. He said that Mr. Poudel focused on the practical aspects of what should be considered and what should be left out – very valuable feedback for a disaster response entity. He assured that all these

comments and suggestions have been noted and recorded as well for documentation purposes. He informed that the compendium chapter is in its final iteration by an expert and is being enriched by professionals from all around the world. The comments and suggestions collected from this seminar will further enrich the document.

Comments & queries from the chatroom and social media

Er. Ram Prasad Bhandari: Very vivid and strong conclusions. Congratulation Ram K Sapkota and the team. Among all, I would like to reiterate on – 1. Disaster-risk-informed development planning, 2. Develop cadre and the roster of DRM.

Disaster-informed analysis in development planning would definitely be a huge leap towards “anticipatory governance”.

Deependra Sharma: The nature of governance in Nepal in relation to DRM can be called as “contingency governance” rather than “anticipatory governance”. This is evident through the focus of government machineries on “rescue” and “relief” functions executed as and when disasters happen. Disaster-informed analysis in development planning would definitely be a huge leap towards “anticipatory governance”. But this is the main challenge in the Nepalese context. The establishment of NDRRMA has created opportunity for the departure of NRA. But there are lots of

challenges mainly originating from the institutional structure, mandate and capacity of the NDRRMA itself. As rightly pointed out by the past Chief Secretary Mr. Leela Mani Poudel, NDRRMA cannot fulfill its mandate by operating as one of the departments of MoHA. A first step towards ensuring the proper functioning of the NDRRMA would be to immediately bring it under the OPMCM. Though we hired engineers and overseers faster than expected, it was very difficult to retain them. The ones that were recruited were fresh engineers lacking experience in dealing with the complexities of reconstruction. Now after four years, we have a pool of experienced and professionally confident engineers. How we are going to engage them after the completion of reconstruction works has now become a big challenge?

Shanker Dhakal: I like the way Mr. Leela Mani Poudel is so straight forward and to the point with so much gravity in his experiences, suggestions and constructive criticism in his observations. I really feel the GoN should bring him again in some policy leadership role soon.



Vijaya Singh: One more thing that we should perhaps analyze that remained unresolved was having a dedicated reconstruction fund managed by NRA. This highly delapidated the capacity of NRA to make the line agencies responsible to NRA.

Remarks by the CEO NRA

In his concluding remarks, the CEO Er. Sushil Gyewali appreciated the comments received especially on the legal provisions; their use and requirement to revisit them to make NRA a smart institution. There should be a balance in the growth of the institution, and the expectations of people should always be the top priority. He agreed that all comments are crucial, very valid and eye opening. In it's given autonomy, NRA has developed many documents, guidelines, policies and formed different committees to fulfill its goals. Altogether 6,000 engineers were trained. Clarity is to be developed that performance evaluation of the Secretary was made by the

committees headed by the Chief Secretary. NRA CEO should have been given this role but that is not the case. Hopefully, NRA will change this provision and the CEO will be responsible for evaluation so that the concerned line agencies/staff will be accountable towards the role they have been given. NRA was started at a time when the local bodies had no representatives, transition was going on from no system to the federal system; three level of elections and the governance system itself was changing. In such a situation, there was always a problem in the development of policy, legal provision, and fund management because every project was going through the Ministry of Finance and NPC. Later, we had better coordination and the problem was resolved. There is a felt realization for the need for more autonomy regarding staff, resources, fund, evaluation and management. If it is managed well, it will be helpful not only to strengthen and speed up the NRA works, but the future organisation NDRRMA will also benefit. He highlighted some efforts that helped to bring faster growth of NRA and achieve the results that we have today. For the leadership role, he mentioned that he took a stand and got some autonomy. NRA's activities were deconcentrated in other terms, and roles and responsibilities were given to the local governments. However, once they formed an advisory council, it played a very important role in creating political consensus and in the management of funds. He reminded that by next July 2021, many things have to be done and documented properly. In addition, some signature activities, like – establishment of the earthquake museum and research & resource center are in the pipeline. The establishment of NDRRMA as the successor of NRA is also an achievement, and the NRA is working hard to complete the mandate and make ready

a package full of process stories, success stories, struggles, achievements, failures, strengths, weaknesses, knowledge, experience and suggestions to transfer from NRA to NDRRMA.

NRA is working hard to make ready a package of stories: struggles, successes, processes, achievements, failures, strengths, weaknesses, knowledge, experiences and suggestions to transfer to NDRRMA.

In addition, he also responded to some comments received from the commentators. He agreed that transparency, accountability and legal documents are necessary to govern an organization and to fulfill the expectation of people; creating a balance between people and organization. Especially, for a hybrid type of organization like NRA, it was and is a massive task in any circumstances, as NRA needed to rely on other government authorities. NRA had the provision to use the human resource from the government ministries, and it was duly done. The roles of CEO and Secretary sometimes got mixed and confused, but that did not affect the decisions and implementation. The delay in the establishment of NRA and the consensus to speed up NRA, both were political decisions. Autonomy on the financial sector, human resources and reconstruction fund were problematic due to the legal provisions, so there is a clear realization to have more autonomy in fund mobilization and human resource management. Lastly, he informed that 100% of the total fund received from the international donors has been fully utilized. Thanking everyone, he concluded with a

note that the outcome of the seminar will be shared through the seminar proceeding report soon.

The commentator Mr. Leela Mani Poudel quickly shared a thought that this seminar report should clearly mention that there must be financial as well as managerial authority for the institution who oversees emergency response to mobilize funds and human resources. Once there is legal and financial authority, only then can institutions like NRA run and get desirable results. In response to the CEO, he mentioned that there is no provision for the Secretary evaluation. He suggested to include in the report that CEO must have the authority for mobilizing funds, legal mandate and human resources, and responsibility for the performance evaluation of

the Secretary. Because there is currently only provision for a performance evaluation system applied up to Joint Secretary level.

Closing remarks by the session Chair

The session Chair Mr. Dhruva P. Sharma began his closing remarks by thanking everyone who had contributed to the session, as well as in the post-presentation discussions. He mentioned that many observations, comments, queries, feedback and suggestions, etc. have been received physically and virtually which are being recorded for the proper documentation. Finally, he thanked everyone again and closed the session.

Seventh Session: Work in Progress of the NRA Impact Study

Presenters: Prof. Dr. Govinda Nepal, Dr. Biswo Poudel and Mr. Rudra Suwal

Commentators: Mr. Kamran Akbar – World Bank with support from Dr. Nigel Fisher and Mr. Bishal Thapa – Economist

Moderator: Dr. Chandra B. Shrestha

Session Chair: Er. Sushil Gyewali



National Reconstruction Authority
Socio-economic impact study
 Pr. Govind Nepal, Dr. Biswo Poudel, Dr. Rudra Suwal

**Evaluation of post-earthquake reconstruction
 and its immediate socio-economic impacts**



National Reconstruction Authority

Earthquake, loss and damage, and reconstruction needs

- An earthquake of magnitude 7.6 struck on 25th April 2015, with several aftershocks following it
- Caused 8979 casualties and injuring 22,309 individuals
- Thirty-one districts out of 75 impacted
- The National Reconstruction Authority (NRA) was formed in December 2015 to plan the reconstruction

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National Reconstruction Authority

Reconstruction needs

PDNA and PDRF estimated Reconstruction need of :

- 790450 private houses including retrofitting of 70k houses
- 7553 public schools
- 1197 health institutions
- 920 archeologically important structures
- 415 government buildings and
- some public roads and other associated public goods

Total cost of the reconstruction -NRs 630 billion

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Major mandates of NRA

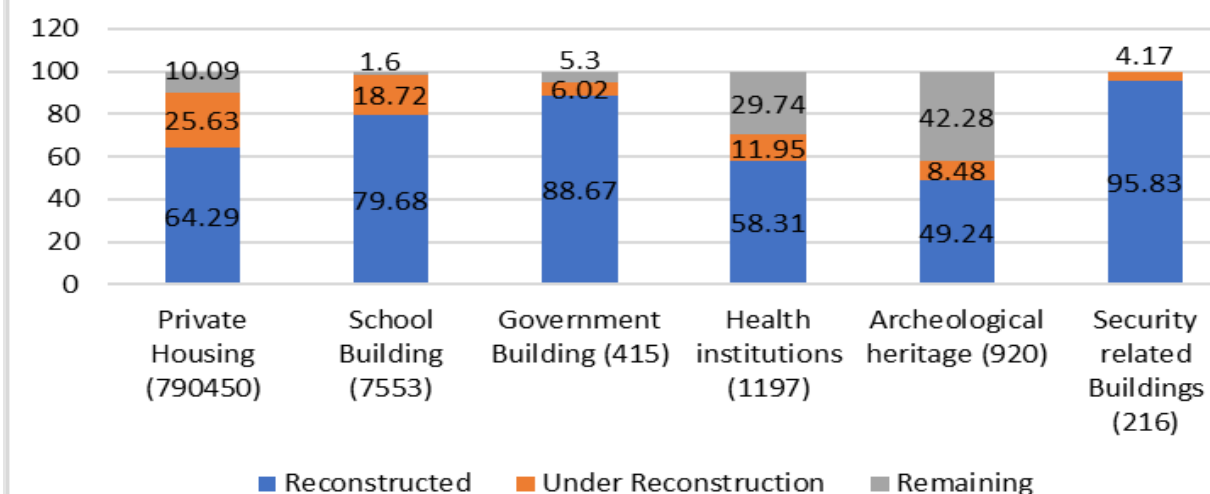
- Reconstruction of private houses, development of integrated settlements, and resettlement of risky residences
- Reconstruction of public health institutions
- Reconstruction of educational institutions
- Reconstruction of archaeologically and historically important heritage
- Reconstruction of government housing
- Reconstruction of any other public structures and infrastructures as considered necessary by the authority

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Physical Progress of Reconstruction until July 15 2020



Source: Presentation made at pre-NDAC Meeting in NPC by NRA

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Need for process and socio-economic impact evaluation

- Record the lessons learned
- Evaluate the NRA performance and immediate impacts of reconstruction on various socioeconomic variables
- Assess contribution of reconstruction to GDP
- Prepare knowledge product that may help government better address post disaster recovery and reconstruction in the future

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Objectives of the study

- The major objective of this study is to prepare a comprehensive evaluation of the works performed by NRA with focus on reconstruction approach and socio-economic contribution.
- Specifically, the study will:
 - Evaluate the total reconstruction efforts against the estimated damages and stated goals of the NRA
 - Document the recovery process
 - Evaluate how reconstruction affected the economic activities in different sectors
 - Evaluate socio-economic impact of reconstruction
 - Quantify the impact of reconstruction efforts on GDP growth and identify channels through which the impact operated
 - Suggest the optimal path for the closure of the NRA

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Study approach / methodology

- Identify indicators of evaluation
- Use primary and secondary data/ information for evaluation
- Use baseline data generated by NRA and others as benchmark information
- Use progress report of NRA, study/evaluation reports, conference articles as literature
- Compare input and output against targets within given institutional setting
- Conduct 3-5 focus group discussion to understand general perception of the earthquake victims
- Carry out field survey to fulfill data/ information gap
- Find out disaggregated impacts of reconstruction against social and income group as well as urban rural settings

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Task 1: evaluate total reconstruction efforts

- i. Summarize damages sector-wise using official documents
- ii. Summarize official mandates provided to NRA
- iii. Summarize the official reports of NRA on task completions
- iv. Summarize budgetary sources (annual or others)
- v. Survey and summarize existing evaluations

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Task 2: document the recovery process

- i. Evaluate early coordination efforts between different agencies
- ii. Evaluate transitional arrangements before reconstruction (sector-wise)
- iii. Evaluate access to finance of the victims during the recovery phase
- iv. Analyze budget release process against budgetary allocation

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Task 3: establish relationship between reconstruction and sector- wise economic activities

- Relationship between reconstruction and agriculture sector activities (how it evolved over time in the last four years)
- Repeat that for different major sectors

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Task 4: establish relationship between reconstruction and other important variables



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- The dependent variables of interest for us will be:
 - School attendance (binary outcome on whether a child goes to school, time taken to go to school, average score of children in school, whether anyone has failed in class)
 - Health Service Outcome (average number of times a hospital is visited in a year by the household members, whether a household member has died of curable disease, time taken to visit the hospital)

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Task 4: establish relationship between reconstruction and other important variables (cont.)



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- Future disaster preparedness (strength of a house, availability of safe and open space)
- Household indebtedness (loan amount owed by the household as a percentage of their annual income)
- Agricultural income, nonagricultural income, possession of technology for farming
- Household violence (whether a violence has occurred, whether police case is reported) this can be verified by ward or district level data prepared by police
- Migration (number of people migrated to Nepali cities, number of household members migrated to other countries)
- Skills (number of households possessing skills (get from NRA on the kind of skills they have trained people on))
- Enterprises formation (binary indicator on household owning business; total annual turn over and profit)
- Employment (number of household currently employed, income of employed household members)

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Sources of treatment

- Reconstruction of private house
- Development of integrated settlement in the ward
- Resettlement of risky settlements in the ward
- Construction of public health institution
- Construction of education institution
- Reconstruction of heritage building
- Reconstruction of government buildings
- Reconstruction of other public infrastructure

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Data

- Using baseline data collected by NRA. Also, all other relevant data collected by other government agencies such as CBS (NLSS etc.)
- Recent data will be collected using enumerators
- Total Districts 31
- Neighboring wards of several of these districts
- Expected samples: 3500-4000, at least 100 from each affected district. Weighted by Population and geography

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Methodology: difference-in- differences

- Proposed method: Define

$$H_i = \begin{cases} 1 & \text{if household } i \text{ is from highly affected region} \\ 0 & \text{else} \end{cases}$$

$$M_i = \begin{cases} 1 & \text{if household } i \text{ is from moderately affected region} \\ 0 & \text{else} \end{cases}$$

$$L_i = \begin{cases} 1 & \text{if household } i \text{ is from mildly affected region} \\ 0 & \text{else} \end{cases}$$

$$T_i^{(v)} = \begin{cases} 1 & \text{if treated with } v \\ 0 & \text{else} \end{cases}$$

$v \in$
{reconstruction of private house for household i , development of integrated settlement, resettlement of risky residences, construction of public health institutions, construction of educational institutions, reconstruction of heritage building, reconstruction of government housing, reconstruction of other public infrastructure in the ward where household i is located}

Model:

Interpretation: β_5 gives the impact of reconstruction activity in highly affected region, β_6 gives the impact in mildly affected region and β_7 is the impact in lowly affected region
We will use the variation of these regression methods to estimate the impact over time as well

$$E[Y_{it} | H_i, M_i, L_i, T_i^v, X_{it}, t] = \beta_0 + \beta_1 H_i + \beta_2 M_i + \beta_3 L_i + \beta_5 H_i T_i^v + \beta_6 M_i T_i^v + \beta_7 L_i T_i^v + \gamma X_{it} + \varphi t$$

Task 5: Impact of reconstruction expenditure on GVA growth by major divisions



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Objective of the economic impact assessment task



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The main objective of the economic impact assessment is to understand the range of major economic impacts and changes occurred from the earthquake reconstruction activities.

Major economic areas

1. Gross value added
2. Gross fixed capital formation (GFCF)
3. Employment and compensation of employees
4. Household and government consumption
5. Household income (disposable income)

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Approaches of economic impact assessment

1. Scoping
2. Baseline profiling and identifying who are impacted
3. Monitoring actual impacts
4. Evaluation and assessment of direct impacts through
 - secondary data analysis of existing data sources
 - primary data collection such as fresh surveys and focus groups
5. **Mixed approach of analysis** (applicable statistical tools: rate, ratios , correlation, regression etc.)

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1. Impact of reconstruction on gross value added (GVA)

- An estimate of Output / GVA by major economic divisions
 - A. Agriculture
(Agriculture and forestry, fishing)
 - B. Industry
(Mining and quarrying , manufacturing, electricity gas and water, construction)
 - C. Services
(Wholesale and retail trade, Hotels and restaurants, Transport communication and storage, Financial intermediation, Real estate renting and business services, public administration and defense, Education, health and social work, other community, social and personal services)

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Methodology of estimation and data sources

Methods

The concept of output/ GVA is based on the National accounts methods(System of National Accounts, SNA)

Indicators are prepared based on the data/information by major economic divisions.

Estimates are derived indirectly using those indicators

Time frame

Base year 2014 (before earthquake)

Estimates for years : 2015, 2016, 2017, 2018, 2019, 2020

Data sources

- Central Bureau of Statistics; National Accounts of Nepal
- National Reconstruction Authority; NRA reports
- Reconstruction Impact Assessment Survey
- NHRP database, MIS

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2. Impact of reconstruction on gross fixed capital formation (GFCF)

Gross fixed capital formation is measured by the total value of acquisitions less disposals of fixed assets during the accounting period.

- Dwellings
- Other buildings and structures (roads, bridges, public buildings, heritage etc.
- Machinery and equipment /valuables
- Major improvements to tangible non-produced assets, including land

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2. Impact of reconstruction on gross fixed capital formation (GFCF) (cont.)

- Estimates of GFCF by types of funding (cost approach)
 - Nepal Government
 - NGOs
 - National
 - International
 - Other government
 - Individual households
 - Others

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2. Impact of reconstruction on gross fixed capital formation (GFCF) (cont.)

Methodology of estimation

- Construction / reconstruction activities (assets) are valued at
 - Cost price (expenditure)
 - Market price (at current prices)

Data sources

- NRA , report on cost of construction
- CLPIU, tranche data progress
- Proposed Reconstruction Impact

Assessment Survey

- NHRP database / MIS
- Financial assistance by INGO

Study period: 2015-2020

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3. Impact of reconstruction on employment

- Estimates of employment and compensation of employees (wages and salaries) generated by reconstruction activities
 - Engineers/ overseers
 - Masons
 - Labors
 - Others

Study period : 2015-2020

Data Sources: NRA bulletins and reports, Proposed Reconstruction Impact Assessment Survey/ study (FGD KII)

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4. Impact of reconstruction on household consumption and government consumption

- Estimates of actual final household consumption by reconstruction activities
- Estimates of government consumption by reconstruction activities

Data Source:

- Proposed reconstruction impact assessment survey
- MoF, Government expenditure details

Study period : 2015-2020

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5. Impact of reconstruction on household income

- The income which can be used for final consumption expenditure and saving includes current transfers disposable income.

- Factor income
- Current transfers

Total income (disposable income)

Data Source : Proposed reconstruction assessment survey

Methods

National Accounts of Nepal , SNA

Time period: 2015 -2020

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Proposed title of the study

Title one: Ex-post evaluation of socio-economic impact of Reconstruction

Comment: Socioeconomic impacts are still emerging, and it may take at least another 5-10 years to become apparently visible

Title Two: Terminal evaluation of socio-economic impact of Reconstruction

Comment: NRA activities will continue next one year and as said above Socio-economic impacts are just emerging

Therefore, we propose:

Title Three: Evaluation of Post -Earthquake Reconstruction and its immediate socio-economic Impacts

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Presentation 11: “Evaluation of post-earthquake reconstruction and its immediate socio-economic impacts, scope and methodological approach”

The presenter Prof. Nepal started his presentation with the background of the evaluation of post-earthquake reconstruction and its immediate socio-economic impacts, scope and methodological approach. He briefed on the 2015 earthquake destruction. The first task they have been given was to evaluate total reconstruction efforts. Task 2 was to document the recovery process, and task 3 was to identify the relationship between reconstruction and sector-wise economic activities. Focusing on this, he said that the title for this project is yet to be finalized. Data regarding the reconstruction needs were presented followed by the major mandates of NRA regarding data of the remaining reconstruction of private houses, school buildings, government buildings, health facilities and cultural heritages. Prof. Nepal stated that they need the records for the future as lessons learned, and to evaluate the NRA performance regarding the immediate impacts of reconstruction on various socio-economic variables, given the major object of the study is to prepare a comprehensive impact evaluation of the works performed by NRA.

In the initial stage, some indicators were developed to shape the collection of primary and secondary data. Consequently, the first-year primary data was collected. With this, Prof. Nepal called upon his colleague Dr. Biswo Poudel to carry forward the presentation

based on task 4, which is focused on identifying the relationship between reconstruction and other important variables, like education, health care, income, household violence, migration, skills, enterprises formation and employment. Dr. Poudel focused on the sources of impact from the reconstruction of private houses, heritage buildings, integration, education, public health institutions and risky settlements at a ward level. The data were based on the NRA baseline and recent data collected from 31 districts and neighboring wards. Then Dr. Poudel handed over to Mr. Rudra Suwal to continue with the presentation for task 5, related to the impact of reconstruction expenditure on GVA growth by major economic divisions. He stated that the main objective of this part was to understand the range of major economic impacts and changes that occurred due to the earthquake reconstruction activities. He added, the approaches of economic impact assessment are scoping, baseline, M & E, and the assessment of direct impact for analysis, using mixed approaches of analysis. According to him, the impact of reconstruction on gross value added (GVA) is an estimated output by the major economic divisions of agriculture, industries, and services. The study also focused on the impact of reconstruction on gross fixed capital formation, and the impact of reconstruction on employment – generated for engineers, overseers, masons, labourers and others. Also intended in this study,

is an evaluation of the impact of reconstruction on household consumption and government consumption, and the impact of reconstruction on household income by consumption expenditure and saving, including current transfer and disposal income.

Looking at the task of NRA, there is a need to have strong decision-making and conflict resolution.

Response by the commentators



The first commentator Mr. Kamran Akbar found the presentation informative. He stated that construction is a leading sector in Nepal. There are about 20-24 areas in infrastructure and construction. He mentioned that Nepal has a difficult terrain, which is why the recent damage has affected the tourism sector so dramatically. In order to educate about disaster management, NRA must work on preparedness as well. He posited that maybe a separate study is needed to explore the status of preparedness for disasters and

emergencies. It is too early to see the impact in such a short period of time, when NRA has not even completed its planned work yet. PDNA was a quick assessment and the damage assessment was purposeful, yet this could not present the whole complex scenario. So, conducting an independent study may be a good idea to know the real state of impact generated by the activities of NRA. For instance, cash retaining capacity, insurance provision and income generation and so on seem essential. He suggested two more points – in the case of education, has the curriculum been affected by the earthquake? He emphasized that from a socio-economic point of view, data needs to be seen from the geographical spatial view. So, he suggested that there is a need to highlight the preparedness of the country to make the development agenda. He also suggested that the damage assessment process needs to look at other factors as well, not only the damage. Every sector study needs to be undertaken either separately or in a combined way to maximise their value. Finally, he mentioned that education facilities are not only associated with bricks, mud and buildings. So, it is also important to look into the damage in the soft learning environment as well as the effects on the curriculum.



The supporting commentator Dr. Nigel Fisher agreed with the comments made by Mr. Kamran Akbar. He congratulated the presenters for proposing an ambitious analysis. He confessed that he has never seen a study of this ambition with such a range of intended lessons learned. Being a social scientist, he said he would comment basically on social impacts mainly at household, family, and community level rather than at the macro-economic level. He inquired if the analysis also looks at conflict resolution in decision making when talking about cultural heritage, as there have been strong disagreements in that sector. Disaggregation of data is required when you calculate the extension of working hours: who lost jobs, and working hours, and who gained working hours. From that, who has gained more income, who lost earning, and how does it affect their livelihoods, are important elements to know, because such data will give us learning for future reference. Looking at the task of NRA, there is a need to have strong decision-making and conflict resolution. Also, the record has to be updated on who lost work and who benefitted.; how many marginalized and vulnerable persons benefitted needs also to be identified. He questioned the title, “long term impact or short-term impact”, especially in relation to what extent will the study be looking at the policy. How are you measuring the impact of diversification? From subsistence farming to occupational farming, are there any examples? He recalled that last June, he had an opportunity to see a paper from Gorkha, and similarly, another paper was about coping strategies. Is there any further study or recommendation from those papers? Of course, there are a series of question that need to be asked on diversification of opportunities. Many victims of such disasters will of course ask many for things but if we boil this down, they

will ask only three main things: education for their kids, medical facilities, and employment. So, these three things are important for the study when planning and undertaking the surveys.



The second commentator Mr. Bishal Thapa started off mentioning that the presentation is amazing in terms of information, but he also stated that it is an ambitious study. He commented that the “immediate” word in the title itself may not be suitable because it limits what the study wants to do. In terms of structure of the study, it seems it is trying to bring two things together that are not necessarily the same: 1. Evaluation, 2. Impact study. An evaluation by definition is being held accountable to certain sets of targets, whereas an impact study is just to look at effects; it is not really an assessment. In that sense, part 4 and 5 were little bit outside an ‘evaluation’ in the strict sense. Also, regarding mandates and goals, the NRA has been given certain mandates that revolves around reconstruction only. However, this study also gives away some sense of goals that discuss about building resilience, which attempts to look at different elements of socio-economic impacts.

Here, a question arises – is this different from a mandate? Was the mandate of NRA to reconstruct or was it to create resilience? Or was it both? Was it to create an economic impact? Thinking about all these questions, were - at any point in the work of NRA – the programmes structured because of the mandate NRA received, to create a specific kind of output beyond ‘just reconstruction’? If the answer to this question is yes, then this correlated solely with the part of the study entitled evaluation, and if the answer if no, then it should not be limited to just evaluation. Secondly, while looking at the evaluation methodology in terms of the metrics that are typically used: 1. The question of relevance, 2. Efficiency, 3. Effectiveness, 4. Impact and 5. Sustainability. Additionality. If the first three parts of the study could be structured around those lines, it makes it easier. Looking at the impacts in part 4 and 5, it is also important to look at the opportunity costs and at the value of constraints. His final query was, to what extent was it crowding out? Sometimes when programmes are announced, they immediately begin to crowd out a lot of other actors and if there is delay, there is a huge impact. So, to what extent did this intended programme result in any crowding out? He shared that he felt the methodology was very robust in terms of what is being presented. He further shared that he wanted to add two things – 1. It is important to capture public perceptions also. What do people really think about NRA? 2. It is also important to look at the communications around NRA as it influences effectiveness. To what extent and how was the media coverage of this initiative? What was the profiling of it? What was the message that everybody was hearing? How were the outcomes and outputs being documented and presented to the national audience?

Presenter’s views on comments



Prof. Nepal, on behalf of the presenters admitted that there were many constructive comments and points from the commentators, especially from Dr. Nigel Fisher who raised the issue about the decision-making process regarding conflict resolution in the reconstruction of cultural heritage. The focus on socio-economic aspects will definitely be improved. Ample focus will also be put into the empirical data. Dr. Biswo said that the study looked into the short-term impacts. For instance: NRA has built 600 health post buildings where we can assess the short-term impacts focusing on whether it has helped to resume the services and how far the services have been covered, etc. Likewise, impacts on the number of school attendants and hospital attendants will be observed. About GDP, there was an investment of about 3 billion USD, so supposedly it may have impacted on the national GDP. Housing is also a measurable component for economic strengthening of the people/households. Most of the investment has been made

in housing from NRA, so it was justifiably taken as the main component for impact assessment. Mr. Suwal agreed to the comments regarding the study being an ambitious one and assured that the comments received are taken positively and they will consider these comments at the next stage in elaborating the assessment.

Comments & queries from chatroom and social media

Er. Ram Prasad Bhandari to the CEO and Prof. Nepal: Thank you for undertaking such a very important study. I am wondering whether the “Socio-economic impact of reconstruction” will also cover the impact to people and the policy makers regarding the understanding and change in the mind-set about Disaster Risk itself? If this is not within the scope of this study, it is very much recommendable to take on that study as well. Otherwise, we risk the syndrome of forgetting, as cited by the EU Commentator in the previous session.

Rabindra Suwal: To the evaluation team, what will be the type of impact evaluation? For example – retrospective, comparative, cause-effect, quasi-experimental? To Prof. Govind Nepal, do you have or have planned to develop a theory of change for the evaluation? if yes, what is the theory of change? To Dr. Biswo Poudel, how are non-disaster affected and disaster affected populations comparable? In this situation, would not the result be biased? Likewise, you have talked about school attendees; are they being measured against the effects of the earthquake or effects of the reconstruction? Lastly, what is the purpose of Focal Group Discussions?

Maggie Stephenson: Very glad to see this research initiative, it will be very helpful. Two remarks: 1. How will your analysis account for the fuel crisis, Covid-19 and other short-term events and long-term trends during the recovery period? 2. To avoid overstating the impact of the earthquake, you may need to look at longer term patterns of change, starting years before 2014, and continuing longer than 2021. Also, continuing research over a longer period of time will also be needed to understand the impact of the earthquake and the recovery programmes. How might the research design plan for a feasible longer monitoring and learning process?

Deependra Sharma: This study is a must and hence a welcome proposal. I see a huge challenge in interpretation of the data due to the overwhelmingly negative impact of Covid-19 on the socio-economic aspects/parameters.

Bishnu Bhandari: This study should have an overall strategy – not only economic but it should deal with social aspects, even the quality of life and number of changes that have taken place in 5 years. This is a post evaluation of socio-economic impact that was suggested by author Dr. Suwal, but I think it should cover socio-economic performance.

Toby Vokuhl to Dr. Govind Nepal: In addition to the quantitative data, you will be collecting in regard to the different indicators, will you also collect qualitative data from building occupiers? Sometimes a gain/loss in ‘capital’ is differently perceived by occupants compared to numerical data. Many thanks.

Youb Raj Paudyal: Excellent presentation and comments. It could be interesting if we analyze people’s understanding and more orientating the study towards disaster response. Are they self-motivated

to use technical persons/trained masons or just for the moment of utilizing the grant system and physical implementation? What are the lessons learned from the overall recovery process to contribute to a disaster resilient society in future?

Nimananda Rijal: Growth in GDP from the housing reconstruction programme of NRA will not indicate proper economic development. Housing reconstruction provides a short-term benefit to the community, particularly for the artisans who are engaging in construction.

Vijaya Singh: I also feel the study focus has to be sharpened as it seems to be going beyond NRA’s mandates. It would be good to focus on unveiling what is not known yet, for example, contributions to the economy and their sustainability as well as making the society more resilient to future risks.

Anonymous: Sampling approach should be applied to study public perceptions.

Closing remarks by the session Chair



The session Chair Er. Sushil Gyewali appreciated and thanked everyone for their participation and contribution. He assured that all the comments will be taken into consideration, and as far as possible they will be incorporated in the study. If in case this will not be possible now, they will be considered for future projects of this nature. This study was focused on socio-economic aspects so there are some areas that are naturally missed out; those areas will be covered in other studies in future. Since the NRA activities are still going on, this study only covers the short-term or immediate term impact. It is for sure that the long-term impacts will emerge in the future. He further mentioned that all the comments received directly or from the chatroom are also received positively. He claimed that this was a very specific study. We will be having another comprehensive study to cover other areas of impact. We will look at such things as why the proposed projects were not implemented, and why the committed resources did not arrive? The mandate of NRA has social, physical and economical dimensions and we have to look into all those aspects. So, there is no clear demarcation of shorter or longer-term resilience impact because such a huge investment in the building sector may produce some immediate impacts and some longer-term impacts as well. Another area to look into the study are the constraints and limitations that NRA experienced. In the next run, we will also look into public perceptions. So far, what we have done is good, and we will continue to cover the other areas in sector-wise studies in near future. With this, the CEO ended his remarks and announced that all the thematic sessions of the seminar have now come to an end.

Closing Session: Summing up and way forward

Session Chair: Er. Sushil Gyewali



The closing session was hosted by Ms. Rita Lamsal. She welcomed all to join in the session and requested Er. Sushil Gyewali – CEO NRA to Chair the session. She further requested all to be seated at their respective seats and prepare for the closing session. She then called upon the ICNR-2021 convener - Dr. Chandra B. Shrestha – to deliver his closing speech.



Dr. Chandra B. Shrestha started his speech by mentioning that he was mesmerized by the participation of all the dignitaries and professionals in the two days NRA seminar. He said that NRA has showcased their work to the world through these presentations, and the discussions held for two days have indeed enriched the papers presented. As the way forward, in terms of livelihood, the fundamental issue that arose was how to link post-disaster livelihoods with on-going livelihood initiatives? In terms of the post-disaster situation, how to interlink established livelihood interventions with disaster related interventions? How to handle sudden surges in demand for employment opportunities, the management of them and the longer-term employment

prospects? Similarly, regarding livelihood vulnerability, what are the real criteria? NRA had taken an administrative solution, but the real solution is yet to be explored. Since the country is moving towards a more socialist state, it is imperative that we think about the poorest of poor. Also, employment through community infrastructure – cash for work and employment that can be generated through community infrastructure is the issue – How far is it successful? How far can we go? What are the limitations? These are the issues that the livelihood chapter has to satisfactorily address. Regarding the rescue & relief chapter, can the current NDRRMA cope with disaster preparedness? This is a common concern among all and demands a serious discourse on how to strengthen NDRRMA. Also, the roles and responsibilities must be clearly defined in rescue and relief. Regarding governance, the question about NRA's establishment as a SPV was the fundamental question, regarding whether NRA should be only a facilitating body or whether it has to be an executing body. Dr. Chandra argued that an institution like NRA must be an executing body. Regarding the impact study, the commentators and audience have passed very thought-provoking comments and queries. Are you clear in terms of scope of this study? What does NRA really want? A huge study could get overwhelming, so perhaps we need to disaggregate all these elements to try to make it implementable. NRA must come up with a satisfactory and meaningful response soon. He concluded by saying that all the commentators and participants are NRA's reference group now, and NRA will be consulting with them time and again to make sure that the compendium chapters are as refined as possible.



Er. Ram K. Sapkota expressed gratitude towards the advisory committee, steering committee, executive committee, the honorable PM K. P. Oli and the CEO of NRA Er. Sushil Gyewali for their visionary leadership, guidance and significant contribution in the Nepalese reconstruction process. He thanked the convener, the session chairs, commentators, presenters and other professionals who contributed in making the NRA seminar a grand success. He also appreciated the role of the Conference Secretariat and development partners for their relentless support. He wished every success to everyone associated in the reconstruction of Nepal.

Er. Sushil Gyewali started his closing speech by congratulating everybody associated with the NRA seminar for a very productive and successful event. He also thanked the European Union for their support and partnership in this programme. He assured that the conclusions and recommendations that came up during the seminar will be addressed in many ways and through many

channels, and of course, through the proceeding report. All the outcomes of the two-day seminar will either go to operation or to policy formulation. They will not be lost. Likewise, some of the suggestions will be applied immediately and some of them will be documented for future reference in disaster preparedness, planning and programmes. He reassured that all the documents will be safely kept within NRA's Institutional Memory. They will be handed over to NDRRMA. Some recommendations will be passed over to the National Planning Commission also, which will be helpful for them to formulate the next planning period. He mentioned that the programme was very fruitful and useful to highlight the activities done so far by NRA, moreover, it was also useful to inform about the rest of the works that have still to be done by NRA, in the next months before its tenure ends. In regard to concerns over the autonomous need of NDRRMA, the next step is to convince the PM. Secondly, the thematic teams will work on the sectoral impacts and will come up with an impact study, so that the compendium chapters can be further enriched. We have to commission specific studies to cover different sectors under the NRA activities, challenges, lessons learned and so on. As we all know, ICNR-2021 will be organized in April 2021 where

these compendium chapters will themselves be presented. Hence, these studies need to be completed way before April so that we can make them ready to share more widely. He thanked everyone once again who were associated with this event directly or indirectly. He then concluded his closing speech and formally closed the seminar.



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